

ST.PHILOMENA'S COLLEGE (AUTONOMOUS), MYSURU

(AFFILIATED TO UNIVERSITY OF MYSORE)

REACCREDITED BY NAAC WITH A GRADE

DEPARTMENT OF CHEMISTRY

POST GRADUATE COURSE – SEMESTER SCHEME

CBCS WITH LEARNING OUTCOME BASED CURRICULUM

Academic year 2018-19 onwards

ST. Philomena's College (Autonomous) Mysore

COURSE -Master of Science in Chemistry

M.Sc., Chemistry

The course will be started under the existing regulations governing two year- four semesters Choice Based Credit System (CBCS) and Continuous Assessment Grading Pattern (CAGP) Master's Degree Programmes under Autonomous Structure.

Guidelines/Regulations

1. Eligibility for Admission

Candidates shall have studied any branch of basic sciences with Chemistry as one of the major/optional subjects securing 45% marks (40% in case of SC/ST candidates) in the aggregate marks in Chemistry of all the semesters at the undergraduate level. The qualification and the percentage of marks for admission shall be as per the guidelines issued by the University of Mysore from time to time.

Note: ***In case of candidates who have taken more than three years to complete their Bachelors Degree, the percentage of mark is arrived as per the guidelines issued by University of Mysore from time to time.***

2. Duration of the Programme

The duration of Programme shall extend over 4 semesters (two academic years) of 20 weeks each including instructions and examinations.

3. Maximum Period for Completion of the Programme

The candidates shall complete the programme within 4 years from the date of admission. The term completing the programmes means passing all the prescribed examinations of the programme to become eligible for the degree. No candidate shall be permitted to appear for the examinations after the prescribed period for completing the programme. Whenever a candidate opts for blank semesters/ dropped papers, he/she have to study the prevailing papers offered by the department when he /she continues his /her studies.

4. Medium of Instruction

The medium of instruction shall be English.

5. Hours of Instruction per Week

There shall be 24-30 hours of instructions per week in subjects without practical/field-work and 28-34 hours of instructions per week in subjects with practical/field-work. These hours may be

distributed for lectures, seminars, tutorials, practicals, project-work and other modes of instruction which individual courses may demand.

6. Attendance

Each paper (theory/practical) shall be treated as an independent MODULE for the purpose of attendance. A student shall attend a minimum of 75% of the total instruction hours in a paper (theory/practical) including tutorials and seminars in each semester. There shall no provision for condonation of shortage of attendance and a student who fails to secure 75% attendance in a paper shall be required to repeat that semester with the payment of semester fees.

7. Guidelines to Implement CBCS & CAGP Master's Degree Programme

Course: Every paper offered will have three components associated with the teaching-learning process, namely
(a) **L** - Lecture (b) **T** -Tutorial (c) **P** - Practical
Where
L -Stands for Lecture session.

T - Stands for Tutorial session consisting participatory discussion/self-study/desk work/ brief seminar presentations by students and such other novel methods that make a student to absorb and assimilate more effectively the contents delivered in the Lecture classes.

P -Stands for Practical session and it consists of Hands on experience / Laboratory Experiments/Field Studies/Case studies that equip students to acquire the much required skill component.

In terms of credits, every one hour session of L amounts to one credit per semester and a minimum of two hour session of T or P amounts to one credit per semester, over a period of one semester of 16 weeks for teaching – learning process. The total duration of a semester is 20 weeks inclusive of semester end examination.

A paper shall have either one or two or all the three components. That means a paper may have only lecture component, or only practical component or combination of any two or all the three components.

The total credit earned by a student at the end of the semester upon successfully completing the course is equal to L + T + P of each paper.

Different papers of study are labeled and defined as follows:

Hard Core Paper

A paper which should compulsorily be studied by a candidate as a core requirement is termed as a **Hard Core Paper**.

Soft Core Paper

If there is a choice or an option for the candidate to choose a paper from a pool of papers from the main discipline subject of study or from a sister/related discipline / subject which supports the main discipline/ subject is termed as a **Soft Core Paper**.

Elective Paper

Generally a paper which can be chosen from a pool of papers and which may be very specific or specialized or advanced or supportive to the discipline / subject of study or which provides an extended scope or which enables an exposure to some other discipline / subject / domain or nurtures the candidate's proficiency / skill is called an Elective Paper.

Elective papers may be offered by the main discipline / subject of study or by sister / related discipline / subject of study. ***A Soft Core paper may also be considered as an elective.***

Open Elective

An elective paper chosen generally from an unrelated discipline / subject, with an intention to seek exposure is called an **open elective. A core paper offered in a discipline / subject may be treated as an elective by other discipline / subject and vice versa**.

Dissertation work

It is a special paper involving application of knowledge in solving / analyzing / exploring a real life situation / difficult problem. A Dissertation work may be a hard core or a soft core as decided by the BOS. It may be of to 4 credits.

8. Scheme of Instruction

8.1 A candidate has to earn a minimum of ***76 credits***, for successful completion of a Master's Degree with a distribution of credits for different papers as given in the following table.

Paper Type	Credits
Hard Core	A minimum of 42, but not exceeding 5
Soft Core	A minimum of 16
Open Elective	A minimum of 08

8.2 A candidate can enroll for a maximum of 24 credits per semester.

8.3 Only such candidates who register for a minimum of 18 credits per semester and complete successfully 76 credits in 4 successive semesters shall be considered for declaration of ranks, medals and are eligible to apply for student fellowship, scholarship, free ships and hostel facilities.

9. Continuous Assessment, Earning of Credits and Award of Grades

The evaluation of the candidate shall be based on continuous assessment. The structure for evaluation is as follows:

9.1 Assessment and evaluation processes happen in a continuous mode. However, for reporting purposes, a semester is divided into 3 distinct components identified as C_1, C_2, and C_3

9.2 The performance of a candidate in a paper will be assessed for a maximum of 100 marks as explained below.

a) The first component (C_1) of assessment is for 15 marks. This will be based on test, assignment, seminar and attendance (Class Participation).During the first half of the semester, the first 50% of the syllabus will be completed. This shall be consolidated during the 8th week of the semester. Beyond 8th week, making changes in C_1 is not permitted. The marks for the class participation - 91-100 % -05 marks, 81-90% - 04 marks and 75-80% -03 marks.

b) The second component (C_2) of assessment is for 15 marks. This will be based on test, assignment, seminar and attendance (Class Participation). The continuous assessment and scores of second half of the semester will be consolidated during the 16th week of the semester. During the second half of the semester, the remaining MODULEs in the

paper will be completed. The marks for the class participation- 91-100 % -05 marks, 81-90% 04 marks and 75-80% -03 marks

c) The outline for continuous assessment activities for Component – I (C_1) and Component – II (C_2) will be proposed by the teacher(s) concerned before the commencement of the semester and will be discussed and decided in the respective Departmental Council. The students should be informed about the modalities well in advance. The evaluated papers / assignments during component - I (C_1) and component - II (C_2) of assessment are immediately returned to the candidates after obtaining acknowledgement in the register maintained by the concerned teacher for this purpose.

d) During the 18th – 20th week of the semester, a semester end examination of 2 hours duration shall be conducted for each paper. This forms the third/final component of assessment (C_3) and the maximum marks for the final component will be 50.

10. Setting Question Papers and Evaluation of Answer Scripts.

a) Question papers in three sets shall be set one by the internal and two by the external examiners. While selecting the examiners the University Guidelines are to be followed. Whenever there are no sufficient internal examiners, the Chairman of Board of Examination [BOE] shall get the question papers set by external examiners.

b) The Board of Examiners shall scrutinize and approve the question papers and scheme of valuation.

c) There shall be single valuation for all theory papers by **external examiners**. A detailed scheme of valuation to be prepared by the department and to be provided to the external examiner along with the answer scripts

d) The examination for Practical Work / Field Work / Project Work will be conducted jointly by internal and external examiners. However, the BOE on its discretion can also permit two internal examiners from the College.

e) If a paper is full of (L = 0): T: (P=0) type, then the examination for C_3 component will be as decided by the BOS concerned.

f) The details of continuous assessment are summarized in the following Table

Component	Syllabus in a paper	Weightage	Period of continuous assessn
C_1	First 50% of the Syllabus	15%	First half of the semester To be consolidated by 8th we
C_2	Remaining 50% of the Syllabus	15%	Second half of the semeste To be consolidated by 16th week
C_3	Semester-end examination (all MODULEs of the paper)	70%	To be completed during 18th – 20th Week
Final grades to be announced latest by 24th week			

g) A candidate's performance from all 3 components will be in terms of scores, and the sum of all three scores will be for a maximum of 100 marks (15 + 15 + 70).

h) **Finally, awarding the grades should be completed latest by 24th week of the Semester.**

11. Minor / Major Project Evaluation

Right from the initial stage of defining the problem, the candidate has to submit the progress reports periodically and also present his / her progress in the form of seminars in addition to the regular discussion with the guide. Components of evaluation are as follows:

Component – I (C_1): Periodic Progress and Progress Reports (15)
Component – II (C_2): Results of Work and Draft Report (15)
Component – III (C_3): Final Viva Voce and evaluation (70). The report evaluation is for 40 and the Viva –voce examination is for 30.

12. In case a candidate secures less than 30% in C_1 and C_2 put together in a paper, the candidate is said to have **DROPPED** that paper, and such a candidate is not allowed to appear for C_3 in that paper.

In case a candidate's class attendance in a paper is less that 75% or as stipulated by the College, the candidate is said to have **DROPPED** that paper, and such a candidate is not allowed to appear for C_3 in that paper.

Teachers offering the papers will place the above details in the P G Department Council meeting during the last week of the semester, before the commencement of C_3 and subsequently a notification pertaining to the above will be brought out by the Principal before the commencement of C_3 examination. A copy of this notification shall also be sent to the office of the Controller of Examinations.

12.1 In case a candidate secures less than 30% in C_3 he/she may choose **DROP/MAKEUP** option.

In case a candidate secures more than or equal to 30% in C_3 but his/her grade (G) = 4, as per section 12.5 below, then he/she may be declared to have been conditionally successful in that paper, provided that such a benefit of conditional clearance based on G = 4 shall not be availed for a maximum of **8credits** for the entire programme of Master's Degree of two years.

A candidate exercising his/her option to MAKEUP examination shall be declared passes if he/she secures more than or equal to 30% in C_3 provided he/she fulfils the conditions mentioned in the Para 12.1 & 12.5. To a candidate who does not pass in MAKE UP examination, no separate MAKEUP examination shall be conducted. Such a candidate has to appear for the examination as and when the C_3 component examination is conducted for Odd & Even semester of that academic year along with the regular candidates..

12.2 A candidate has to re-register for the DROPPED paper when the paper is offered again by the department if it is a hard core paper. The candidate may choose the same or an alternate core/elective in case the dropped paper is soft core/ elective paper. A candidate who is said to have DROPPED project work has to re-register for the same subsequently within the stipulated period. **The details of any dropped paper will not appear in the grade card.**

12.3 The tentative / provisional grade card will be issued by the Registrar (Evaluation) at the end of every semester indicating the papers completed successfully. This statement will not contain the list of DROPPED papers.

12.4 Upon successful completion of Master's degree a final grade card consisting of grades of all papers successfully completed by the candidate will be issued by the Registrar (Evaluation).

12.5 The Grade (G) and the Grade Point (GP) earned by the candidate in the subject will be as given below.

P	G	GP = V x G
90 – 100	10	V x 10
80 – 89.99	9	V x 9
70 – 79.99	8	V x 8
60 – 69.99	7	V x 7
50 – 59.99	6	V x 6
40 – 49.99	5	V x 5
30 – 39.99	4	V x 4
00 – 29.99	0	V x 0

Here, P is the percentage of marks P= $[(C_1+C_2) + C_3]$ secured by a candidate in a paper which is rounded to nearest integer. V is the credit value of paper. G is the Grade and GP is the Grade Point.

12.6 A candidate can withdraw any paper within ten days from the date of notification of final results of that semester. Whenever a candidate withdraws a paper, he/she has to register for the same paper in case it is hard core paper, the same paper or an alternate paper if it is soft core/open elective.

A DROPPED paper is automatically considered as a paper withdrawn.

12.7 The Semester Grade Point Average (SGPA) of a candidate after successful completion the required number of credits (76) is given by

$$\textbf{SGPA} = \frac{\sum GP}{Total\ number\ of\ credits}$$

12.8 The Final Semester Grade Point Average (SGPA) of a candidate after successful completion the required number of credits (76) is given by

$$\textbf{CGPA} = \frac{\sum GP\ of\ all\ the\ four\ Semesters}{\sum\ Credits\ of\ all\ the\ Semesters}$$

13. Classification of results

The Final Cumulative Grade Point (FGP) to be awarded to the student is based on CGPA secured by the candidate and is given as follows:

CGPA	FGP	
	Numerical Index	Qualitative Index

4< = CGPA < 5	5	SECOND CLASS
5< = CGPA < 6	6	
6< = CGPA < 7	7	FIRST CLASS
7< = CGPA < 8	8	
8< = CGPA < 9	9	DISTINCTION
9< = CGPA < 10	10	

Overall percentage = 10 x CGPA or is said to be 50% in case CGPA < 5

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Preamble

The M.Sc course in general chemistry was started in the college in the year 2008 with the syllabus prescribed by the university as the department had no freedom to revise the syllabus. The college became autonomous in the year 2011 and took the initiative to consult the relevant industries, students graduated from the department, reputed research institutions and revised CBCS syllabus based on the inputs given by them in the year 2014.

As the department enrolls students from different Indian States and also from Foreign Countries, the proposed syllabus is designed after making intensive discussion with the industry professionals, senior faculty and feedbacks from the students to make it more amenable to cater the needs of students and society in the present day context. Keeping in mind these facts the syllabus was revised again in the 2016.

The present proposed revised syllabus for the year 2018 gives more thrust in preparing the students for the professional scenario with relevance to practical needs and requirements. A supplementary approach includes handling the projects, visiting industries and research institutions to have a fruitful discussion with expert to overcome the bridge between syllabus and industry. The syllabus is competitive enough to meet the global standards and thus fills the gap between the basic knowledge and advanced terminologies with a learning structure.

The syllabus gives the students to choose a soft core of their choice in all the four semesters. The concept of choosing open elective papers from other departments gives them an opportunity to learn the subjects of their choice other than chemistry. The courses credits have been increased from 76 to 80 to give them an opportMODULEy to choose softer core papers.

The revised syllabus covers various aspects of chemistry that includes

I. HARD CORE

- Fundamentals of chemical analysis
- Concepts and models of inorganic chemistry
- Stereochemistry, reaction mechanisms and heterocyclic chemistry
- Chemical thermodynamics, chemical kinetics and electrochemistry
- Separation techniques
- Co-ordination chemistry
- Synthetic organic chemistry and molecular rearrangements
- Quantum chemistry, nuclear chemistry and photochemistry
- Instrumental methods of analysis
- Organic spectroscopy

- Bio-inorganic chemistry
- Application of quantum chemistry and group theory

II. SOFT CORE

- Air and Water Pollution
- Organo metallic chemistry
- Chemical kinetics
- Synthetic organic chemistry
- Solid state chemistry and semiconductors: biophysical chemistry
- Inorganic polymers and industrial inorganic chemistry
- Food and Pharmaceutical Analysis
- Soil, Fuel, Biomedicak and Forensic Analysis
- Bio-Molecules
- Kinetics, automated and radiometric methods of analysis
- Statistical mechanics and phase rule studies; solid state chemistry
- Structural methods in inorganic chemistry
- Photochemistry, pericyclic reaction and organometallic chemistry

III. OPEN ELECTIVE

- English Grammar and Usage
- Business Management and Communication Skill
- Human Resource Management
- Disaster Management
- Web Designing & Net Working
- Fundamental of Mathematics
- Discrete Mathematics
- Personality Development
- Basics of Accounting
- Freelance Journalism

To encourage research, the students are provided with all the facilities in the department to do their project work under the guide ship of the faculty. The outcome of their research work is being published in reputed international research journals. The students regularly attend conferences, workshops, seminars, science competitions, visit industries and research institutions to study the application part of the syllabus. The department has established a 'Chemical Club' to encourage the students in take part in different co curricular activities. The club is managed by the students. It conducts departmental competitions, intercollegiate science competitions, organizes the guest lectures and maintains a wall magazine to exploit the students to exhibit their creative and innovative ideas.

VISION AND MISSION OF THE COLLEGE

VISION:

The college is guided by the visionary zeal of providing value- based education to everyone irrespective of religion, caste, creed or sex by which the character is formed, intellect is explained and one can stand on his/her feet**.**

MISSION:

To transform young men and women who come to learn not from books, but also from life and to share the experience of working and playing together, this inculcates life skills to become good citizens with integrity and discipline.

VISION AND MISSION OF THE DEPARTMENT

VISION:

To develop the department into a full-fledged knowledge center for learning chemistry professionally, ethically and to make students successful employed entrepreneurial.

MISSION:

1. To provide a vibrant and innovative environment for students to acquire knowledge necessary for developing themselves into qualified professionals.
2. To continuously update curriculum in tune with emerging trends with interdisciplinary approach.
3. To inculcate the spirit of innovative thinking among the students, train them to meet the challenges needs of the stakeholders and enable the learners implementing the concepts to address the real world problems.

PO No.	Programme Educational Objectives (PEOs)
PEO-1	**PROFESSIONAL DEVELOPMENT** To train the students to acquire knowledge in their chosen programme and apply professionally and ethically with responsibility towards the need of the society
PEO-2	**CORE PROFICIENCY** To expertise the students to organize, understand, apply, evaluate, and solve problems by providing hands on experience through modern tools necessary for practice.
PEO-3	**TECHNICAL ACCOMPLISHMENTS** To equip the students with the talent to interpret in core applications by building up a multi-disciplinary concept.
PEO-4	**PROFESSIONALIS**M To train the students to acquire the significance of self-discipline, communication skills, professional attitude, holistic personality development, responsibility and team work to be better entrepreneur.
PEO-5	**LEARNING ENVIRONMENT** To provide an environment for life-long learning to inculcate the importance of research, creativity, invention and leadership to become a successful entrepreneur

Mapping of Mission of the Department with Programme Educational Objectives

Mission	Programme Educational Objectives (PEOs)				
	PEOs-1	PEOs-2	PEOs-3	PEOs-4	PEOs-5
M1	*			*	
M2		*			*
M3			*	*	

PO No.	Programme Outcomes (POs) Upon completion of the Programme the student will be able -
PO-1	To impart in-depth knowledge in discipline with special emphasis on use of technology in chemical applications.
PO-2	To develop more advanced epistemological beliefs, enhanced critical thinking ability and meta cognitive skills, and an understanding of the relations among perspectives derived from interdisciplinary approach.
PO-3	To provide training in observing, formulating problems, hypothesizing, designing experiments, analyzing data, drawing conclusions, and communicating skills.
PO-4	To inculcate research culture leading to publication of review articles and research article from the projects.
PO-5	To equip the students with specialised scientific and technical approach in professional attitude towards team work, responsibility and accountability for a better career.

PSO No.	Programme Specific Outcomes PSOs Upon completion of the Programme the student will acquire -
PSO-1	Knowledge in depth in Fundamentals, Organic reaction mechanisms, Structural and Molecular Organic Chemistry, Coordination chemistry, **Bio-inorganic chemistry,** Quantum chemistry and group theory, Nuclear chemistry, Molecular spectroscopy and Organometallics.
PSO-2	Innovative ideas and skills on Materials Science and Pharmacology of Plant Products through interdisciplinary departments
PSO-3	The laboratory experiences include enhancing mastery of analytical, organic, inorganic and physical experiments, develops practical skills and improving team work abilities.
PSO-4	The dissertation work/internship provides training in handling research tools to write and publish review and research articles.
PSO-5	Knowledge on Separation Techniques, Advanced Spectroscopy, Synthetic Organic Chemistry and Molecular Rearrangements, Advanced Physical Chemistry, Instrumental Methods of Analysis, Good Laboratory Practices, Pharmaceutical chemistry, Renewable Energies, Green Chemistry, Industrial chemistry, Food and Pharmaceutical Analysis, Agricultural chemistry and Water remediation and conservation studies provide jobs in

	industries and research organizations.

Mapping of Programme Educational Objectives with Program Outcomes and Programme Specific outcomes

	Program Outcomes					**Program Specific Outcomes**				
Programme Educational Objectives	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PSO-1**	**PSO-2**	**PSO-3**	**PSO-4**	**PSO-5**
PEOs-1	*		*		*	*		*	*	*
PEOs-2		*	*	*	*		*	*	*	
PEOs-3			*		*	*		*	*	*
PEOs-4		*			*		*			*
PEOs-5	*			*	*				*	*

SEMESTER-WISE DISTRIBUTION
HARDCORE, SOFT CORE, OPEN ELECTIVES AND CREDITS
FIRST YEAR

Sl. No	Subject Code	QP Code	Paper Title	Type	L	T	P	Credits	Total Credits
FIRST SEMESTER									
1.	A0210	55001	Fundamentals of Chemical Analysis	HC	3	0	0	3	
2.	A0220	55002	Concepts and Models of Inorganic Chemistry	HC	3	0	0	3	
3.	A0230	55003	Stereochemistry, Reaction Mechanisms and Heterocyclic Chemistry	HC	3	0	0	3	
4.	A0240	55004	Chemical Thermodynamics, Chemical Kinetics and Electrochemistry	HC	3	0	0	3	
ANY TWO OF THE FOLLOWING TO BE CHOSEN									
5.	AP210	NA	Analytical Chemistry Practical	HC	0	0	4	2	
6.	AP220	NA	Inorganic Chemistry Practical	HC	0	0	4	2	
7.	AP230	NA	Organic Chemistry Practical	HC	0	0	4	2	
8.	AP240	NA	Physical Chemistry Practical	HC	0	0	4	2	
ANY ONE OF THE FOLLOWING SC TO BE CHOSEN									**22**
9.	A0310	55005	Air and Water Pollution	SC	2	0	0	2	
10.	A0320	55006	Organometallic Chemistry	SC	2	0	0	2	
11.	A0330	55007	Synthetic Organic Chemistry	SC	2	0	0	2	
12.	A0340	55008	Chemical Kinetics	SC	2	0	0	2	
13.	A0350	55009	Solid State Chemistry and Semiconductors: Biophysical Chemistry	SC	2	0	0	2	
14.			Inorganic Polymers and Industrial Inorganic						
OE FROM OTHER DEPARTMENTS									
13.			Offered by other Department	OE	3	1	0	4	
SECOND SEMESTER									
14.	B0210	55101	Separation Techniques	HC	3	0	0	3	
15.	B0220	55102	Coordination Chemistry	HC	3	0	0	3	
16.	B0230	55103	Synthetic Organic Chemistry and Molecular Rearrangements	HC	3	0	0	3	
17.	B0240	55104	Quantum Chemistry; Nuclear Chemistry and Photo Chemistry and Microwave and Infrared	HC	3	0	0	3	
ANY TWO OF THE FOLLOWING PRACTICALS TO BE CHOSEN **Note:** The practical chosen in the previous semester shall not be repeated.									
18.	BP230	NA	Organic Chemistry Practical	HC	0	0	4	2	
19.	BP240	NA	Physical Chemistry Practical	HC	0	0	4	2	**23**
20.	BP210	NA	Analytical Chemistry Practical	HC	0	0	4	2	
21.	BP220	NA	Inorganic Chemistry Practical	HC	0	0	4	2	
ANY ONE OF THE FOLLOWING INTERDISCIPLINES TO BE CHOSEN									
22.	B0520	96502	Materials Science	SC	0	0	3	3	
23.	B0530	96503	Pharmacology of Plant Products	SC	0	0	3	3	
OE FROM OTHER DEPARTMENTS									
24.			Offered by other Department	OE	3	1	0	4	

SECOND YEAR

Sl .No	Subject Code	QP Code	Paper Title	Type	L	T	P	Credits	Total Credits
THIRD SEMESTER									
1.	C0210	55201	Instrumental Methods of Analysis	HC	4	0	0	4	
2.	C0220	55202	Organic Spectroscopy	HC	4	0	0	4	
3.	C0230	55203	Chemistry of Natural Products	HC	4	0	0	4	
ANY TWO OF THE FOLLOWING TO BE CHOSEN **Note:** The practical chosen in the previous semesters shall not be repeated.									
4.	CP330	NA	Organic Chemistry Practical	SC	0	0	4	2	
5.	CP340	NA	Physical Chemistry Practical	SC	0	0	4	2	
6.	CP310	NA	Analytical Chemistry Practical	SC	0	04	4	2	**18**
7.	CP320	NA	Inorganic Chemistry Practical	SC	0	0	4	2	
ANY ONE OF THE FOLLOWING SC TO BE CHOSEN									
8.	C0310	55204	Food and Pharmaceutical Analysis	SC	2	0	0	2	
9.	C0320	55205	Soil, Fuel, Biomedical and Forensic Analysis	SC	2	0	0	2	
10.	C0330	55206	Bio-Molecules	SC	2	0	0	2	
FOURTH SEMESTER									
11.	D0210	55301	Bioinorganic Chemistry	HC	4	0	0	4	
12.	D0220	55302	Molecular Symmetry and Group Theory; Statistical Mechanics, Phase Rule Studies; Radiation Chemistry, Polymers and Electrophoresis.	HC	4	0	0	4	
13.	DP210	NA	Dissertation	HC	0	0	4	4	
ANY TWO OF THE FOLLOWING PRACTICALS TO BE CHOSEN **Note:** The practical chosen in the previous semesters shall not be repeated.									
14.	DP330	NA	Organic Chemistry Practical	SC	0	0	4	2	
15.	DP340	NA	Physical Chemistry Practical	SC	0	0	4	2	
16.	DP310	NA	Analytical Chemistry Practical	SC	0	0	4	2	**18**
17.	DP320	NA	Inorganic Chemistry Practical	SC	0	0	4	2	
ANY ONE OF THE FOLLOWING SC TO BE CHOSEN									
18.	D0310	55304	Kinetics, Automated and Radiometric methods of Analysis	SC	2	0	0	2	
19.	D0320	55305	Statistical Mechanics and Phase Rule Studies; Solid State Chemistry	SC	2	0	0	2	
20.	D0330	55306	Structural Methods in Inorganic Chemistry	SC	2	0	0	2	
21.	D0340	55307	Photochemistry, Pericyclic Reactions and Organometallic Chemistry	SC	2	0	0	2	
		Total Credits (HC-56+SC-17+ OE- 08)							**81**
	HC= HARD CORE PAPER. SC= SOFT CORE PAPER. OE = OPEN ELECTIVE PAPER								

FIRST SEMESTER

Course Title	FUNDAMENTALS OF CHEMICAL ANALYSIS							
Course Type	Hard Core-Theory		Total hours	48	Hours/Week	03	Credits	03
Course Code	A0210	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To introduce some important analytical terms.
CO-2	Understand the technique in sampling
CO-3	Calculations of statistical parameters such as errors, accuracy and precision, sensitivity and others
CO-4	To understand the standardization and calibration of analytical methods
CO-5	Theory behind acid-base, precipitation, redox and complexometric titration
CO-6	Quantitative application of titration in the real samples

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Apply the analytical methods and techniques.	PSO-1	Remember, Understand, Apply and Analyse
CLO-2	Finding errors and treatment of analytical data	PSO-1	Apply, Analyse and Evaluate
CLO-3	Designing and Implementing the sampling plan	PSO-1	Remember, Apply, and Analyse
CLO-4	Select, calculate and apply the quantitative analysis of organic and inorganic compounds	PSO-1	Understand, Apply and Analyse
CLO-5	Differentiate the role of solvent in acid-base titrations	PSO-1	Understand, Apply and Analyse
CLO-6	Analyse the titrations based on acid-base, precipitation and complex metric reactions	PSO-1	Understand, Apply, Analyse and Evaluate

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **Language of analytical chemistry** - Analysis, determination and measurement. Classifying analytical techniques. Selecting an analytical method - accuracy, precision, sensitivity, selectivity, robustness and ruggedness. Scale of operation, equipment, time and cost. Making the final choice **04 hrs**

1.2 **Errors and treatment of analytical data:** Limitations of analytical methods – Error: determinate and indeterminate errors, minimization of errors. Accuracy and precision, distribution of random errors, the normal error curve. Statistical treatment of finite samples -measures of central tendency and variability, mean, **08 hrs**

median, range, standard deviation and variance. Student's t-test, confidence interval of mean. Testing for significance - comparison of two means and two standard deviations. Comparison of an experimental mean and a true mean. Criteria for the rejection of an observation - Q-test. Propagation of errors: determinate errors and indeterminate errors.

1.3 **Standardization and calibration**: Comparison with standards - direct comparison and titrations. External standard calibration - the least squares methods, regression equation and regression coefficient. Internal standard methods and standard-addition methods. Figures of merit of analytical methods - sensitivity and detection limit, linear dynamic range. **04 hrs**

Key word: Errors, standardization, Analytical calibration

MODULE-2 **16 hrs**

2.1 **Obtaining and preparing samples for analysis**: Importance of sampling, designing a sample plan-random, judgement, systematic- judgment, stratified and convenience sampling. Type of sample to collect - grab and composite samples. *Insitu* sampling. Size of sample and number of samples. Implementing the sampling plan - solutions, gases and solids. Bringing solid samples into solution - digestion and decomposing. **05 hrs**

2.2 **Quantitative applications** – selecting and standardizing a titrant, inorganic analysis - alkalinity, acidity and free CO2 in water and waste waters, nitrogen, sulphur ammonium salts, nitrates and nitrites, carbonates and bicarbonates. Organic analysis - functional groups like carboxylic acid, sulphonic acid, amine, ester, hydroxyl and carbonyl. Air pollutants like SO_2. Quantitative calculations. Characterization applications - equivalent weights and equilibrium constants. **06 hrs**

2.3 **Titrimetric analysis:** An overview of titrimetry. Principles of titrimetric analysis. Role of solvent in acid-base titrations, solvent systems, differentiating ability of a solvent, some selected solvents, titrants and standards, titration curves, effect of water, determining the equivalence point, typical applications - determination of
Carboxylic acids, phenols and amines. **05 hrs**

Key word: Sampling, titration-application

Module-3 **16 hrs**

3.1 **Acid-base titrations:** Titration curves. Titrations based on acid-base reactions - titration curves for strong acid and strong base, weak acid and strong base and weak base and strong acid titrations. Selecting and evaluating the end point. Finding the end point by visual indicators, monitoring *p*H and temperature. **03 hrs**

3.2 **Precipitation titrations:** Titration curves, feasibility of precipitation titrations, factors affecting shape - titrant and analyte concentration, completeness of the reaction, titrants and standards, indicators for precipitation titrations involving silver nitrate, the Volhard, the Mohr and the Fajan's methods, typical applications. **03 hrs**

3.3 **Complexometric titrations:** Complex formation reactions, stability of complexes, stepwise formation constants, chelating agents, EDTA - acidic properties, complexes with metal ions, equilibrium calculations involving EDTA, conditional formation constants, derivation of EDTA titration curves, effect of other complexing agents, factors affecting the shape of titration curves - completeness of reaction, indicators for EDTA titrations - theory of common indicators, titration methods employing EDTA - direct, back and displacement titrations, indirect determinations, titration of mixtures. **05 hrs**

3.4 **Redox titration:** Balancing redox equations, calculation of the equilibrium constant of redox reactions, calculating titration curves, detection of end point, **05 hrs**

visual indicators and potentiometric end point detection. Quantitative applications - adjusting the analyte's oxidation state, selecting and standardizing a titrant. Inorganic analysis - chlorine residuals, dissolved oxygen in water, water in non-aqueous solvents. Organic analysis - chemical oxygen demand (COD) in natural and waste waters, titrations of mercaptans and ascorbic acid with I_3 and titration of organic compounds using periodate.

Key words: Acid-base, precipitation, complexometric and redox titration

30% Theory and 70% Problems

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern Analytical chemistry	David Harvey	Mcgraw Hill	1st	2000
4	Analytical Chemistry 2.0	David Harvey	McGraw-Hill Companies, Depauw	1st	2018
5	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6/E, 6th	2008
6	Calculations in Chemistry	John olusina obimakinde, Samuel Oluwaseun Obimakinde	IK international publishing house	1st	2014

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Course Title	CONCEPTS AND MODELS OF INORGANIC CHEMISTRY							
Course	Hard Core-		Total	48	Hours/Wee	03	Credits	03
Course Code	A0220	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	Understanding of the principles of VSEPR model and molecular orbital theory
CO-2	Understand the delocalized π-bonding of molecular orbital.
CO-3	Understand the structure and bond properties
CO-4	Demonstrate and understanding of the basic principles of acid – base chemistry and nonaqueous solvents
CO-5	Understand the charachteristic structures of ionic solids

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students	PSOs Addressed	Cognitive Level
CLO-1	Able to predict the structures of ionic compounds and calculation of lattice enthalpy	PSO-1	Understand, Apply and Analyse
CLO-2	Correlate the structural type with radius ratio	PSO-1	Analyse and Evaluate
CLO-3	Perovskite and spinel structures	PSO-1	Understand, Apply
CLO-4	Characteristics of Bronsted acids and applications of acid-base chemistry	PSO-1	Remember, Understand
CLO-5	Predicting the shape Structure and bond properties of polyatomic molecules using VSEPR model	PSO-1	Understand, Analyse, Apply
CLO-6	Understand the chemistry of lanthanides and actinides with extraction, color and magnetic properties	PSO-1	Remember, Understand

COURSE CONTENTS

	MODULE-1	16 hrs
1.1	Structures and energetics of ionic crystals: Introduction, MX (NaCl, CsCl, ZnS) and MX2 (fluorite, rutile, β-crystobalite and cadmium iodide) types. The perovskite and spinel structures. Thermodynamics of ionic crystal formation. Lattice energy, Born-Haber cycle, Born-Lande equation. Applications of lattice energetics. Ionic radii, factors affecting the ionic radii, radius ratio rules. Types of imperfections-point defect-schottky defect and frenkel defect.	8 hrs

1.2	Structures and energetics of inorganic molecules: Introduction, Energetics of hybridization. VSEPR model for explaining structure of AB_2, AB_2E, AB_3E, AB_2E_2, ABE_3, AB_2E_3, AB_4E_2, $AB_5E$and AB_6 molecules. M.O. treatment of homonuclear (Li_2, Be_2, B_2, C_2, N_2, O_2, F_2) and heteronuclear diatomic molecules (CO, NO, HF, HCl) .M.O. treatment involving delocalized π-bonding (CO_3^{2-} , NO_3^- , $NO_2^-$and N_3^-), M.O. correlation diagrams (Walsh) for triatomic molecules. *Keywords: Ionic crystals, lattice energy, MOT*	**8 hrs**
MODULE–2		**16 hrs**
2.1	**Modern concept of acids and bases**: Lux-Flood and Usanovich concepts, solvent system and leveling effect. Principles of Hard-Soft Acids and Bases.	**6 hrs**
2.2	**Non-aqueous solvents**: Classification of solvents, Properties of solvents (dielectric constant, donor and acceptor properties) protic solvents (anhydrous H_2SO_4, HF and glacial acetic acid) aprotic solvents (liquid SO_2, BrF_3 and N_2O_4). Solutions of metals in liquid ammonia, hydrated electron. Super acids. **Keywords:** HSAB	**10 hrs**
MODULE–3		**16 hrs**
3.1	**Electron deficient compounds:** Diborane and its reactions, higher boranes, (preparations, properties, structure, bonding and classification). Wade's rules and metallocarboranes.	**4hrs**
3.2	**Lanthanides**-Electronic configuration, oxidation states, lanthanide contraction; extraction of a mixture of lanthanides from monazite sand, separation of lanthanides- ion-exchange, solvent extraction, chromatographic methods, colour, spectra and magnetic properties. Uses: lanthanides as shift reagents, super	**8hrs**
3.3	**Actinides**-electronic configuration, oxidation states, separation of actinides; spectral and magnetic properties, Trans-uranium elements. ***Keywords: Boranes, lanthanoids, actinoids***.	**4hrs**

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
2	Basic Concepts of Inorganic Chemistry	D.N. Singh	Pearson Education,	2nd	2010
3	Inorganic chemistry	Mark Weller, TinaOverton	Oxford University press	6th	2010
4	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
5	Advance Inorganic Chemistry	F. Albert Cotton, Geoffrey, Wilkinson	John Wiley and Sons	6th	2004
6	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004
7	Concise Inorganic Chemistry	J.D. Lee	Blackwell Science Ltd.,	5th	2003

8	Inorganic Chemistry	Catherine E. *Housecroft* And G. Sharpe Alan	Pearson Education	2nd	2003
9	Basic Inorganic Chemistry	F.A. Cotton, G. Wilkinson	John Wiley and Sons	3rd	2002
10	Inorganic Chemistry	D.F. Shriver, P.W. Atkins	Oxford University Press	2nd	1994

RECOMMENDED BOOKS

1	Inorganic Chemistry	James E. Huheey, Ellen	Pearson Education	4th	2013
2	Basic Concepts of	D.N. Singh	Pearson Education,	2nd	2010
3	Inorganic chemistry	Mark Weller,	Oxford Uiversity	6th	2010
4	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
5	Advance Inorganic	F. Albert Cotton,	John Wiley and	6th	2004
6	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004

Course Title	**STEREOCHEMISTRY, REACTION MECHANISMS AND HETEROCYCLIC CHEMISTRY**								
Course Type	**Hard Core-Theory**		**Total Hours**	48	**Hours/Week**	03	**Credits**	03	
Course Code	**A0230**	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100	
			External	**Duration**	03 hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To Identify and differentiate chirality at centers, axis, planes and determine the absolute configuration
CO-2	To know the stability of different Conformations
CO-3	To convey the role of temperature, solvents, and catalysts in organic reactions
CO-4	Elucidate reaction mechanisms and Intermediates formed
CO-5	To understand the preparation and applications of Heterocyclic compounds

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	R/S, E/Z - prediction for molecules	PSO-1	Apply Analyse And Evaluate
CLO-2	Determine Stereoselectivity of various organic transformations	PSO-1	Apply and Analyse

CLO-3	Evaluate the stability of various conformers of acyclic and cyclic systems using steric, electronic and stereoelectronic effects.	PSO-1	Understand, Apply and Analyse
CLO-4	Envisage the organic reaction mechanism and product identification	PSO-1	Understand and Apply
CLO-5	Formation and Identification of Reaction Intermediates.	PSO-1	Understand and Apply
CLO-6	Preparation and applications of various heterocyclic compounds.	PSO-1	Understand

COURSE CONTENTS

MODULE- 1 **16hrs**

1.1	**Stereoisomerism:** Projection formulae [Fly wedge, Fischer, Newman and Saw horse], enantiomers, diasteroisomers, racemic mixture and their resolution, configurational notations of simple molecules, *DL* and *RS* configurational notations.	**3 hrs**
1.2	**Stereoselectivity:** Stereoselective reactions, diasteroselective reactions, stereospecific reactions, regioselective and regiospecific reactions.	**2 hrs**
1.3	**Optical isomerism:** Conditions for optical isomerism, optical isomerism due to chiral centers and molecular dissymmetry, allenes and biphenyls, criteria for optical purity.	**2 hrs**
1.4	**Geometrical isomerism:** Due to C=C, C=N and N=N bonds, E, Z conventions, determination of configuration by physical and chemical methods.	**2 hrs**
1.5	**Conformational isomerism:** Elementary account of conformational equilibria of ethane, butane and cyclohexane.	**2 hrs**
1.6	**Conformational analysis:** Conformation of cyclic compounds such as cyclopentane, cyclohexane, cyclohexanone derivatives and decalins. Conformational analysis of 1,2, 1,3, and 1,4-disubstitutedcyclohexane derivatives and *D*-Glucose, Effect of conformation on the course/rate of reactions.	**5 hrs**

Key words- Confirmational and Constitutional isomers

MODULE- 2 **16 hrs**

2.1	**Structure and reactivity:** Acids and bases, Structural effect on acidity and basicity, hydrogen bonding resonance, inductive and hyper conjugation effects.	**2 hrs**
2.2	**Basics of organic reactions:** Meaning and importance of reaction mechanism, classification and examples for each class.	**2 hrs**
2.3	**Substitution reactions:** Mechanism of nucleophilic substitution reactions – Kinetics, mechanism and stereo chemical factors affecting the rate of S_N1, S_N2, S_Ni reactions, neighbouring group participation.	**4 hrs**
2.4	**Aromatic nucleophilic substitution**: S_N1, S_N2 and benzyne mechanism, Bucherer reaction. Aromatic electrophilic substitution: Mechanism of nitration, halogenation, sulphonation, Friedel-Crafts alkylation and acylation, Mannich reaction, chloromethylation, Vilsmeier-Haack reaction.	**4 hrs**
2.5	**Reaction Intermediates:** Formation, structure, stability, detection and reactions of carbocations (classical and non-classical), carbanions, free radicals, carbenes, nitrenes, nitrile oxides, nitrile imines, nitrile ylides and	**4 hrs**

arynes.

Key words- Substitution reaction mechanisms and reactive intermediates

MODULE-3 **16 hrs**

3.1 **Heterocyclic chemistry:** Nomenclature of heterocyclic systems, Structure, reactivity, synthesis and reactions (minimum three) of furan, pyrrole, thiophene, indole, pyridine, quinoline, isoquinoline, pyrazole, imidazole, pyrone, coumarin, chromones, pyrimidines and purines. **16 hrs**

Key words- Heterocyclic compounds, Preparation, applications

20% Problems and 80% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Organic Chemistry	Robert. N. Boyd and Robert Thornton Morrison	Pearson	6th	2016
2.	Stereochemistry: Confomation and Mechanism	Kalsi P. S	New Age International Private Limited	9th	2017
3.	Stereochemistry of Organic Compounds: Principles and Applications	D. Nasipuri	New Age International Private Limited	3rd	2018
4.	Stereochemistry of Organic Compounds	E.L. Eliel and S.H. Wilen	John Wiley and Sons, New York.	-	1994
5.	A Guide Book to Mechanism in Organic Chemistry	Petersykes	Pearson	6th	2003
6.	Advanced Organic Chemistry	J. March	Wiley	7th	2013
7.	Organic Reaction Mechanism.	R.K. Bansal,	by Wiley Eastern Limited, New Delhi,		1993
8.	Organic Chemistry	Marye Anne Fox, James K. Whitesell	Jones & Bartlett Learning	3rd illustrated	2004
9.	Heterocyclic Chemistry	Raj K. Bansal	New Age International	3rd	1999
10.	Heterocyclic Chemistry	Joule & Smith	Wiley-Blackwell	5th	2013
11.	Principles of Modern Heterocyclic Chemistry	L.A. Pacquette	Benjamin-Cummings	-	1978

12.	Comprehensive Heterocyclic Chemistry	Kartritzky series	Elsevier	-	2008
RECOMMENDED BOOKS					
1	Introduction to Stereochemistry	Kurt Mislow	Dover Publications	6th	2002
2	Principles of Organic Synthesis,	ROC Norman and J. M. Coxon	CRC Press	6th	1996.
3	Advanced Organic Chemistry	F. A. Carey and R. J. Sundberg,	Plenum	-	1990
4	Basic Principles of Heterocyclic Chemistry	Alan R. Katritzky and J.M. Lagowski	Elsevier	-	1968

Course Title	**CHEMICAL THERMODYNAMICS, CHEMICAL KINETICS AND ELECTROCHEMISTRY (HC)**							
Course Type	Hard Core- Theory		**Total Hours**	48	Hours/Week	03	Credits	03
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Correlating the concepts of kinetics through demonstration of experiments.
CO-2	Explanation of different thermodynamic variables by solving the problems.
CO-3	Illustrate the different numerical concepts involved in electrochemistry.
CO-4	Understand the role of thermodynamic cycles, availability and irreversibility.
CO-5	To state the basis for the the collision model of chemical kinetics.

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Understand the role of internal energy, enthalpy, entropy and thermodynamic properties.	PSO-1	Understand, Apply and Analyse
CLO-2	Recognize and understand the different forms of pure substances.	PSO-1	Apply and understand
CLO-3	Identify and use MODULEs and notations in thermodynamics**.**	PSO-1	Apply, Analyse and Evaluate
CLO-4	To describe how rate laws are determined.	PSO-1	Understand, Apply and evaluate
CLO-5	To describe an activation complex or transition state.	PSO-1	Understand and Analyse
CLO-6	To identify the compounds being oxidized and reduced.	PSO-1	Analyse and understand

COURSE CONTENTS

MODULE-1 **16 hrs**

1.1 **Concepts of entropy and free energy:** Entropy as a measure of unavailable energy. Entropy change during spontaneous process. Helmholtz and Gibbs free energies (Definition and equation).. Variation of free energy with temperature and pressure. Maxwell's relations, Nernst heat theorem .Third law of thermodynamics - calculation of absolute entropies. **06 hrs**

1.2 **Partial molar properties:** Partial molar volumes and their determination by intercept method and density measurements. Chemical potential and its significance. Variation of chemical potential with temperature and pressure. Formulation of the Gibbs Duhem equation. Derivation of Duhem-Margules equation. **04 hrs**

1.3 **Fugacity:** Determination of fugacity of gases. Variation of fugacity with temperature and pressure. Activity and activity coefficients. Variation of activity with temperature and pressure. Determination of activity co-efficient by vapor pressure, depression in freezing point and solubility measurements by electrical methods **06 hrs**

Key words- Entropy, Free energy, Temperature, Activity.

MODULE-2 **16 hrs**

2.1 **Kinetics of complex reactions**: Parallel, consecutive and reversible reactions. Determination of order of reaction. Arrhenius equation, energy of activation and its experimental determination. Potential energy surface, theoretical calculation of energy of activation. Simple collision theory - mechanism of bimolecular reaction. Lindeman's theory, Hinshelwood's theory for unimolecular reaction. Activated complex theory of reaction rate, Kinetics of reactions in solution - Salt effect, effect of dielectric constant (single sphere and double sphere model), effect of pressure, volume and entropy change on reaction rates. Kinetics of heterogeneous reactions - Langmuir's theory, unimolecular and bimolecular surface reactions. **10 hrs**

2.2 **Fast reactions:** Study of kinetics by flow techniques, equation for contact time, stopped flow and continuous flow methods. Relaxation method, equation for relaxation time, temperature jump and pressure jump methods, flash photolysis, pulse radiolysis and shock tube method. **06 hrs**

Key words- Kinetics , Rate equation, Collision, Theories.

MODULE – 3 **16 hrs**

3.1 **Electrochemistry:** Arrhenius theory of strong and weak electrolytes and its limitations, Debye-Huckel theory of strong electrolytes, Debye Huckel-Onsager equation, Debye-Huckel limiting equation for activity co-efficient, Debye-Huckel equation for appreciable concentrations. Liquid junction potential and its determination. Transport Number - Determination of transport number by Hittorf method and E.M.F method. True and apparent transport numbers. Abnormal transport numbers, effect of temperature and concentration on transport number. **08 hrs**

3.2 **Irreversible electrode process:** Introduction, reversible and irreversible electrodes, Potential energy surface, theoretical calculation of energy of activation. reversible and irreversible cells. Polarization, over voltage - ohmic, concentration, activation, experimental determination of over voltage. Hydrogen and Oxygen over voltage. Effect of temperature, current density and *p*H on over voltage. Equations for concentration over potential, diffusion current – stationary current, potential curves, thickness of diffusion layer, diffusion controlled current – potential curves at a dropping mercury electrode, polarography, half wave potential, application in qualitative and quantitative analysis. Butler-Volmer equation, Tafel equation. **08 hrs**

60% Theory and 40% Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Thermodynamics for Chemists	S. Glasstone	Read books	--	2007
2	Chemical Kinetics	K.J. Laidler	Pearson	3rd	2003
3	Kinetics and Mechanism of Chemical	J. Rajaram and J.C. Kuriacose	Macmillan Publishers	Reprint	2000
RECOMMENDED BOOKS					
1	Physical Chemistry	P.W. Atkins	ELBS, edition, Oxford	4th	1990
2	Physical Chemistry	Arun Bhal and B.S. Bhal	S. Chand	13th	1963

Course Title	**AIR AND WATER POLLUTION**								
Course Type	Soft Core- Theory		**Total Hours**	32	**Hours/Week**	02	**Credits**	02	
Course Code	A0310	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100	
			External	Duration	03 hrs	C3	70 Marks		

Course Objectives:

CO No.	Course Objectives
CO-1	Air, water pollution
CO-2	Air quality monitoring
CO-3	Pollution control
CO-4	Water pollution
CO-5	Analysis of pollutants

MAPPING CLO'S WITH PSO's AND CD's:

CO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Air pollution, analysis and control	PSO-1	Understand
CLO-2	Air quality monitoring	PSO-5	Apply, Analyse
CLO-3	Air Pollution control	PSO-5	Apply, Analyse
CLO-4	Control devices for particulates	PSO-5	Understand, Apply and Analyse
CLO-5	Water pollution and analysis	PSO-1	Understand, Apply and Analyse
CLO-6	Qualitative and quantitative analysis of water and air samples	PSO-5	Evaluate

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1	**Air pollution, analysis and control:** Historical overview - global implications of air pollution, sources of pollutants, classification of pollutants. Sources and effects of particulates, carbon monoxide, sulphur oxides, nitrogen oxides, hydrocarbons and photochemical oxidants on human health, vegetation and materials. Standards for air pollutants.	**4hrs**
1.2	**Air quality monitoring:** Sampling methods and devices for particulates and gaseous pollutants. SO2: ambient air measurements and stack gas measurements - Turbidimetric, colorimetric, conductometric and coulometric methods, NOX: Griess-ilosvay and Jacobs-Hockheiser colorimetric methods, chemiluminiscent technique, CO: NDIR, amperometric, FID and catalytic oxidation methods. Hydrocarbons: total and individual hydrocarbons by gas chromatography. Oxidants and ozone: colorimetric, coulometric, titrimetric and chemiluminescence methods.	**5hrs**
1.3	**Air Pollution control:** Atmospheric cleaning processes, approaches to contaminant control detection and control at source.	**2hrs**
1.4	**Control devices for particulates:** Gravitational settlers, centrifugal collectors, wet collectors, electrostatic precipitation and fabric filtration.	**2hrs**
1.5	**Control devices for gaseous pollutants:** adsorption, absorption, condensation and combustion processes. Automative emission control-catalytic converters.	**3hrs**

MODULE- 2 **16hrs**

2.1	**Water pollution and analysis:** Water resources, origin of wastewater, types of water pollutants; their sources and effects, chemical analysis for water pollution control - objectives of analysis, parameters of analysis, sample collection and preservation. Environmental and public health significance and measurement of colour, turbidity, total solids, acidity, alkalinity, hardness, chloride, residual chlorine, chlorine demand, sulphate, fluoride, phosphates and different forms of nitrogen in natural and waste/polluted waters, heavy metal pollution - public health significance of Pb, Cd, Cr, Hg, As, Cu, Zn and Mn, general survey of the instrumental techniques for the analysis of heavy metals in aquatic systems, organic loadings – significance and measurement of DO, BOD, COD, TOD, and TOC, phenols, pesticides, surfactants and tannin and lignin as water pollutants and their	**16hrs**

80% Theory and 20% field work

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of Publication
01	Environmental chemistry.	Anil K De,	New age International Pvt Ltd	6th	2006
02	Environmental chemistry.	B.K. Sharma	Krishna Prakashan Media	14th	2011

03	Environmental chemistry	H. Kaur	Pragati prakashan	1st	2010
04	Environmental chemistry	Balram Pani	I.K. International publishing house	1st	2007

Course Title	**INORGANIC POLYMERS AND INDUSTRIAL INORGANIC CHEMISTRY**							
Course	**Soft core - Theory**		**Total**	**3**	**Hours/Week**	**02**	**Credits**	**02**
Course Code		**Evaluation**	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	knowledge regarding polymer science
CO-2	Economic importance on inorganic fibres
CO-3	Applications of Phosphate containing polymers

MAPPING CLO'S WITH PSO's AND CD's:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Knowledge regarding polymer science	PSO-1	Understand, Analyse
CLO-2	Economic importance on inorganic fibres	PSO-1	Understand, Analyse
CLO-3	Applications of Phosphate containing polymers	PSO-1	Remember, Understand

MODULE – 1		**16 hrs**
1.1	Introduction, classification, phosphate containing polymers, preparation, properties, structure. Polymers containing boron- preparation, properties, structure, degree of polymerization	**3 hrs**
1.2	**Borazine** - Preparation, properties, structure, derivatives of borazine	**1 hr**
1.3	**Heterocyclic inorganic ring system**: Sulphur-nitrogen ring, nitrogen-	**2 hrs**
1.4	**Phosphonitrilic or Phosphazine polymers:** Preparation, properties, structure and applications.	**2 hrs**
1.5	**Silicates:** Structure, classification – silicates with discrete anions, silicates containing chain anion, silicates with layer structure, silicones with three dimensional net work and applications.	**3 hrs**
1.6	**Silicone:** General methods of preparation, properties. Silicone polymers - silicone fluids, silicone greases, silicone resins, silicone rubbers and their applications.	**3 hrs**
1.7	**Iso and heteropoly acids** – of molybdenum, tungsten and vanadium, industrial applications of iso and heteropoly acids. ***Key Words: Borazine, Phosphazine polymers, Silicone.***	**2 hrs**

MODULE – 2		16 hrs
2.1	**Inorganic fibers**: Introduction, properties, classification, asbestos fibers, optical fibers, carbon fibers, Applications.	**2 hrs**
2.2	**Zeolites:** Introduction, types of zeolites, manufacture of synthetic zeolites and	**2 hrs**
2.3	**Mineral fertilizers**: **Phosphorous containing fertilizers** - Economic importance, importance of superphosphate, ammonium phosphates and their synthesis.	**3 hrs**
2.4	**Nitrogen containing fertilizers** - Importance and synthesis of ammonium sulfate, ammonium nitrate and urea.	**2 hrs**
2.5	**Potassium containing fertilizers** - Economic importance and manufacture of potassium sulfate. **Inorganic pigments**: General information and economic	**2 hrs**
2.6	**White pigments** – titanium dioxide pigments, zinc oxide pigments.	**2 hrs**
2.7	**Colored pigments** – Iron oxide, chromium oxide, mixed-metal oxide pigments and ceramic colorants. Corrosion protection pigments, luster pigments, luminescent pigments, magnetic pigments. ***Key Words: Zeolites, Inorganic Pigments.***	**3 hrs**

REFERANCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
2	Basic Concepts of Inorganic Chemistry	D.N. Singh	Pearson Education,	2nd	2010
3	Inorganic chemistry	Mark Weller, TinaOverton	Oxford Uiversity press	6th	2010
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9	Basic Inorganic Chemistry	F.A. Cotton, G. Wilkinson	John Wiley and Sons	3rd	2002
10	Inorganic Chemistry	D.F. Shriver, P.W. Atkins	Oxford University Press	2nd	1994

RECOMMANDED BOOKS

1	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
2	Basic Inorganic Chemistry	F.A. Cotton, G. Wilkinson	John Wiley and Sons	3rd	2002
3	Inorganic Chemistry	D.F. Shriver, P.W. Atkins	Oxford University Press	2nd	1994
4	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005

Course Title	**ORGANOMETALLIC CHEMISTRY**							
Course Type	**Soft Core- Theory**		**Total Hours**	32	**Hours/Week**	02	**Credits**	02
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	**Duration**	03 hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	Apply the concepts of organometallic chemistry, coordination chemistry and material chemistry to catalysis.
CO-2	Survey on homogenously catalysed reaction and heterogeneous reactions
CO-3	Interpret the influence of chemical behaviours on catalytic cycle.
CO-4	Predicting the stability, structure and the number of M-M bond in the complex
CO-5	Biological and Medicinal Applications

MAPPING CLO'S WITH PSO's AND CD's:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Predicting the stability, structure and the number of M-M bond in the complex	PSO-1	Understand, Analyse
CLO-2	Biological and Medicinal Applications	PSO-1	Understand
CLO-3	Learn about the language of catalysis	PSO-1	Understand, Remember
CLO-4	Interpret the influence of chemical behaviours on catalytic cycle.	PSO-1	Understand, Remember
CLO-5	Ziegler-Natta, a heterogeneous polymerization of catalysis.	PSO-1 PSO-5	Remember, Analyse, Apply.

MODULE – 1 **16 hrs**

1.1	**Fundamental concepts:** Introduction 16 and 18 electron rule Classification of organometallic compounds by bond type, nomenclature, the effective atomic number rule, complexes that disobey the EAN rule, common reactions of organometallic compounds.	**5 hrs**
1.2	**Organometallics of transition metals:** Preparation, bonding and structures of nickel, cobalt, iron and manganese carbonyls. Preparation and structures of metal nitrosyls in organometallics.	**5 hrs**
1.3	**Ferrocene and ruthenocene**: Preparation, structure and bonding.	**3 hrs**
1.4	**Metals containing alkene, alkyne, arene and allyl ligands**: preparation, structure and Bonding. The isolobal principles. ***Keywords: Metal carbonyls, Ferrocene***	**3 hrs**
MODULE – 2		**16 hrs**
2.1	**Homogeneous catalysis - Industrial Applications:** Alkene hydrogenation, hydroformylation, The Wacker's process, Monsanto acetic acid process and L-DOPA synthesis, alkene oligomerizations, water-gas shift reactions.	**6 hrs**
2.2	**Heterogeneous catalysis - Commercial Applications:** Alkene polymerization: Ziegler-Natta catalysis, Fischer-Tropsch carbon chain growth.	**3 hrs**
2.3	**Biological and Medicinal Applications:** Organomercury, boron, silicon and arsenic	**3 hrs**
2.4	**Zeolites as catalysts for organic transformation:** Uses of ZSM - 5	**1 hrs**
2.5	**Alkene metathesis**, hydroboration, arylation or vinylation of olefins (Heck reaction) ***Keywords: Homogenous, heterogeneous catalysis, Industrial applications.***	**3 hrs**

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Basic Organometallic Chemistry	B.D. Gupta and A.J. Elias	Universities Press	1st	2010
3	Concise Coordination Chemistry	R.Gopalan and V.Ramalingam.			
2	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
3	Organometallic Chemistry	R.C. Mehrothra and A. Singh	New Age International	2nd	2006
4	The Organometallic Chemistry of transition metals	Robert H. Crabtree	Wiley Interscience,	4th	2005
5	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
6	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004
7	Advanced Inorganic Chemistry	F. A. Cotton and G. Wilkinson	Wiley-Eastern Company	6th	1990

RECOMMANDED BOOKS

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Basic Organometallic Chemistry	B.D. Gupta and A.J. Elias	Universities Press	1st	2010
2.	Concise Coordination Chemistry	R.Gopalan and V.Ramalingam.			
3.	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
4.	Organometallic Chemistry	R.C. Mehrothra and A. Singh	New Age International	2nd	2006
5.	The Organometallic Chemistry of	Robert H. Crabtree	Wiley Interscience,	4th	2005
6.	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005

Course Title	SYNTHETIC ORGANIC CHEMISTRY							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code	A0330	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To understand the concept of rearrangements in organic reactions
CO-2	To know the concepts of reaction mechanism and its applications
CO-3	To Identify suitable reagent for various Organic Reactions
CO-4	To understand the concepts of protection- deprotection and its applications
CO-5	To introduce range of key reactions for application in organic syntheses.
CO-6	Develop basic skills for the multi-step synthesis of organic compounds

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Knowledge of key reactions in organic chemistry including reactions involving enols and enolates.	PSO-5	Understand
CLO -2	Rationalisation of the outcome of organic reactions using mechanistic reasoning	PSO-5	Understand and Apply
CLO -3	Ability to apply protection –deprotection strategies to plan organic syntheses	PSO-5	Understand and apply
CLO -4	Understanding of Reagents in synthesis	PSO-5	Apply

CLO -5	Knowledge of the retrosynthetic approach to planning organic syntheses.	PSO-5	Apply
CLO -6	Plan synthesis of desired product.	PSO-5	Analyze and Create

COURSE CONTENTS:

MODULE-1 **16hrs**

Molecular rearrangements: Introduction

1.1 **Carbon to Carbon migration**: Pinacol-pinacolone, Wagner-Meerwein, Benzidine, Demjanov, benzylic acid, Favorskii, Arndt-Eistert synthesis, Fries rearrangement, Steven's rearrangement, dienophine rearrangement **6 hrs**

1.2 **Carbon to Nitrogen migration**: Hofmann, Curtius, Lossen, Schmidt and Beckmann rearrangement. **4 hrs**

1.3 **Miscellaneous rearrangements**: Sommelet-Hauser, Neber, Japp-Klingemann rearrangement, Meisenheimer rearrangements, Bayer-Villeger rearrangement, Allylic rearrangements **6 hrs**

Key words- Carbocation, carbanion, Imines.

MODULE-2 **16 hrs**

2.1 **Reagents in organic synthesis:** Use of following reagents in organic synthesis and functional group transformations: Lithium disopropylamide (LDA), Gilmann reagent, dicyclohexyl carbodimide (DCC), dichloro dicyano quinone (DDQ), trialkyl silyl halides, trimethyl silyl cyanide, phase transfer catalyst, crown ethers, cyclodextrins, Fenton's reagent, Ziegler-Natta catalyst, diazomethane, tributyl tinhydride, stannous chloride, Sharpless epoxidation, Woodward and Prevost hydroxylation, Stark enamine reaction. **16 hrs**

Microwave induced organic synthesis, ionic liquids in organic synthesis, polymer supported reagents and synthesis, the use of ultra sound in organic synthesis.

Key words- Oxidation, Dihydroxylation, Reduction, Hydrogenation

MODULE-3 **16 hrs**

3.1 **Protecting groups:** Protection of hydroxyl, carboxyl, carbonyl, thiol and amino groups. Illustration of protection and deprotection in synthesis. **3 hrs**

3.2 **Name reactions:** Keto-enol tautomerism, mechanism and synthetic applications of aldol condensations, Claisen reaction, Schmidt reaction, Perkin reaction, Knoevenagel, benzoin and Stobbe condensation, Darzens glysidic ester condensation, Cannizaros reaction, Tischenko reaction. Michael addition, Robinson's annulation reaction. **6 hrs**

3.3 **Retrosynthesis:** Introduction to disconnection approach: Basic principles and terminologies used in disconnection approach. One group C-X and two group C-X disconnections. Synthons and synthetic equivalents. Retrosynthesis and synthesis of benzofurans, *p*-methoxy acetophenone, saccharine, α-bisabolene, nuciferal, penicillin-V. **7 hrs**

Key words- Protection-Deprotection, Retroanalysis, Enolate ion.

70% Problems (in the form of Predicting products) and 30% Theory

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Organic Chemistry	Robert. N. Boyd and Robert Thornton Morrison	Pearson	6th	2016
2.	A Guide Book to Mechanism in Organic Chemistry	Petersykes	Pearson	6th	2003
3.	Advanced Organic Chemistry	J. March	Wiley	7th	2013
4.	Organic Reaction Mechanism.	R.K. Bansal,	Wiley Eastern Limited, New Delhi,		1993
5.	Organic Chemistry	Marye Anne Fox, James K. Whitesell	Jones & Bartlett Learning	3rd illustrated	2004
6.	Mechanism and Structure in Organic Chemistry	E.S. Gould H. Pine, Hendrickson, Cram and Hammond	Mac Grow Hill, New York		1987
7.	Reactions, Rearrangements and Reagents	Somorendra Nath Sanyal	Bharati Bhavan	4th	2019
RECOMMENDED BOOKS					
1.	Principles of Organic Synthesis,	ROC Norman and J. M. Coxon	CRC Press	6th	1996
2.	Advanced Organic Chemistry	F. A. Carey and R. J. Sundberg,	Plenum		1990
3.	Basic Principles of Heterocyclic Chemistry	Malcolm Sainsbury	RSC	-	2001

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Course Title	CHEMICAL KINETICS (SC)							
Course Type	Soft Core- Theory		Total Hours	48	Hours/Week	02	Credits	02
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Correlating the concepts of kinetics through demonstration of experiments.
CO-2	Explanation of different thermodynamic variables by solving the problems.
CO-3	Illustrate the different numerical concepts involved in electrochemistry.
CO-4	Understand the role of thermodynamic cycles, availability and irreversibility.
CO-5	To state the basis for the collision model of chemical kinetics.

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Understand the role of internal energy, enthalphy, entropy and thermodynamic properties.	PSO-1	Understand, Apply and Analyse
CLO-2	Recognise and understand the different forms of pure substances.	PSO-1	Apply and understand
CLO-3	Identify and use MODULEs and notations in thermodynamics.	PSO-1	Apply, Analyse and Evaluate
CLO-4	To describe how rate laws are determined.	PSO-1	Understand, Apply and evaluate
CLO-5	To describe an activation complex or transition state.	PSO-1	Understand and Analyse
CLO-6	To identify the compounds being oxidized and reduced.	PSO-1	Analyse and understand

COURSE CONTENTS

MODULE-1 **16 hrs**

1.1 Concepts of entropy and free energy: Entropy as a measure of unavailable energy. Entropy change during spontaneous process. Helmholtz and Gibbs free energies (Definition and equation).. Variation of free energy with temperature and pressure. Maxwell's relations, Nernst heat theorem .Third law of thermodynamics - calculation of absolute entropies. **07 hrs**

1.2 Partial molar properties: Partial molar volumes and their determination by intercept method and density measurements. Chemical potential and its significance. Variation of chemical potential with temperature and pressure. Formulation of the Gibbs Duhem equation. Derivation of Duhem-Margules equation. **07 hrs**

1.3	Fugacity: Determination of fugacity of gases. Variation of fugacity with temperature and pressure. Activity and activity coefficients. Variation of activity with temperature and pressure. Determination of activity co-efficient by vapor pressure, depression in freezing point and solubility measurements by electrical methods ***Key words***- Entropy, Free energy, Temperature, Activity.	**02 hrs**

MODULE-2 **16 hrs**

2.1	**Kinetics of complex reactions**: Parallel, consecutive and reversible reactions. Determination of order of reaction. Arrhenius equation, energy of activation and its experimental determination. Potential energy surface, theoretical calculation of energy of activation. Simple collision theory - mechanism of bimolecular reaction. Lindeman's theory, Hinshelwood's theory for unimolecular reaction. Activated complex theory of reaction rate, Kinetics of reactions in solution - Salt effect, effect of dielectric constant (single sphere and double sphere model), effect of pressure, volume and entropy change on reaction rates. Kinetics of heterogeneous reactions - Langmuir's theory, unimolecular and bimolecular surface reactions. **Fast reactions:** Study of kinetics by flow techniques, equation for contact time, stopped flow and continuous flow methods. Relaxation method, equation for relaxation time, temperature jump and pressure jump methods, flash photolysis, pulse radiolysis and shock tube method. ***Key words- Kinetics , Rate equation, Collision Theories***	**08 hrs**
2.2	**Electrochemistry:** Arrhenius theory of strong and weak electrolytes and its limitations, Debye-Huckel theory of strong electrolytes, Debye Huckel-Onsager equation, Debye-Huckel limiting equation for activity co-efficients, Debye-Huckel equation for appreciable concentrations. Liquid junction potential and its determination. Transport Number - Determination of transport number by Hittorf method and E.M.F method. True and apparent transport numbers. Abnormal transport numbers, effect of temperature and concentration on transport number. **Irreversible electrode process:** Introduction, reversible and irreversible electrodes, Potential energy surface, theoretical calculation of energy of activation. reversible and irreversible cells. Polarization, over voltage - ohmic, concentration, activation, experimental determination of over voltage. Hydrogen and Oxygen over voltage. Effect of temperature, current density and pH on over voltage. Equations for concentration over potential, diffusion current – stationary current, potential curves, thickness of diffusion layer, diffusion controlled current – potential curves at a dropping mercury electrode, polarography, half wave potential, application in qualitative and quantitative analysis. Butler-Volmer equation, Tafel equation. ***Key words: Debye-Huckel, Polarization, over voltage.***	**08 hrs**

60% Theory and 40% Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication

1	Thermodynamics for Chemists	S. Glasstone	Read books	--	2007
2	Chemical Kinetics	K.J. Laidler	Pearson	Third	2003
3	Kinetics and Mechanism of Chemical	J. Rajaram and J.C. Kuriacose	Macmillan Publishers	Reprint	2000
RECOMMENDED BOOKS					
1	Physical Chemistry	P.W. Atkins	ELBS, edition, Oxford	fourth	1990
2	Physical Chemistry	Arun Bhal and B.S. Bhal	S. Chand	13th	1963

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Course Title	**SOLID STATE CHEMISTRY AND SEMICONDUCTORS; BIOPHYSICAL CHEMISTRY(SC)**								
Course Type	**Soft Core- Theory**		**Total Hours**	32	Hours/Week	02	Credits	02	
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100	
			External	Duration	02Hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To discuss the fundamental topics of solid state chemistry and electrokinetics
CO-2	To understand various concepts of conductors and its applications
CO-3	Exploration of the fundamental relationship between electronic structure, chemical bonding, and atomic order
CO-4	Utilize band theory to describe the operation of modern semiconductor devices.
CO-5	Specify atomic planes, directions, and families of planes and directions within a given crystal structure using Miller indices

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Will learn the unique properties organic and inorganic solids have with respect to compounds in solution or in the gas phase	PSO-5	Understand, Apply and Analyse
CLO-2	Will learn the methods used to prepare, purify, and crystallize that are specific for organic and	PSO-5	Apply and understand

	inorganic solids.		
CLO-3	Use of spectroscopic, diffraction, microscopic, thermal, and magnetic methods to characterize organic and inorganic solids	PSO-5	Apply, Analyse and Evaluate
CLO-4	Describe solid state phase relations, their chemical synthesis, and thermodynamical and kinetic parameters reaction kinetics as well as characterization methods	PSO-5	Understand, Apply and evaluate
CLO-5	Design and development of materials with pre-required properties based on the structure of solids.	PSO-5	Understand and Analyse
CLO-6	To identify the compounds being oxidized and reduced.	PSO-5	Analyse and understand

COURSE CONTENTS

MODULE-1
16 hrs

1.1	Solid state chemistry: Types of imperfections, classification of imperfections, point defects, Schottky defects, Frenkel defects, disordered crystals, line defects, dislocation types, plane defects, small-angle and large-angle boundaries, stacking faults, crystal growth and twinning	**7 hrs**
1.2	Semiconductors: Band theory, energy bands, intrinsic and extrinsic semiconductors. Conductivity: electrons and holes, temperature dependence on conductivity, Optical properties: absorption spectrum, photoconductivity, photovoltaic effect and luminescence. Junction properties: metal-metal junctions, metal-semiconductor junctions, p-n junctions, transistors, industrial applications of semiconductors: Mixed oxides, spinels and other magnetic materials.	**7 hrs**
1.3	Superconductors: Meissner effect, type I and II super conductors, isotope effect, basic concepts of BCS theory, manifestations of the energy gap, Josephson devices	**2 hrs**

Key words- Crystal, semiconductor, defects

MODULE-2 **16 hrs**

2.1	Electrokinetic phenomena: Electrophoresis - principles of free electrophoresis, zone electrophoresis, gel electrophoresis and its applications in qualitative and quantitative study of proteins. Determination of isoelectric point of a protein. Electro osmosis and streaming potential and its biological significance. Biological significance of Donnan membrane phenomenon. Micelles and its involvement during digestion and absorption of dietary lipids. Diffusion of solutes across bio membranes and its application in the mechanism of respiratory exchange. "Salting In" and "Salting Out" of proteins. Osmotic behavior of cells and osmo-regulation and its application in the evolution of excretory systems of organisms. Effect of temperature and pH on the viscosity of biomolecules	**16 hrs**

(albumin solution). Significance of viscosity in biological systems - mechanism of muscle contraction, detection of intrastrand disulfide bonds in proteins, polymerization of DNA and nature of blood flow through different vessels. Effect of temperature, solute concentration (amino acids) on surface tension. Biological significance of surface tension - stability of Alveoli in lungs, interfacial tension in living cells (Danielli and Davson model). Application of sedimentation

Key words: Electro osmosis, Micelles, bio membranes, streaming potential.

60% Theory and 40% Problems

REFERENCES

Sl. No	**Title of the book**	**Name of the author**	**Name of the publisher**	**Edition**	**Year of publicatio**
1	Applied Biopharmacokinetics and Pharmacokinetics	Leon Shargel, Andrew YuPrentice	Hall international, Inc	Fourth	-
2	Chemical Kinetics	K.J. Laidler	Pearson	Third	2003
3	Kinetics and Mechanism of	J. Rajaram and J.C. Kuriacose	Macmillan Publishers	Reprint	2000
RECOMMENDED BOOK					
1	Essentials of Physical Chemistry and Pharmacy	H.J. Arnikar, S.S. Kadam, K.N. Gujan	Orient Longman, Bombay	-	1992

Course Title	**ANALYTICAL CHEMISTRY PRACTICALS**								
Course Type	**Hard Core-** Practical		**Total Hours**	96	**Hours/Week**	04	**Credits**	**02**	
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100	
			External	Duration	03Hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	**Course Objectives**
CO-1	Quantitative analysis of water samples by EDTA titrations
CO-2	Determination of stoichiometry of certain basic reactions
CO-3	Potentiometric and conductometric determination
CO-4	Chromatographic determination and detection
CO-5	Absorption and emission method of analysis

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	**Course Learning Outcomes On Completion of the Course the Students will Learn to**	**PSOs Addressed**	**Cognitive Level**

CLO-1	Quantitative analysis using precipitation titration	PSO-3	Apply Analyse and Evaluate
CLO-2	Quantitative analysis using spectrophotometric method	PSO-3	Apply, Analyse and Evaluate
CLO-3	Quantitative analysis by complexometric titration	PSO-3	Apply, Analyse and Evaluate
CLO-4	Quantitative analysis by acid base titration	PSO-3	Apply Analyse and Evaluate
CLO-5	Quantitative analysis by flame photometric method	PSO-3	Apply Analyse and Evaluate

COURSE CONTENTS:

Part I	
	1. Determination of total acidity of vinegar and wines by acid-base titration. 2. Determination of purity of a commercial boric acid sample, and Na_2CO_3 content of washing soda. 3. Determination of replaceable hydrogen and relative molecular mass of a weak organic acid by titration with NaOH. 4. Periodate determination of ethylene glycol and glycerol (Malprade reaction). 5. Determination of carbonate and bicarbonate in a mixture by *p*H-metric titration and comparison with visual acid-base titration 6. Determination of the *p*H of hair shampoos and *p*H determination of an unknown soda ash. 7. Analysis of water/waste water for acidity by visual, *p*H metric and conductometric titrations. 8. Analysis of water/waste water for alkalinity by visual, *p*H metric and conductometric titrations. 9. Determination of ammonia in house-hold cleaners by visual and conductometric titration. 10. Potentiometric determination of the equivalent weight and Ka for a pure unknown weak acid. 11. Spectrophotometric determination of creatinine and phosphorus in urine. 12. Flame emission spectrometric determination of sodium and potassium in river/lake water
Part II	
	1. Determination of percentage of chloride in a sample by precipitation titration - Mohr, Volhard and Fajan's methods. 2. Determination of silver in an alloy and Na_2CO_3 in soda ash by Volhard method. 3. Determination of total hardness, calcium and magnesium hardness and carbonate and bicarbonate hardness of water by complexation titration using EDTA. 4. Determination of calcium in calcium gluconate/calcium carbonate tablets/injections and of calcium in milk powder by EDTA titration. 5. Analysis of commercial hypochlorite and peroxide solution by iodometric titration. 6. Determination of copper in an ore/an alloy by iodometry and tin in stibnite by iodimetry. 7. Determination of ascorbic acid in vitamin C tablets by titrations with $KBrO_3$ and of vitamin C in citrus fruit juice by iodimetric titration. 8. Determination of iron in razor blade by visual and potentiometric titration using sodium metavanadate.

	9. Determination of iron in pharmaceuticals by visual and potentiometric titration using cerium(IV) sulphate. 10. Cation exchange chromatographic separation of cadmium and zinc and their estimation by EDTA titration. 11. Separation and determination of chloride and bromide on an anion exchanger. 12. Thin layer chromatographic separation of amino acids.

100% practical

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern Analytical chemistry	David Harvey	Mcgraw Hill	1st	2000
4	Analytical Chemistry 2.0	David Harvey	McGraw-Hill Companies, Depauw	1st n	2018
5	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6/E, 6th	2008
6	Vogel's Textbook of Quantitative Chemical Analysis	J. Mendham, R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd., New Delhi	6th, Third Indian Reprint,	2003.
7	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
8	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
9	Instant Notes of Analytical Chemistry by	Kealey and Haines,	Viva Books Pvt. Ltd.,		2002.
10	Principles and Practice of Analytical Chemistry	by F. W. Fifield and Kealey,	Blackwell Sci., Ltd. Malden, USA,	5th	2000
11	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
12	Biochemical analysis	Irwin H Segel	Wiley	2nd	2004
13	Spectroscopy	H. Kaur`	Pragathi prakashan	11th	2016

14	Instrumental Methods of Analysis	Supriya S. Mahajan	Popular Prakashan Private Ltd	1st	2010
15	Analytical Chemistry	S. M. Khopkar	New age international publishers	1st	2013
16	Elements of Spectroscopy	Guptha, Kumar, Sharma	Pragati Edition	28th	2016
17	Calculations in Chemistry	John olusina obimakinde, Samuel Oluwaseun obimakinde	IK international publishing house	1st	2014

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Course Title	INORGANIC CHEMISTRY PRACTICALS							
Course Type	**Hard Core-Practical**		**Total Hours**	**96**	**Hours/Week**	**04**	**Credits**	**02**
Course Code	AP220	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duratio	03Hr	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	To impart advanced knowledge on the classification and further analysis of mixture of metal to ions qualitatively.
CO-2	Identify individual ions present in water and mixture of salt and the chemistry behind it
CO-3	To acquire knowledge of laboratory techniques, inorganic synthesis and characterization.
CO-4	To impart the knowledge about redox, complexometric and precipitation reactions
CO-5	Quantitative analysis of metal ions in alloy/mixture and Estimation of various ions by titrimetry and spectrophotometric methods.

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcome	PSOs Addressed	Cognitive Level
CLO-1	Ensures the students to understand, acquire knowledge and have hands on experience in multistep inorganic compound synthesis and analysis by using spectroscopic techniques. Separation techniques.	PSO-3	Analyze, Apply Evaluate
CLO-2	Gravimetric analysis	PSO-3	Analyze, Apply
CLO-3	Laboratory techniques.	PSO-3	Analyze, Apply Evaluate
CLO-4	To impart the knowledge about redox, complexometric and precipitation reactions.	PSO-3	Analyze, Apply Evaluate
CLO-5	Spectrophotometric determinations	PSO-3	Analyze, Apply Evaluate

Part I	
	1. Determination of iron in haematite using cerium(IV) solution (0.02M) as the titrant, and gravimetric estimation of insoluble residue. 2. Estimation of calcium and magnesium carbonates in dolomite using EDTA titration, and gravimetric analysis of insoluble residue. 3. Determination of manganese dioxide in pyrolusite using permanganate titration. 4. Quantitative analysis of copper-nickel in alloy/mixture: i. Copper volumetrically using KIO_3. ii. Nickel gravimetrically using DMG 5. Determination of lead and tin in a mixture: Analysis of solder using EDTA titration. 6. Quantitative analysis of chloride and iodide in a mixture:

	i. Iodide volumetrically using KIO_3 . ii. Total halide gravimetrically 7. Gravimetric analysis of molybdenum with 8-hydroxyquinoline. 8. Spectrophotometric determinations of: a. Titanium using hydrogen peroxide. b. Chromium using diphenyl carbazide in industrial effluents c. Iron using thiocyanate/1,10-phenanthroline method in commercial samples d. Nickel using dimethylglyoxime in steel solution 9. Micro-titrimetric estimation of : a) Iron using cerium (IV). b) Calcium and magnesium using EDTA 10. Quantitative estimation of copper(II), calcium(II) and chloride in a mixture. 11. Circular paper chromatographic separation of: (Demonstration) a. Iron and nickel. b. Copper and nickel
Part II	
	Semi micro qualitative analysis of mixtures containing **TWO** anions and **TWO** cations (excluding sodium, potassium and ammonium cations) and **ONE** of the following less common cations: W, Mo, Ce, Th, Ti, Zr, V, U and Li.

REFERENCES

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	A Text Book of Quantitative Inorganic Analysis	A.I. Vogel	ELBS	3rd	
2	Vogel's Text Book of Quantitative Chemical Analysis	J.Mendham,R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd	6th	2003
3	Spectrophotometric Determination of Elements	Z. Marczenko.	Ellis Horwood	-	1976
4	Vogel's Qualitative Inorganic Analysis	– Svelha.	Pearson Education	**7th**	2008
5	Quantitative Chemical Analysis	Daniel C. Harris,	-	7th	2006
6	Practical Inorganic chemistry	Gurdeep Raj	GOEL Publishing house	26th	2015
7	Quantitative Analysis	R.A. Day and A	Prentice Hall, Inc. New Delhi.	6th	1993

Course Title	ORGANIC CHEMISTRY PRACTICALS							
Course Type	**Hard Core**- Practical		**Total Hours**	96	**Hours/Week**	04	Credits	02
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	To understand the concept of oxidation and reduction in organic reactions
CO-2	To understand the concepts of reaction mechanism and its applications
CO-3	To Identify suitable reagent for various chemical conversion
CO-4	Develop basic skills for the multi-step synthesis of organic compounds
CO-5	To Practice identification and separation of organic Mixtures.

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	The ability to apply reagent knowledge Practically	PSO-3	Understand
CLO-2	Understanding the solution of problems related to the synthesis of organic target molecules	PSO-3	Apply and Analyse
CLO-3	Practical skills in handling organic chemistry reactions	PSO-3	Apply
CLO-4	Evaluate the risks associated with an experiment	PSO-3	Analyse

COURSE CONTENTS

PART – I	
1. 1	Preparation of *p*-nitro aniline from acetanilide.
1.2	Preparation of n-butyl bromide from n-butanol.
1.3	Preparation of *p*-nitroiodobenzene from paranitroaniline.
1.4	Preparation of aniline from nitrobenzene.
1.5	Preparation of s-*D*-Glucose Penta acetate.
1.6	Preparation of Phenoxy acetic acid
1.7	Preparation of cyclohexanone from cyclohexanol.
1.8	Preparation of chalcone.
1.9	Preparation of *S*-Benzylthiuronium chloride.
1.10	Condensation of anthracene and maleic anhydride (Diels-Alder reaction).
1.11	Preparation of m-nitrobenzoic acid from methyl benzoate.
Part –II	
2.1	Qualitative analysis: Separation of binary mixtures, identification of functional groups and Preparation of suitable solid derivatives. ***Key words- Organic Preparation, Binary Mixture Analysis***

60% Problems (in the form of predicting products) and 40% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Comprehensive practical organic chemistry	V. K. Ahluwalia and Renu Agarwal	Universities Press	-	2001
2.	An Introduction to Practical Organic Chemistry	Caroline Frances Cornwallis	BiblioBazaar	-	2009
3.	Advanced Practical Organic Chemistry	By John Leonard, Barry Lygo, Garry Procter	Stanley Thrones Publishing	2nd	1994
4.	Manual of Organic Chemistry	Dey and Seetharaman	Allied Publishers,	-	1957
5.	Laboratory Manual of Organic Chemistry	R.K. Bansal	New Age International (P) Ltd. London	3rd	1996
6.	A Text Book of Practical Organic Chemistry by Vol.III	A.I. Vogel,	Longman Scientific & Technical	-	1989
RECOMMENDED BOOKS					
1.	Practical Organic Chemistry	Mann & Saunders	Orient Longman	4th	2004
2.	Semi micro Qualitative Organic Analysis	Cheronis, Entrikin and Hodnet.	R.E. Krieger Publishing Company,	3, reprint	1983

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Course Title	**PHYSICAL CHEMISTRY PRACTICALS**							
Course Type	Hard Core- Theory		**Total Hours**	48	Hours/Week	04	Credits	02
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To demonstrate an ability to conduct experiments in the above sub-disciplines with mastery of appropriate techniques and proficiency using core chemical instrumentation and modeling methods
CO-2	To develop skills in quantitative modeling of static and dynamic chemical systems.
CO-3	To develop laboratory competence in relating chemical structure to spectroscopic phenomena
CO-4	To demonstrate the ability to perform accurate quantitative measurements with an understanding of the theory and use of contemporary chemical instrumentation, interpret experimental results, perform calculations on these results and draw reasonable, accurate conclusions

CO-5	To demonstrate broad knowledge of descriptive Chemistry

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Think critically and analyze chemical problems.	PSO-3	Understand, Apply and Analyse
CLO-2	Present scientific and technical information resulting from laboratory experimentation in both written and oral formats	PSO-3	Apply and understand
CLO-3	Work effectively and safely in a laboratory environment.	PSO-3	Apply, Analyse and Evaluate
CLO-4	Use technologies/instrumentation to gather and analyze data.	PSO-3	Understand, Apply and evaluate
CLO-5	Work in teams as well as independently	PSO-3	Understand and Analyse
CLO-6	Apply modern methods of analysis to chemical systems in a laboratory setting	PSO-3	Analyse and understand

COURSE CONTENTS

PART – I	
1. 1	Study of kinetics of hydrolysis of an ester using HCl/H_2SO_4 at two different temperatures and concentrations, determination the rate constants and energy of activation.
1. 2	Study of kinetics of reaction between $K_2S_2O_8$ and KI, first order, determination of rate constant and *E*a.
1.3	Conductometric titration of a mixture of HCl and $ClCH_2COOH$ against NaOH.
1.4	Potentiometric titration of KI against $KMnO_4$ solution.
1.5	Determination of dissociation constant of weak acid by potentiometric method.
1.6	Potentiometric titration of $AgNO_3$ against KBr.
1.7	Verification of Beer's law and calculation of molar extinction coefficient for $CuSO_4$
1.8	Spectrophotometric titration of $FeSO_4$ against $K_2Cr_2O_7$.
1.9	Determination of heat of solution of Organic acid by variable temperature method.
1.11	Kinetics of photodegradation of indigocarmine (IC) using ZnO as photocatalyst and study the effect of [ZnO] and [IC] on the rate of photo degradation.
1.12	Potentiometric titration of KI against $KMnO_4$ solution.
1.13	Determination of dissociation constant of weak acid by potentiometric method.
1.14	Potentiometric titration of $AgNO_3$ against KBr.
1.15	To study the salt effect on kinetics of reaction between $K_2S_2O_8$ and KI.
1.16	Study of kinetics of reaction between CAT and indigocarmine spectrophotometrically and determination of rate constant.

60% Theory and 40% Problems

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Practical Physical Chemistry	-	-	-	-
2	Experimental Physical Chemistry	-	-	-	-
3	Experiments in Physical Chemistry	Yadav	Geol Publishing House	-	-
RECOMMENDED BOOKS:					
1	Experiments in Chemistry	D.V. Jahagirdar	Himalaya Publishing House	-	-
2	Experimental Physical Chemistry	R.C. Das and B. Behera	Tata Mc Graw Hill	-	-

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SECOND SEMESTER

Course Title	SEPARATION TECHNIQUES								
Course Type	Hard Core-Theory		Total Hours	48	Hours/Week	03	Credits	03	
Course Code	B0210	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100	
			External	Duration	03Hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Understand the theory behind the chromatographic separations
CO-2	Separation based on the nature of the stationary and mobile phase
CO-3	Solvent extraction for preparative organic chemistry
CO-4	Advantages of the solid phase micro extraction over the solvent extraction
CO-5	Electrophoretic methods of separation based on the interim charge associated with bimolecular system

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Chromatographic separation-mathematical equations	PSO-5	Remember, Understand, Apply , Analyse and evaluate
CLO-2	Separation of synthetic and natural samples by chromatographic methods	PSO-5	Apply, Analyse and Evaluate
CLO-3	Separation of samples-based on charge, molecular mass, lock-key mechanism	PSO-5	Apply, Analyse and Evaluate
CLO-4	Solvent extraction methods- extraction systems such as acid, metal complexes, solid phase extraction systems	PSO-5	Remember, Understand, Apply and Analyse
CLO-5	Electrophoretic methods-separation based on electrophoresis and electro-osmotic flow	PSO-5	Understand, Apply, Analyse and evaluate
CLO-6	Problems related to the chromatographic separation	PSO-5	Evaluate

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **Chromatography:** Definition, principles and mechanism of separation, classification of chromatographic techniques. General descriptions of column chromatography - frontal analysis, displacement analysis and elution analysis. General theory of column chromatography: characterizing a chromatogram - retention time, retention volume and baseline width. Chromatographic resolution, capacity factor, column selectivity. Column efficiency – band broadening - rate theory and plate **10 hrs**

theory. Peak capacity, non ideal behavior. Optimizing chromatographic separations using capacity factor, column selectivity and column efficiency - Van Deemter equation, and its modern versions, Golay equation and Huber-Knox equation.

1.2	**Gas chromatography (GC):** Principles, instrumentation - mobile phase, chromatographic columns, stationary phases, sample introduction, temperature control, and detectors for gas chromatography. Quantitative and qualitative	**2 hrs**
1.3	**High performance liquid chromatography (HPLC):** Principles, instrumentation – columns (analytical and guard columns), stationary phases, mobile phases, choosing a mobile phase, isocratic *vs* gradient elution, HPLC plumbing, sample introduction. Detectors for HPLC - spectroscopic, electrochemical and others, quantitative applications.	**4 hrs**

Keywords: Chromatography-Introduction, Mathematical equations, Gas chromatography, High performance liquid chromatography

MODULE - 2 **16 hrs**

2.1	**Ion exchange chromatography (IEC):** Definitions, requirements for ion-exchange resin, synthesis and types of ion-exchange resins, principle, basic features of ion- exchange reactions, resin-properties-ion-exchange capacity, resin selectivity and factors affecting the selectivity, applications of IEC in preparative, purification and recovery processes. IEC with eluent suppressor columns. Single column ion chromatography.	**5 hrs**
2.2	**Size-exclusion chromatography:** Theory and principle of size-exclusion chromatography, experimental techniques of gel-filtration chromatography (GFC) and gel-permeation chromatography (GPC), materials for packing - factors governing column efficiency, methodology and applications.	**3 hrs**
2.3	**Thin layer chromatography:** Principle, apparatus and methodology, applications, HPTLC.	**1 hrs**
2.4	**Affinity chromatography:** Definitions, separation-mechanism-matrices, matrix activation, role of spacer arms and applications.	**1 hrs**
2.5	**Supercritical fluid chromatography (SFC):** Properties of supercritical fluids, instrumentation and operating variables, comparison of SFC with other types of chromatography, applications.	**3 hrs**
2.6	**Supercritical fluid extraction:** Advantages of supercritical fluid extraction, instrumentation, supercritical fluid choice, off-line and on-line extractions, typical applications of supercritical fluid extraction.	**3 hrs**

Key word: IEC, SEC, TLC, AC, SFC, SCF

Module-3 **16hrs**

3.1	**Electrophoretic methods** - Electrophoresis & Capillary Electrophoresis**:** Theory - electrophoretic mobility, electroosmotic mobility, electroosmotic flow velocity, total mobility, migration time,	**8hrs**

efficiency, selectivity and resolution. Instrumentation - capillary tubes, hydrodynamic and electrokinetic methods of sample injection, applying electric field and detectors. Capillary electrophoresis methods - capillary zone electrophoresis, micellar electrokinetic capillary chromatography, capillary gel electrophoresis and capillary electro chromatography.

3.2 **Solvent extraction:** Theory - Nernst partition law, efficiency and selectivity of extraction. **1hr**

3.3 **Extraction systems:** Extraction of covalent neutral molecules, extraction of uncharged metal chelates and synergic extraction, extraction of ion-association complexes - non chelated complexes, chelated complexes and oxonium systems. Use of salting out agents. Methods of extraction - batch and continuous extractions. applications. **4hrs**

3.4 **Solid Phase Extraction (SPE):** Principles, apparatus and instrumentation. Solid phase sorbents, extraction formats - Automated solid phase extraction. Solid phase micro extraction (SPME). Applications of SPE and SPME. **3hrs**

Key words: Solvent extraction, electrophoresis, solid phase extraction systems

80% Theory and 20% Problems

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern chemistry Analytical	David Harvey	Mcgraw Hill	1st	2000
4	Analytical 2.0 Chemistry	David Harvey	McGraw-Hill Companies, Depauw	1st	2018
6	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
7	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
8	Instant Notes of Analytical Chemistry	Kealey and Haines,	Viva Books Pvt. Ltd.,		2002.
9	Principles and Practice of Analytical Chemistry	by F. W. Fifield and Kealey,	Blackwell Sci., Ltd. Malden, USA,	5th	2000

10	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007

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Course Title	COORDINATION CHEMISTRY								
Course Type	Hard Core-Theory		Total Hours	48	Hours/Week		03	Credits	03
Course Code	B0220	Evaluation	Internal	C1+C2 = 15+15			30 Marks		100
			External	Duration	03Hrs	C3	70 Marks		

Course Objectives:

CO No.	Course Objectives
CO-1	Structural arrangements of the ligands around the metal atom and constitution of and geometry of higher coordination number.
CO-2	Thermodynamics and stability of complex formation
CO-3	Understand the electronic structures
CO-4	Elucidate the electronic spectras
CO-5	Understand the concept of magnetism

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Ligand field splitting parameter, Calculation of the LFSE	PSO-1	Understand, Apply and Analyse
CLO-2	Kinetic consequences of Reaction pathways	PSO-1	Understand, Analyse
CLO-3	Inferring the electron configuration from a magnetic moment.	PSO-1	Apply, Analyse and Evaluate
CLO-4	Understand the pi bonding and the selection rule and calculation of magnetic properties.	PSO-1, PSO-5	Understand, Apply, Analyse, Evaluate
CLO-5	Term symbols of d systems, electronic spectra, Racah parameters	PSO-1, PSO-5	Analyse, Apply, Evaluate

MODULE – 1		**16 hrs**
1.1	**Preparation of coordination compounds:** Introduction, Preparative methods - simple addition reactions, substitution reactions, oxidation-reduction reactions, thermal dissociation reactions, photochemical, thermal decomposition and trans	**4 hrs**
1.2	**Detection of complexation**- colour change, conduction, pH, solubility change, precipitate formation, IR Spectra and magnetic methods. Geometries of metal complexes of higher coordination numbers (2-12).	**3 hrs**
1.3	**Stability of coordination compounds:** Introduction- thermodynamic and kinetic stability; stability constants (formation constants) -stepwise (K_n) and overall stability (β_n)constants, Trends in stepwise stability constants, factors influencing the stability of metal complexes with reference to the nature of metal ion and ligands, chelate effect, macrocyclic effect, the Irving-William series.	**6 hrs**

1.4	**Determination of stability constants:** Theoretical aspects of determination of stability constants of metal complexes by Spectrophotometric, *p*H metric (Bjerrums)and polarographic methods. **Keywords:** Preparative methods, Stability constants	**3 hrs**
MODULE – 2		**16 hrs**
2.1	**Crystal field theory:** Salient features of CFT, d-orbital splitting in octahedral, tetrahedral, square planar and tetragonal complexes, Jahn-Teller distortions, measurement of 10 Dq and factors affecting it. Evidences for metal-ligand [illegible]	**4 hrs**
2.2	**Molecular Orbital Theory:** MOT to octahedral, tetrahedral and square planar complexes without and with pi-bonding. MO energy diagrams for octahedral complexes with sigma ligands having pi-systems.	**4 hrs**
2.3	**Electronic spectra**: Introduction, selection rules and intensities, electronic spectra of octahedral and tetrahedral complexes, Term symbols for dn ions, Orgel and Tanabe-Sugano diagrams, charge-transfer spectra.	**4 hrs**
2.4	**Magnetic properities**: Introduction, magnetic susceptibility and its measurements, spin and orbital contributions to the magnetic moment, the effects of temperature on μeff, spin-cross over, ferromagnetism, antiferromagnetism and ferrimagnetism. ***Keywords: MOT, Electronic spectra, Magnetic properties.***	**4 hrs**
MODULE - 3		**16hrs**
3.1	**Reaction and Mechanisms:** Introduction, Substitution reactions - Inert and labile compounds, mechanisms of substitution. Kinetic consequences of Reaction pathways - Dissociation, interchange and association. Experimental evidence in octahedral substitution - Dissociation, associative mechanisms, the conjugate base mechanism, the kinetic chelate effect.Stereochemistry of reactions- Substitution in trans and its complexes, isomerization of chelaterings. Substitution reactions of square-planar complexes - kinetics and stereochemistry of square-planar substitutions, evidence for associative reactions, explanations of the trans effect.	**10hrs**
3.2	**Transfer processes:** Inner-sphere mechanism and outer-sphere mechanism, conditionsfor high and low oxidation numbers.	**3hrs**
3.3	**Metal-metal bonding:** Evidences and factors favoring of M-M bonding, Wade's-Mingo's- Lauher rules, bi, tri, tetra, penta and hexa nuclear metal clusters. **Keywords:** Reaction mechanism	**3hrs**

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Physical Inorganic Chemistry- A Coordination Chemistry Approach	S.F.A. Kettle	Spektrum, Oxford		1996
2	Concise Coordination Chemistry	R. Gopalan and V. Ramalingam.			
3	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
4	Basic Concepts of Inorganic Chemistry	D.N. Singh	Pearson Education,	2nd	2010
5	Inorganic chemistry	Mark Weller, TinaOverton	Oxford Uiversity press	6th	2010
6	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
7	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004
8	Concise Inorganic Chemistry	J.D. Lee	Blackwell Science Ltd.,	5th	2003
9	Inorganic Chemistry	Catherine E. *Housecroft* And G. Sharpe Alan	Pearson Education	2nd	2003
10	Concepts and Models of Inorganic Chemistry	Douglas, B., McDaniel, D. H., and Alexander, J. J	John Wiley & sons, New York,	2nd	2006.
11	*Modern Inorganic Chemistry*	Jolly, W. L.,	McGraw Hill New York	2nd	1991

RECOMMANDED BOOKS

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Physical Inorganic Chemistry- A Coordination Chemistry Approach	S.F.A. Kettle	Spektrum, Oxford		1996
2	Concise Coordination Chemistry	R.Gopalan and V.Ramalingam.			
3	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter	Pearson Education	4th	2013
4	Inorganic chemistry	Mark Weller, Tina Overton	Oxford Uiversity press	6th	2010
5	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
6	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004
7	Advanced Inorganic Chemistry	F. A. Cotton and G. Wilkinson	Wiley-Eastern Company	6th	1990
8	Basic Organometallic Chemistry: Concepts, Syntheses, and Applications of Transition Metals	Gupta, B. D., and Elias, Anil. J	Universities Press, CRC Press	1st	2010.

Course Title	**SYNTHETIC ORGANIC CHEMISTRY AND MOLECULAR REARRANGEMENTS**							
Course Type	**Hard Core-Theory**		**Total Hours**	**48**	**Hours/Week**	03	Credits	03
Course Code	B0230	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To understand the concept of rearrangements in organic reactions
CO-2	To know the concepts of reaction mechanism and its applications
CO-3	To Identify suitable reagent for various Organic Reactions
CO-4	To understand the concepts of protection- deprotection and its applications
CO-5	To introduce range of key reactions for application in organic syntheses.
CO-6	Develop basic skills for the multi-step synthesis of organic compounds

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students	PSOs Addressed	Cognitive Level

	will Learn to		
CLO-1	Knowledge of key reactions in organic chemistry including reactions involving enols and enolates.	**PSO-5**	Understand
CLO -2	Rationalisation of the outcome of organic reactions using mechanistic reasoning	**PSO-5**	Understand and Apply
CLO -3	Ability to apply protection –deprotection strategies to plan organic syntheses	**PSO-5**	Understand and apply
CLO -4	Understanding of Reagents in synthesis	**PSO-5**	Apply
CLO -5	Knowledge of the retrosynthetic approach to planning organic syntheses.	**PSO-5**	Apply
CLO -6	Plan synthesis of desired product.	**PSO-5**	Analyze and Create

COURSE CONTENTS

MODULE 1 **16hrs**

	Molecular rearrangements: Introduction	**6 hrs**
1.1	**Carbon to Carbon migration**: Pinacol-pinacolone, Wagner-Meerwein, Benzidine, Demjanov, benzylic acid, Favorskii, Arndt-Eistert synthesis, Fries rearrangement, Steven's rearrangement, dienophine rearrangement	
1.2	**Carbon to Nitrogen migration**: Hofmann, Curtius, Lossen, Schmidt and Beckmann rearrangement.	**4 hrs**
1.3	**Miscellaneous rearrangements**: Sommelet-Hauser, Neber, Japp-Klingemann rearrangement, Meisenheimer rearrangements, Bayer-Villeger rearrangement, Allylic rearrangements	**6 hrs**
	Key words- Carbocation, carbanion, Imines.	

MODULE 2 **16 hrs**

2.1	**Reagents in organic synthesis:** Use of following reagents in organic synthesis and functional group transformations: Lithium disopropylamide (LDA), Gilmann reagent, dicyclohexyl carbodimide (DCC), dichloro dicyano quinone (DDQ), trialkyl silyl halides, trimethyl silyl cyanide, phase transfer catalyst, crown ethers, cyclodextrins, Fenton's reagent, Ziegler-Natta catalyst, diazomethane, tributyl tinhydride, stannous chloride, Sharpless epoxidation, Woodward and Prevost hydroxylation, Stark enamine reaction. Microwave induced organic synthesis, ionic liquids in organic synthesis, polymer supported reagents and synthesis, the use of ultra sound in organic synthesis.	**16 hrs**
	Key words- Oxidation, Dihydroxylation, Reduction, Hydrogenation	

Module 3 **16 hrs**

3.1	**Protecting groups:** Protection of hydroxyl, carboxyl, carbonyl, thiol and amino groups. Illustration of protection and deprotection in synthesis.	**3 hrs**
3.2	**Name reactions:** Keto-enol tautomerism, mechanism and synthetic applications of aldol condensations, Claisen reaction, Schmidt reaction, Perkin reaction, Knoevenagel, benzoin and Stobbe condensation, Darzens glysidic ester condensation, Cannizaros reaction, Tischenko reaction. Michael addition, Robinson's annulation reaction.	**6 hrs**
3.3	**Retrosynthesis:** Introduction to disconnection approach: Basic principles and terminologies used in disconnection approach. One group C-X and two group C-X disconnections. Synthons and synthetic equivalents. Retrosynthesis and synthesis of benzofurans, *p*-methoxy acetophenone, saccharine, α-bisabolene, nuciferal, penicillin-V.	**7hrs**
	Key words- Protection-Deprotection, Retroanalysis, Enolate ion.	

70% Problems (in the form of Predicting products) and 30% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Organic Chemistry	Robert. N. Boyd and Robert Thornton Morrison	Pearson	6th	2016
2.	A Guide Book to Mechanism in Organic Chemistry	Petersykes	Pearson	6th	2003
3.	Advanced Organic Chemistry	J. March	Wiley	7th	2013
4.	Organic Reaction Mechanism.	R.K. Bansal,	Wiley Eastern Limited, New Delhi,		1993
5.	Organic Chemistry	Marye Anne Fox, James K. Whitesell	Jones & Bartlett Learning	3rd illustrated	2004
6.	Mechanism and Structure in Organic Chemistry	E.S. Gould H. Pine, Hendrickson, Cram and Hammond	Mac Grow Hill, New York		1987
7.	Reactions, Rearrangements and Reagents	Somorendra Nath Sanyal	Bharati Bhavan	4th	2019
RECOMMENDED BOOKS					
12.	Principles of Organic Synthesis,	ROC Norman and J. M. Coxon	CRC Press	6th	1996
13.	Advanced Organic Chemistry	F. A. Carey and R. J. Sundberg,	Plenum		1990
14.	Basic Principles of Heterocyclic Chemistry	Malcolm Sainsbury	RSC	-	2001

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Course Title	QUANTUM CHEMISTRY; NUCLEAR CHEMISTRY AND PHOTO CHEMISTRY AND MICROWAVE AND INFRARED SPECTROSCOPY(HC)							
Course Type	Hard Core- Theory		Total Hours	48	Hours/Week	03	Credits	03
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Understand the level of mathematics as a tool to understand atomic and molecular structure and properties as well as chemical reactivity.
CO-2	To acquire hands on experience with a quantum mechanics software package.
CO-3	Synthesis and Characterisation of polymers.
CO-4	Technological developments in commodity and advanced polymers.
CO-5	To link thermodynamics to the micro description used in statistical mechanics.

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Evaluating the different parameters like entropy enthalpy and free energies.	PSO-1	Evaluate and Apply
CLO-2	Kinetics, mechanism of free radical polymerization and methodology used of control molecular weight of polymers.	PSO-1	Understand and apply
CLO-3	Different techniques of polymerization of polymers.	PSO-1	Apply, and Analyse
CLO-4	Conceptual understanding of quantum mechanics of atoms and molecules.	PSO-1	Understand
CLO-5	Applying the concept and solving the problems on different aspects.	PSO-1	Understand and Apply
CLO-6	Defines the importance of phase diagram in the field of material science and engineering	PSO-1	Analyse and apply

COURSE CONTENTS

Module-1 **16 hrs**

1.1	Wave-particle duality of material particles, de Broglie equation, Heisenberg Uncertainty principle, Concept of operators (operator–operand), Algebra of operators, commutative and non-commutative operators, linear operator, Laplacian operator, Hamiltonian operator, Eigen value, Eigen function, class Q function, Hermitian operator, Wave equation for stretched strings, Schrodinger wave equation for particles, Eigen values and Eigen functions, Postulates of quantum mechanics. Application of Schrodinger equation to a free particle and to a particle trapped in a potential field (one dimension and three dimensions). Wave equation for H-atom, Application of Schrodinger equation to rigid rotator and harmonic oscillator. Approximate methods – Necessity of approximate methods, perturbation method, the theory of perturbation method – first order and second order correction, Variation theorem: statement and proof.	**16 hrs**
	Keywords - Functions, Particle, one, two dimensions.	
Module-2		**16 hrs**
2.1	**Nuclear chemistry**: Radioactive decay – General characteristics, decay kinetics, parent –daughter decay growth relationships, determination of half-lives, Nuclear stability – packing fraction, binding energy, Brief survey of alpha, beta and gamma decays. Nuclear reactions – Bethe's notation, types of nuclear reactions – specific nuclear reactions, photonuclear reactions, Oppenheimer – Phillips process, spallation reactions. Definition of Curie and related calculations, Szilard-Chalmers process. Geiger-Muller counters – G.M. Plateau, dead time, coincidence loss, determination of dead time.	**10 hrs**
2.2	**Photochemistry**: Introduction to photochemistry, quantum yield and its determination, factors affecting quantum yield, Actinometry - Uranyloxalate and potassium ferrioxalate actinometers, acetone and diethylketone actinometers. Term symbols and significance. Photosensitization: by mercury, dissociation of H2. Photochemical kinetics of: Decomposition of CH_3CHO, formation of HCl. Photo degradation: Photo catalyst – ZnO, TiO_2, principle, application of ZnO/TiO_2 in the photo degradation of dyes (IC), pesticides (DDT) and in industrial effluents. Effect of photo degradation on COD value.	**6 hrs**
	Key words: Photo catalyst, Nuclide, Half life, Nuclear reactions.	
Module– 3		**16 hrs**
3.1	**Microwave spectroscopy:** Rotation spectra of diatomic Molecules - rigid and non rigid rotator model. Rotational quantum number and the selection rule. Effect of isotopic substitution on rotation spectra. Relative intensities of the spectral lines. Classification of polyatomic molecules based on moment of inertia - Linear, symmetric top, asymmetric top and spherical molecules. Rotation spectra of polyatomic molecules (OCS CH_3F and BCl_3). Moment of inertia expression for linear tri-atomic molecules. Experimental techniques - Microwave spectrometer. Stark effect in rotation spectra and determination of dipole moments.	**7 hrs**
3.2	**Vibration spectroscopy*:*** Vibration of diatomic molecules, vibrational energy curves for simple harmonic oscillator. Effects of an harmonic oscillation. Vibration - rotation spectra of carbon monoxide. Expressions for fundamental and overtone frequencies. Parallel and perpendicular vibrations (CO_2 and H_2O), fundamental, overtone, combination and difference bands. Fermi resonance. Force constant and its significance.	**5 hrs**
3.3	**ESR spectroscopy:** Theory, presentation of the spectrum, hyperfine coupling, the g value and factors affecting the magnitude of the g value. Zero-field splitting and Kramer's' degeneracy. Application to simple inorganic and organic free radicals and to metal complexes.	**4 hrs**

60% Theory and 40% Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Physical Chemistry	P.W. Atkins	ELBS, Oxford University Press	4th	1990
2	Quantum Chemistry	R.K. Prasad	New Age International Publishers		1996
3	Vibration Spectroscopy Theory and Applications	D.N. Satyanarayana	New Delhi.	----	--
RECOMMENDED BOOKS					
1	Physical Chemistry	P.W. Atkins	ELBS, edition, Oxford	fourth	1990
2	Fundamentals of Molecular Spectroscopy	C.N. Banwell and E.M. Mc Cash	New Delhi.	---	--

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Course Title	**MATERIALS SCIENCE (ID)**							
Course Type	**Soft Core- Theory**		**Total Hours**	48	**Hours/Week**	03	**Credits**	03
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

TEACHING METHOD: Lecture method and working model to make the concepts clear.

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To learn the principles of materials science and engineering though lab investigation
CO-2	To learn the basic skills required to properly use materials science instruments
CO-3	To learn to organize the lab results into a logic, concise and accurate report
CO-4	To describe the different types of bonding in solids, and the physical ramifications of these differences.
CO-5	To review physics and chemistry in the context of materials science & engineering

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Given a type of material, be able to qualitatively describe the bonding	PSO-5	Understand, Apply and Analyse

	scheme and its general physical properties, as well as possible applications.		
CLO-2	Given a type of bond, be able to describe its physical origin, as well as strength.	PSO-5	Apply and understand
CLO-3	Be able to qualitatively derive a material's Young's modulus from a potential energy curve.	PSO-5	Apply, Analyse and Evaluate
CLO-4	Given the structure of a metal, be able to describe resultant elastic properties in terms of its 1D and 2D defect	PSO-5	Understand, Apply and evaluate
CLO-5	Given a simple set of diffraction data, be able to index the peaks and infer the structure.	PSO-5	Understand and Analyse
CLO-6	Be able to do simple diffusion problems.	PSO-5	Analyse and understand

COURSE CONTENTS

MODULE-1 **16**

1.1	Introduction to materials [Ref.1-3]	**04 hrs**
1.1 1	Classification of materials - levels of structure, structure-property relationship in materials. crystalline and amorphous materials	
1.1 2	High Tc superconductors, alloys and composites, semiconductors	
1.1 3	Solar energy materials, luminescent and optoelectronic materials, polymer	
1.14	Liquid crystals and quasi crystals, ceramics-types, properties and applications	
1.2	Structure of solids [Ref.1-3]	
1.2.1	The crystalline and non-crystalline states	
1.2.2	Covalent solids, metals and alloys, ionic solids, the structure of silica and silicates	

MODULE-2

2.1	Preparation techniques of materials [Ref.1-3]	**04 hrs**
2.11	Single crystal growth, zone refining, epitaxial growth	
2.12	Melt-spinning and quenching methods, sol-gel, polymer processing	
2.13	Preparation of ceramic materials, fabrication, control and growth modes of organic and inorganic thin films: different techniques of thin film preparations	
2.2	Synthesis of nanomaterial	
2.21	Top down and bottom up approaches of synthesis of nano-structured materials	
2.22	Nano rods, nanotubes/wires and quantum dots	
2.23	Fullerenes and tubules, single wall and multiwall nanotubes	

MODULE-3

3.1	Phase Transition	**04 hrs**
3.11	Solid phases and phase diagrams [Ref.1-3]	
3.12	Single and multiphase solids, solid solutions and Hume-Rothery rules	
3.13	Intermediate phase, the intermetallic and interstitial compounds	
3.14	Properties of alloys: solid solutions and two component alloy systems	
3.15	Phase diagram, Gibbs phase rule, lever rule- first, second and third order phase	
3.16	Some typical phase diagrams: Pb-Sn and Fe-Fe2O3, eutectic, eutectoid,	
3.17	Order disorder phenomenon in binary alloys, long range order, super lattice,	
3.2	Phase transformation	
3.21	Time scale for phase changes	
3.22	Nucleation and growth, nucleation kinetics, growth and overall transformation	
3.23	Applications - transformation in steel- precipitation processes, solidification	

60% Theory and 40% Problems

REFERENCES:

Sl. No	**Title of the book**	**Name of the author**	**Name of the publisher**	**Edition**	**Year of publication**
1	Elements of Materials Science and Engineering	Van Vlack L. H.,	Addison Wesley	6th Edn	1989.
2	Materials Science and Engineering	Raghavan V	Prentice Hall of India,	5th Edn	2009.
3	Engineering Materials	Budinski K. G. and Budinski M. K.,	Prentice-Hall of India Pvt. Ltd,	8th Edn	2004
RECOMMENDED BOOKS					
1	Materials Science and Engineering	Raghvan V	Prentice Hall of India,	5th Edn	2009.
2	Engineering Materials	Budinski K. G. and Budinski M. K.,	Prentice-Hall of India Pvt. Ltd,	8th Edn	2004

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Course Title	PHARMACOLOGY OF PLANT PRODUCTS							
Course Type	Soft Core- Theory		Total Hours	48	Hours/Week	03	Credits	03
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03 hrs	C3	70 Marks	

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO -1	Learn the constituent present in crude drugs responsible for Pharmacological action.	PSO-2	Understand
CLO-2	To educate and encourage the Herbal Medicines	PSO-2	Understand and Apply
CLO-3	Knowledge of Pharmacological studies	PSO-2	Understand and apply
CLO-4	Understanding of Clinical Trials and their application	PSO-2	Understand and apply
CLO-5	Importance and application of Ayush and Ayurveda	PSO-2	Understand & Apply

COURSE CONTENTS

Module 1 **Introduction to basic concepts of pharmacology** **8hrs**

1.1 Definition

1.2 Sources of drugs & routes of drugs administration

1.3 Therapeutic drug monitoring system-Basic principles of TDM

1.4 Therapeutic index

1.5 Prodrug concepts

Module 2 **Plant products as drugs** **8hrs**

2.1 **Historical aspects of plant usage**

2.1.1 Whole plants

2.1.2 Plant extracts

2.1.3 Tinctures

2.1.4 Teas

2.1.5 Portions

2.1.6 Ayurveda and plants

Module 3 **Secondary Metabolites and their Roles in Plant Physiology and as alternative Medicine** **8hrs**

3.1 Terpenes

3.2 Phenols

3.3 Tannins

3.4 Flavonoids

3.5 Alkaloids

3.6	Nitrogenous Compounds	
Module 4	**Extraction and characterization of bioactive components from**	**5hrs**
4.1	**Biochemical action of some phytochemicals**	
4.1.1	Quercetin	
4.1.2	Ellagic Acid	
4.1.3	Curcumin	
4.2	**Plants for treatment of diseases**	
4.2.1	Diabetes	
4.2.2	CVD	
4.2.3	Alzheimers	
4.2.4	Cancers	
Module-5		**3hrs**
5.	**Drug delivery systems**	
5.1	sustained release	
5.2	enteric coated formulations liposome and nanoparticles	
Module 6	**Pharmacokinetics**	**8hrs**
6.1	Basics of pharmacokinetics	
6.2	calculation of pharmacokinetic estimates	
6.3	Compartment models used in pharmacokinetics (~~oral and~~ intravenous)	
6.4	Compartment fitting (one comp & two comp)	
6.5	Pharmcodynamic /pharmacokinetic (PK/PD) correlation	
Module-7	**Pharmacodynamics**	**8hrs**
7.1	Types of action	
7.2	Mechanism of drug action	
7.3	**Drug receptor interaction with examples**	
7.3.1	Agonist	
7.3.2	Antagonist	
7.3.3	Partial Agonist	
Module 8	**Clinical Trials**	**8hrs**
8.1	Clinical evaluation of new drug	
8.2	Phases of clinical trail	
8.3	Ethics and protocol	
8.4	Preparation of clinical trail	
8.5	New drug development process and drugs registration	
Module 9	**AYUSH and ayurvedic drugs**	
Module 10	**Pre-clinical pharmacokinetic and dynamic studies**	**8hrs**
10.1	Lipinski's rule for drug like molecule	
10.2	High throughput screening (*invitro and invivo*) for pre-clinical pharmacokinetic and pharmacodynamic studies	

15% Problems and 85% Theory

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Textbook of Pharmacology	Barar F S K	S Chand & Company	1st	2012
2.	Text Book of Phytochemistry	Iqbal A	-		1993
3.	Pharmacognosy and Phytochemistry:	Vinod D. Rangari	-		2009
4.	Medicinal Plants	V. K. Gupta , Dr. Anpurna Kaul & Surjeet Singh	-	2nd	2012
5.	Phytochemistry, Pharmacology and Therapeutics		-		
6.	Medical Pharmacology	Padmaja Udaykumar	-		2016

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SECOND YEAR

THIRD SEMESTER

Course Title	**INSTRUMENTAL METHODS OF ANALYSIS**							
Course Type	**Hard Core- Theory**		**Total Hours**	48	Hours/Week	03	Credits	03
Course Code	C0210	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Qualitative and quantitative analysis based on flame excitation methods such as FES and AAS
CO-2	Molecular luminescence techniques for scattering systems
CO-3	Electrochemical methods based on electron sensitive sensors
CO-4	Applications such as amperometry, coulometry, voltammetry, electrogravimetry
CO-5	Thermal method study such as TGA, DTA, DSC and enthalpimetry

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Atomic absorption methods- AAS, FES introduction and application	PSO-5	Remember, Understand, Apply and Analyse
CLO-2	Luminiscence- Fluoroscence and phosphorescence- theory and application	PSO-5	Remember, Apply, and Analyse
CLO-3	Electrochemical methods- theory and development of sensors	PSO-5	Remember, Apply, Analyse and evaluate
CLO-4	Coulometry and voltammetry-diffusion methods-applicability	PSO-5	Remember, Understand, Apply, and Analyse
CLO-5	Thermal methods- TGA, DTA-problematic approach	PSO-5	Remember, Analyse and apply
CLO-6	Thermal methods- DSC and enthalpimetric analysis-problems with solutions	PSO-5	Remember, apply and analyse

COURSE CONTENTS:

MODULE -1 **16 hrs**

1.1 **Flame photometry and Atomic absorption spectrometry:** Energy level diagrams – atomic absorption spectra. Flame characteristics. Flame atomizers and electrothermal atomization. Comparison of spectral interferences, chemical and physical interferences in FP and AAS. Use of organic solvents. Quantitative techniques - calibration curve procedure and the standard addition technique. Typical commercial instruments for FP and AAS, applications. Qualitative analysis and quantitative evaluations. Relative detectabilities of atomic absorption and flame emission spectrometry. **8 hrs**

1.2	**Molecular luminescence spectrometry:** Theoretical basis for fluorescence and phosphorescence. Singlet and triplet excited states. Variables affecting luminescence – quantum efficiency, transition types, structure and structural rigidity, temperature and solvent effects, effect of *p*H, dissolved oxygen and concentration effect. Excitation spectra *vs* emission spectra. Fluorescence instrumentation - fluorometers and spectrofluorometers. Sensitivity and selectivity. Modification necessary to measure phosphorescence. General scope of applications of luminescence.	**6 hrs**
1.3	**Nephelometry and turbidometry:** Principles, instrumentation and applications. ***Key words: Flame and atomic absorption, luminescence, Nephelometry and turbidometry***	**2 hrs**

MODULE-2 **16 hrs**

2.1	**Classification of electrochemical methods:** Controlling and measuring current and potential potentiometers, galvanostats and potentiostats. Potentiometric methods of analysis. Potentiometric electrochemical cells. The Nernst equation. Liquid junction potentials. Reference electrodes - SHE, calomel electrode and silver/silver chloride electrode. Metallic indicator electrodes - electrodes of first kind and second kind. Redox electrodes. Membrane electodes – membrane potential, selectivity of membranes. Glass ion selective electrodes. Crystalline solid state ion selective electrodes. Liquid-based ion selective electrodes. Gas sensing electrodes. Potentiometric biosensors. Quantitative applications. Activity *vs* concentration. Quantitative analysis using external standards and the method of standard additions. Measurement of *p*H. Clinical and environmental applications.	**6 hrs**
2.2	**Electrogravimetric analysis:** Theory, apparatus, cell processes, deposition and separation, electrolytic separation of metals, applications.	**2 hrs**
2.3	**Coulometric methods of analysis:** General discussion, coulometry at controlled potential, apparatus and general technique, applications, coulometric titrations (amperometric coulometric) - principles, apparatus, comparison of coulometric titrations with conventional titrations, automatic coulometric titrations, applications.	**4 hrs**
2.4	**Voltammetry:** Fundamentals of voltammetry. Cyclic voltammetry**:** Principles and applications. Stripping analysis: Stripping voltammetry - basic principles, electrodes used for stripping analysis, apparatus for stripping analysis, applications, determination of lead in water voltammetry with micro electrodes. ***Keywords: Electrodic sensors, Electrogravimetric methods, coulometry, voltammetry, diffusion methods***	**4 hrs**

MODULE-3 **16 hrs**

3.1	**Thermal method of analysis:** Introduction	**1 hr**
3.2	**Thermogravimetric analysis (TGA):** types of thermogravimetric analysis, principles. Factors affecting the results - heating rate, furnace, instrument control/data handling. Applications - purity and thermal stability, evaluation of correct drying temperature, analysis of complex mixture and determination of kinetic parameters of thermal degradation.	**4 hrs**

3.3 **Differential thermal analysis (DTA):** Theory - variables affecting the DTA curves. Differences between TGA and DTA. General principles. Instrumentation. Applications - analysis of the physical mixtures and thermal behaviour study. Determination of melting point, boiling point and decomposition point. **3 hrs**

3.4 **Differential scanning calorimetry (DSC):** Basic principle. Differences between DTA and DSC. Instrumentation - power compensated DSC, Heat flux DSC. Applications - studies of thermal transitions and isothermal crystallization. Pharmaceutical industry for testing the purity of the samples. Thermomechanical analysis. Dynamic mechanical analysis. **5 hrs**

3.5 **Enthalpimetric analysis:** Thermometric titrimetry and direct injection enthalpimetry – principle, instrumentation, applications. **3 hrs**

Key words: Thermal methods-TGA, DTA, DSC, enthalpimetric analysis

60% Theory and 40% Problems

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern Analytical chemistry	David Harvey	Mcgraw Hill	1st	2000
4	Analytical Chemistry 2.0	David Harvey	McGraw-Hill	1st	2018
5	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House, Companies, Depauw	5th	2002.
6	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
7	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
8	Spectroscopy	H. Kaur`	Pragathi prakashan	11th	2016
9	Instrumental Methods of Analysis	Supriya S. Mahajan	Popular Prakashan Private Ltd	1st	2010
10	Analytical Chemistry	S. M. Khopkar	New age international publishers	1st	2013

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Course Title	ORGANIC SPECTROSCOPY							
Course Type	Hard Core- Theory		Total Hours	64	Hours/Week	04	Credits	04
Course Code	C0220	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To identify the functional group by using IR spectra
CO-2	To know the various ionization and mass separating techniques used in mass spectroscopy
CO-3	To develop an understanding of the significance of the number, positions, intensities and splitting of signals in nuclear magnetic resonance spectra and to be able to assign structures to simple molecules on the basis of nuclear magnetic resonance spectra
CO-4	To know the significance of 13C and 2D NMR Spectroscopy
CO-5	Elucidate reaction mechanisms and Intermediates formed

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Arrange components of IR & UV spectroscopy device, Explain working principles and taking spectrum of IR spectroscopy.	PSO-5	Understand and Apply
CLO-2	Identification of organic molecules using IR & UV Spectra.	PSO-5	Apply and Analyse
CLO-3	Apply mass spectroscopy (exact mass, and fragmentation patterns) to organic structural analysis.	PSO-5	Understand, Apply and Analyse
CLO-4	Interpretation of the types of organic spectra commonly used in the research setting: Mass Spectra, Infrared Spectra, 1D and 2D Nuclear Magnetic Resonance Spectra.	PSO-5	Understand, Apply and Analyse
CLO-5	The ability to explain common terms in NMR spectroscopy such as Chemical shift, coupling constant and anisotropy and describe how they are affected by molecular structure.	PSO-5	Remember, Analyse and apply
CLO-6	Learn how to use spectra to elucidate structures of organic compounds and how to solve chemical and structural problems in a systematic manner	PSO-5	Remember, apply and analyse

COURSE CONTENTS:

MODULE -1 **16 hrs**

1.1 UV Visible spectroscopy: Introduction, electronic transitions, simple chromophoric groups - systems of extended conjugation - aromatic systems - types of auxochromes - Functions of auxochromes - absorption and intesity shift - types of trasitions - transition probability - types of absorption bonds - solvent effects and choice of solvent - effect of polarity on various type of bonds Woodwards empirical rules for predicting the wavelength of maximum absorption for conjugated dienes, cyclic trienes and polyenes, α,s-unsaturated aldehydes and ketones, benzene and substituted benzene rings. **10 hrs**

1.2 Optical rotatory dispersion and circular dichroism: Theory of optical rotatory dispersion, octant rule, haloketone rule, Cotton effect and plain curves. Relation between optical rotatory dispersion and structure. **6 hrs**

Key words- UV-Vis spectroscopy, optical rotatory dispersion, circular dichroism

MODULE -2 **16 hrs**

2.1 IR spectroscopy: Introduction, instrumentation, sample handling, modes of vibrations, Hooks law, Characteristic group frequencies and skeletal frequencies. Finger print region, Identification of functional groups - alkenes, aromatics, cabonyl compounds (aldehydes and ketones, esters and lactones), halogen compounds, sulphur and phosphorus compounds, amides, lactums, amino acids and amines. Factors affecting group frequencies and band shapes, conjugation, resonance and inductance, hydrogen bonding and ring strain. Tautomerism, *Cis-trans* isomerism. Applications of IR spectroscopy **10 hrs**

2.2 Raman spectroscopy: Introduction, Raman and Rayleigh scattering, Stokes and anti-Stokes lines, polarization of Raman lines, depolarization factor, polarizability ellipsoid. Theories of Raman spectra - classical and quantum theory. Rotation-Raman and vibration-Raman spectra. Comparison of Raman and IR spectra, rule of mutual exclusion principle. Advantages of Raman spectra. **6 hrs**

Key words- Vibrational (IR and Raman) spectroscopy

MODULE 3 **16 hrs**

3.1 Nuclear Magnetic Resonance Spectroscopy **16 hrs**

General introduction and definition, magnetic properties of nuclei (magnetic moment, g factor) and theory of nuclear resonance. Larmor precession frequency, resonance condition and relaxation processes.

Chemical shift: Standards employed in NMR, factors affecting chemical shift, electronegativity, shielding and deshielding mechanism, van der Waals deshielding, H-bonding, diamagnetic and paramagnetic anisotropics. Spin-spin coupling, chemical shift values and correlation for protons bonded to carbon and other nuclei. Instrumentation and sample handling. Equivalence and magnetic equivalence proton exchange reactions, effects of chiral center, complex spin-spin interaction, stereochemistry, hindered rotation, Karplus curve - variation of coupling constants with dihedral angles.

Structural elucidation of simple organic molecules

Key words- Resonance, Chemical Shift, Anisotropy

MODULE-4 **16 hrs**

4.1 **Advanced NMR:** Simplification of complex spectra: isotopic substitution, increasing magnetic field strength, double resonance, spin decoupling, contact shift reagents, FTNMR: Principle and applications, variable temperature profile, Nuclear Overhauser Effect (NOE). **5 hrs**

4.2 **^{13}C-NMR spectroscopy**: Comparison of 1H-NMR and ^{13}C-NMR. Multiplicity – proton decoupling, noise decoupling, off resonance decoupling, selective proton decoupling, noise decoupling by FT mode, chemical shift, application of ^{13}C-NMR. NMR of ^{31}P, ^{11}B, ^{19}F and ^{15}N. **3 hrs**

4.3 **Two dimensional NMR spectroscopy**: COSY, NOESY and magnetic resonance imaging (MRI) **3 hrs**

4.4 **Integrated structural elucidation methods:** Structural elucidation by integrated spectroscopic methods involving the spectra of UV-Vis, IR, NMR, Mass **5 hrs**

Key words-2D NMR, ^{13}C-NMR, NOE

60% Theory and 40% Problems

References:

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013

2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern Analytical chemistry	David Harvey	Mcgraw Hill	1st	2000
4	Analytical Chemistry 2.0	David Harvey	McGraw-Hill	1st	2018
5	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House, Companies, Depauw	5th	2002.
6	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
7	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
8	Spectroscopy	H. Kaur`	Pragathi prakashan	11th	2016
9	Instrumental Methods of Analysis	Supriya S. Mahajan	Popular Prakashan Private Ltd	1st	2010
10	Analytical Chemistry	S. M. Khopkar	New age international publishers	1st	2013

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Course Title	**CHEMISTRY OF NATURAL PRODUCTS**							
Course Type	Hard Core- Theory		**Total Hours**	48	Hours/Week	03	Credits	03
Course Code	C0210	**Evaluation**	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	A brief introduction to plant systematics
CO-2	Emphasis on classification, nomenclature, structure, biosynthesis, occurrence, analysis and synthesis of Natural products
CO-3	Important classes of compounds (secondary metabolites) in and from nature will and stress will be put on pharmaceutical perspectives
CO-4	Emphasis on general methods of structural elucidation of compounds of natural origin
CO-5	Enhance their understanding of biological and biochemical sciences.

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Understand isolation, purification and characterization of simple chemical constituents from the natural source	PSO-1	Apply and Analyse
CLO-2	Learn the different types of alkaloids & steroids etc and their chemistry and medicinal importance.	PSO-1	Understand and Apply
CLO-3	Understand the importance of natural	PSO-1	Understand and

	compounds as lead molecules for new drug discovery		Apply
CLO-4	Elaborate general methods of structural elucidation of compounds of natural origin	PSO-1	Apply and Analyse
CLO-5	Learn advanced methods of structural elucidation of compounds of natural origin	PSO-1	Apply and Analyse
CLO-6	Explain vitamins Chemistry and Physiological significance of Vitamin.	PSO-1	Understand

COURSE CONTENTS:

MODULE -1 **16 hrs**

1.1 **Steroids:** Introduction, Structure and synthesis of cholesterol. Ergosterol and its irradiation products. Biological importance of bile acids, estrone, progesterone, testosterone, androsterone and corticosterone.

Key words- Cholesterol, Bile Acids, Estrone, Progesterone, Testosterone, Androsterone and Corticosterone

MODULE -2 **16 hrs**

2.1 **Alkaloids:** Introduction, classification, isolation and general methods of structural elucidation. Biological importance of alkaloids. Structure and synthesis of quinine, morphine, reserpine and lysergic acid. Biosynthesis of alkaloids (nicotine and morphine).

Key words- Quinine, Morphine, Reserpine, Nicotine and Lysergic Acid

MODULE -3 **16 hrs**

3.1 **Vitamins**:

Introduction, constitution and synthesis of thiamine, riboflavin, pyridoxine, biotin, ascorbic acid, vitamin A, E and K groups. Biosynthesis of vitamin C.

Key words- Thiamine, Riboflavin, Pyridoxine, Biotin, Ascorbic acid

40% Problems (Structural Elucidation) and 60% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Essentials of physiological chemistry	Anderson	John Wiley & Sons, New		1953
2	Principles of Biochemistry	K. Albert, L. Lehninger, D.L. Nelson, M.M. Cox	CBZ	1st	1993
3	Harper's Biochemistry	R.Harper	Prenctic Hall Press	22nd	1990
4	Organic Chemistry Vol. I & II	I. L. Finar	ELBS Longmann		1984
5	The Alkaloids	K.W. Bently	John Wiley & Sons	-	1966
6	Introduction to Alkaloids	George Albert Swan	Blackwell Scientific	-	1967
RECOMMENDED BOOKS					

1	The Chemistry and Biochemistry of Steroids	Leland J. Chinn	Geron-X	-	1969
2	The Chemistry of Natural Products	R.H. Thomson	Chapman and Hall	2nd	1993

THIRD SEMESTER

Course Title	**FOOD AND PHARMACEUTICAL ANALYSIS**								
Course Type	Soft Core- Theory		**Total Hours**	32	Hours/Week	02	Credits	02	
Course Code	C0310	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100	
			External	Duration	03Hrs	C3	70 Marks		

Course Objectives:

CO No.	Course Objectives
CO-1	Sampling of food
CO-2	Sampling of beverages and preservation
CO-3	Detection and determination of contents of food and beverages
CO-4	Quality control of drugs
CO-5	Analytical assay methodology for common drugs such as analgesics, antibiotics, Antiallergies, Antimalarials, Antituberculosists

MAPPING CLO'S WITH PSO's AND CD's:

CO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Sampling of food	PSO-5	Remember, and Understand
CLO-2	Qualitative and quantitative analysis of food	PSO-5	Remember, Apply, and Analyse
CLO-3	Sampling of drugs	PSO-5	Remember, and Understand
CLO-4	Qualitative and quantitative analysis of pharmaceutical samples	PSO-5	Remember, Apply, and Analyse

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **Food analysis:** Objectives of food analysis. Sampling procedures. Detection and determination of sugars and starch. Methods for protein determination. Oils and fats and their analysis – iodine value, saponification value and acid value. Rancidity - detection and determination (peroxide number). Tests for common edible oils. Analysis of foods for minerals - phosphorus, sodium, potassium and calcium. General methods for the determination of moisture, crude fibre and ash contents of food. Analysis of milk for fat and added water. **08 hrs**

1.2 Non-alcoholic beverages -determination of chicory and caffeine in coffee; - caffeine and tannin in tea. Alcoholic beverages -methanol in alcoholic drinks and chloral hydrate in toddy. Food additives - chemical, preservatives - inorganic preservatives sulphur dioxide and sulphites, their detection and determination. Organic preservatives - benzoic acid and benzoates, their detection and determination. Flavouring agents - detection and determination of vanilla and vanillin. Coloring matters in foods - classification, certified colors, detection of water soluble dyes, color in citrus fruits, beet dye in tomato products, mineral color. Pesticide residues in foods - determination of chlorinated organic pesticides. Control food quality - codex **08 hrs**

alimentarius, Indian standards.

Key workds: oils and fats, flavouring agents

MODULE- 2 **16 hrs**

2.1 **Drugs and pharmaceutical analysis:** Importance of quality control; drugs and pharmaceuticals. Sources of impurities in pharmaceutical chemicals. Analytical quality control in finished/final products. **08 hrs**

2.2 Common methods of assay. Analysis of common drugs; Analgesics - aspirin, paracetamol; Anthelmintics - mebendazole; Antiallergies - chlorpheneramine malleate; Antibiotics - penicillin, chloromecytin; Anti-inflammatory agents - oxyphenbutazone; Antimalarials - primaquine phosphate; Antituberculosists - INH; Narcotics - nicotine, morphine; Expectorants - Benadryl; Sedative - diazepam; Vitamins - A. C, B_1, B_2, B6, niacin and folic acid. Estimation of drug residues in biological samples. **08 hrs**

Key words: drugs, pharmaceuticals, vitamins

100% theory

References:

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
01	Analytical Agricultural Chemistry	S.L. Chopra and J.S. Kanwar	Kalyani Publishers, New	1st	1999
02	Pharmaceutical Analysis	T. Higuchi and E.B. Hanssen	John Wiley and Sons	1st	1997
03	Quantitative Analysis of Drugs in Pharmaceutical Formulations	P. D. Sethi	CBS Publishers and Distributors by New Delhi	3rd	1997
04	A First Course in Food Analysis	A.Y. Sathe	New Age Internationals (P) Ltd, Bangalore	1st	1999

Course Title	**SOIL, FUEL, BIOMEDICAL AND FORENSIC ANALYSIS**							
Course Type	**Soft Core- Theory**		**Total Hours**	32	Hours/Week	02	Credits	02
Course Code	C0320	Evaluation	**Internal**	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

CO No.	Course Objectives
CO-1	Sampling of soil and fuel
CO-2	Analysis of soil and fuel
CO-3	Analysis of bio components in blood
CO-4	Analytical methods for physiological fluids
CO-5	Estimation of poisonous metals in blood

MAPPING CLO'S WITH PSO's AND CD's:

CO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to Course Outcomes	PSOs Addressed	Cognitive Level

CLO-1	Sampling of soil and fuel	PSO-1	Remember, Understand and apply
CLO-2	Qualitative and quantitative analysis of soil	PSO-5	Apply, Analyse and Evaluate
CLO-3	Qualitative and quantitative analysis of fuel	PSO-5	Apply, Analyse and Evaluate
CLO-4	Sampling of biological samples	PSO-5	Remember, Understand and Apply
CLO-5	Qualitative and quantitative analysis components of blood	PSO-1	Apply, Analyse and Evaluate

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **7 hrs**

Soil Analysis: Inorganic and organic components of soil, collection and preparation of soil samples for analysis. Measurement of soil *p*H and conductivity. Determination of organic carbon, total nitrogen, 53 available nitrogen, ammonia nitrogen, nitrate nitrogen and nitrite nitrogen. Available phosphorus and sulphur-their determination. Analysis of soil for sodium, potassium and calcium and magnesium. Micronutrient elements and their analysis. Pesticide residues in soil, their separation and determination.

1.2 **9 hrs**

Fuel analysis- Fuels and their classification. Solid fuels and their classes - natural, artificial and industrial solid fuels. Coal and its analysis - proximate analysis and ultimate analysis. Liquid fuels and their types. Aniline point, flash point and fire point and their determination, octane number of liquid fuels. Gaseous fuels and their classes, advantages. Combustion of a carbonaceous fuel – flue gas. Analysis of flue gas or automobile exhaust for CO_2, CO, O_2 and N_2 by Orsat's apparatus. Calorific value of fuel - net and gross calorific values. Determination of calorific value of solid and liquid fuels by bomb calorimeter method.

Key words: Soil and fuel analysis

MODULE – 2 **16 hrs**

2.1 **06 hrs**

Biomedical and forensic analysis: Composition of body fluids and detection of abnormal levels of certain constituents leading to diagnosis of disease. Sample collection and preservation of physiological fluids.

2.2 **10 hrs**

Analytical methods for the constituents of physiological fluids (blood, serum, urine). Blood - estimation of glucose, cholesterol, urea, haemoglobin and bilirubin. Urine - urea, uric acid, creatinine, calcium phosphate, sodium, potassium and chloride. Biological significance, analysis and assay of enzymes (pepsin, monoaminoxidase, tyrosinase); and hormones (progesterone, oxytocin, insulin). Chemical, instrumental and biological assays to be discussed wherever necessary. General discussion of poisons with special reference to mode of action of cyanide, Carbonmonoxide, organophosphates and snake venom. Estimation of poisonous materials such as lead, mercury and arsenic in biological materials.

Key words: Biomedical and forensic analysis

100% theory

References:

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6th	2008
4	Vogel's Textbook of Quantitative Chemical Analysis	J. Mendham, R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd., New Delhi	6th	2003.
5	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
6	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
7	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
8	Principles and Practice of Analytical Chemistry	F.W. Fifield and Kealey,	Blackwell Sci., Ltd. Malden, USA,	3rd	2000
9	Clinical chemistry and molecular diagnostics	Carl A Burtis and David E Bruns	Elsevier	7th	2015

Course Title	**BIO-MOLECULES**								
Course Type	Soft Core- Theory		**Total Hours**	32	Hours/Week	02	Credits	02	
Course Code		Evaluation	**Internal**	C1+C2 = 15+15			30 Marks		100
			External	Duration	03Hrs	C3	70 Marks		

Course Objectives:

CO No.	Course Objectives
CO-1	To discuss the fundamental and general topics of protein, amino acids and carbohydrates
CO-2	To convey about important biomolecules essential to life.
CO-3	To study the structures of proteins and carbohydrates.
CO-4	Elucidate reaction mechanisms and Intermediates formed
CO-5	To discuss the aspects of the principles of organic chemistry in the structure and function of biomolecules.

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Understand Organic Chemical principles in Life Processes	**PSO-5**	Understand
CLO-2	Skilled at composition/structure and functions of biomolecules.	**PSO-5**	Apply and Analyse
CLO-3	Understand important biological processes like Protien and carbohydrate biosynthesis, DNA Replication and sequencing	**PSO-5**	Understand, Apply and Analyse
CLO-4	Apply fundamental synthetic chemistry to biological molecule preparation	**PSO-5**	Understand and Apply
CLO-5	Follow new developments in the nutrition involving carbohydrates and fats	**PSO-5**	Understand

COURSE CONTENTS

MODULE 1 **16hrs**

1.1 **Amino Acids:** General structure, Physiological properties **2 hrs**

1.2 **Peptides:** Structure and conformation of peptide bond, peptide synthesis: Solution phase and Merrifield's solid phase synthesis, Racemization and use of HOBt, Synthesis of oxytocin and vasopressin, biological importance of insulin, selective cleavage of polypeptide bonds (chemical and enzymatic). **4 hrs**

1.3 **Proteins:** Structure determination: C and N terminal residue determination, primary, secondary, tertiary and quartenary structure determination, denaturing and renaturing of proteins. **3 hrs**

1.4 **Nucleic acids:** Introduction, structure and synthesis of nucleosides and nucleotides, protecting groups for hydroxy group in sugar, amino group in the base and phosphate functions. Methods of formation of internucleotide bonds: DCC, phosphodiester approach and phosphoramide methods. Solid phase synthesis of oligonucleotides. Structure of RNA and DNA, Crick-Watson model, role of nucleic acids in the biosynthesis of proteins. Genetic code, replication of DNA. **7 hrs**

Key words- Protiens,Amino acids, Nucleic acids

MODULE 2 **16 hrs**

2.1 **Carbohydrates:** Introduction, Ring size determination of monosaccharides, configuration and conformations of monosaccharides, anomeric effect, Hudson's rules, epimerization and mutarotation. Synthesis, industrial and biological importance of glycosides, amino sugars, sucrose, maltose and lactose. **7 hrs**

2.2 **Polysaccharides:** General methods of structure elucidation, importance of cellulose, starch, glycogen, dextran, hemicellulose, pectin, agar-agar. Photosynthesis and biosynthesis of carbohydrates **4 hrs**

2.3 **Lipids:** Nomenclature, classification, purification, synthesis of lipids, phospholipids, sphingolipids, biological importance of lipids: Lecithin, sphingolipids, oils and fats **3 hrs**

2.4 **Prostaglandins:** Introduction, classification and biological importance, constitution of PGE1. **2hrs**

Key words- Carbohydrates, Lipids, prostaglandins

100% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Principles of Biochemistry	K. Albert, L. Lehninger, D.L. Nelson, M.M. Cox	CBZ	1st	1993
2.	Harper's Biochemistry	R.Harper	Prenctic Hall Press	22nd	1990
3.	Organic Chemistry Vol. I & II	I. L. Finar	ELBS Longmann		1984
RECOMMENDED BOOKS					
1	Harper's Review of Biochemistry	P.W. Martin, P.A. Mayer and V.W. Rodfwell	Maurzen	15th	1981

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THIRD SEMESTER

OPEN ELECTIVE

Course Title	FUNDAMENTALS OF ISOLATION, SEPARATION, PURIFICATIONAND CHARACTERIZATION TECHNIQUES							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Understand principles of Separation techniches
CO-2	Understand the principle of chromatography
CO-3	Understanding the importance of purification techniques
CO-4	Able to understand the biomolecules and its function in our body
CO-5	Understand the structure and function of important biological molecules such as DNA, RNA and some enzymes.

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Apply the analytical methods and techniques.	PSO-1	Understand, Apply and Analyse
CLO-2	Understand the structure and function of important biological molecules such as DNA, RNA and some enzymes.	PSO-1	Remember, Understand
CLO-3	Chromatographic separation- mathematical equations	PSO-5	Understand, Apply
CLO-4	UV- visible spectroscopy and its applications	PSO-5	Remember, Understand, Apply
CLO-5	NMR spectroscopy and mass spectroscopy.	PSO-1	Understand, Analyse, Apply

MODULE-1		**16hrs**
1.1	**Isolation and separation techniques**: Solvent extraction, continuous extraction, chromatography [principles of TLC, paper chromatography, column chromatography, ion exchange chromatography, gas chromatography (GC) and HPLC] and electrophoresis.	**7 hrs**
1.2	**Purification**: Crystallization, sublimation, fractional crystallization, distillation techniques (simple distillation, steam distillation, distillation under reduced pressure, fractional distillation).	**6 hrs**
1.3	**Chemistry of biomolecules:** A brief survey of alkaloids, terpenes, vitamins, amino acids, peptides, proteins and nucleic acids. ***Keywords: chromoatography, vitamins, alkaloids.***	**3 hrs**
MODULE – 2		**16 hrs**
2.1	**Spectroscopy:** Introduction, electromagnetic radiation	**2 hrs**
2.2	**UV-visible spectroscopy** – modes of electronic excitations, simple chromophoric-auxochrome theory, solvent effect and choice of solvent, applications.	**3 hrs**

2.3	**IR spectroscopy** – principle, characteristic group frequencies and skeletal frequencies, finger print region, identification offunctional groups.	**3 hrs**
2.4	**NMR spectroscopy** – general introduction, magnetic properties of nuclei, Larmor precession frequency, resonance condition and relaxation processes. Chemical shift: spin-spin coupling, chemical shift values and correlation for protons bonded to carbon and heteroatoms.	**5hrs**
2.5	**Mass spectroscopy** – principle, ionization process, molecular ion, base peak, identification of compounds from fragmentation patterns. ***Keywords: Spectroscopy, UV visible, molecular ion, resonance***	**3hrs**

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Elementary Practical Organic Chemistry	Arthur I Vogel	CBS Publishers and Distributors		
2	ELBS Longmann	I.L. Finar, Organic Chemistry		Vol. I and II,	1984.
3	Advanced General Organic Chemistry	S K. Ghosh	Book and Allied (P)		1998
4	. Organic Spectroscopy	William Kemp	English Language Book society, Macmillan		1987.
5	Application of Absorption Spectroscopy of	John R. Dyer	Prentice Hall of India Private Ltd., New Delhi		
6	Spectrometric Identification of Organic Compounds	Robert M. Silverstein, G. G. Clayton Bassler and Terence C. Morrill	John Wiley & Sons, New York	4th	1981.

OPEN ELECTIVE

Course Title	BASIC CONCEPTS OF CHEMISTRY OPEN ELECTIVE							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Understand the Instrumental methods of analysis
CO-2	Importance of analytical chemistry in fundamental research
CO-3	Understand the periodicity of the elements.
CO-4	Comparitive study with reference to electronic configuration.
CO-5	Understand General characteristics of organic compounds, importance of organic chemistry,

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Predict the stable geometry of inorganic molecules	PSO-1	Understand
CLO-2	Precise knowledge about chemical bonding.	PSO-1	Remember, Understand
CLO-3	To convey the basic concepts of organic reactions	PSO-1	Understand, Apply
CLO-4	Nomenclature, classification, isomerism,	PSO-1	Remember, Understand
CLO-5	Symmetry elements and symmetry operations with examples of simple molecules	PSO-5	Understand, Analyse, Apply

MODULE-1	16 hrs

1.1	**Analytical objectives:** Meaning of analytical chemistry. Origin of analytical chemistry. Qualitative and quantitative analysis. The analytical process – steps in an analysis: obtaining a representative sample, preparing the sample for analysis, performing necessary chemical separations, performing the measurement. Methods of analysis – classical methods, principles of titrimetric and gravimetric analysis, typical examples of applications. Instrumental methods of analysis, their classification, their advantages and disadvantages over classical methods. Comparison of different analytical methods. Importance of analytical chemistry in fundamental research, product development, product quality control, monitoring and control of pollutants, assay and medical and clinical studies.	**9 hrs**
1.2	**Periodic properties:** Atomic size, ionic radii, ionization potential, electron affinity and electronegativity. Applications of electronegativity.	**2 hrs**
1.3	**General characterisctics of s-, p-, d- and f-block elements:** Comparitive study with reference to electronic configuration, oxidation states, chemical properties, spectra and magnetic properties of d- and f-block elements.	**3 hrs**
1.4	**Chemical bonding:** Ionic bond – size effect and solubility. Covalent bond – simple binary systems. Hydrogen bond – water and in biological systems.	**2 hrs**
MODULE - 2		**16 hrs**
2.1	**Organic Chemistry:** General characteristics of organic compounds, importance of organic chemistry, tetravalency of carbon, hybridization, bond length, bond angle, bond energy, representation of organic compounds, nomenclature, classification, isomerism, inductive effect, resonance effect, hyperconjugation, organic reactions and their classification.	**5 hrs**
2.2	**Applications of Physical Chemistry:** Concepts of entropy and free energy. Partial molar volume and its determination by density measurements. Symmetry elements and symmetry operations with examples of simple molecules. X-ray diffraction. Bragg equation and Miller indices. Order of a reaction and its determination. Energy of activation and its determination. Assumptions of activated complex theory. Fast reactions with examples. Polymers and their classification. Arrhenius theory of strong and weak electrolytes. Assumptions of Debye-Huckel theory of strong electrolytes. Types of electrodes. Corrosion and its prevention. Laws of photochemistry. Quantum yield and its determination. Photodegradation.	**11 hrs**

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and	Saunders College Publishing	8th	2005
2	Analytical Chemistry	G.D. Christian	John Wiley & Sons, Inc	5th	2001
3	. Basic Inorganic Chemistry	F.A. Cotton, G. Wilkinson and P.L . Gaus	JohnWiley and Sons	3rd	2002
4	Inorganic Chemistry Principles of Structure and	James E. Huheey, Ellen A. Keiter, Richard L. Keiter, Okhil K. Medhi	Delhi University, New Delhi		2006
5	Elements of Physical Chemistry	Lewis and Glasstone.			2001
6	Physical Chemistry	P.W. Atkins	Oxford University Press	4th	1990.
7	. Physical Chemistry	G.M. Barrow	McGraw Hill International Service		1988

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Course Title	**ANALYTICAL CHEMISTRY PRACTICALS**								
Course Type	Soft Core- Practical		Total Hours	96	Hours/Week	04	Credits		**02**
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks		100
			External	Duration	03Hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Quantitative analysis of water samples by EDTA titrations
CO-2	Quantitative analysis based on Flame emission methods
CO-3	Potentiometric and conductometric determination
CO-4	Analysis of waste water for DO and COD
CO-5	Spectrophotometric methods of analysis

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Quantitative analysis using precipitation titration	PSO-3	Apply Analyse and Evaluate
CLO-2	Quantitative analysis using spectrophotometric method	PSO-3	Apply, Analyse and Evaluate

CLO-3	Quantitative analysis by complexometric titration	PSO-3	Apply, Analyse and Evaluate
CLO-4	Quantitative analysis by acid base titration	PSO-3	Apply Analyse and Evaluate
CLO-5	Quantitative analysis by flame photometric method	PSO-3	Apply Analyse and Evaluate
CLO-6	Quantitative analysis using potentiometry and conductometry	PSO-3	Apply Analyse and Evaluate

COURSE CONTENTS:

Part III

1. Determination of calcium in limestone by redox, acid-base and complexation titrations.
2. Determination of mercury in an algaecide by EDTA titration; and arsenic in ant control preparation by redox titration.
3. Determination of aluminium and magnesium in antacids by EDTA titration.
4. Determination of zinc in a sample of foot powder and thallium in a sample of rodenticide by EDTA titration.
5. Determination of saccharin in tablets by precipitation titration.
6. Determination of iodine value and saponification value of edible oils.
7. Analysis of a mixture of iron(II) and iron(III) by EDTA titration using *p*H control.
8. Potentiometric titration of a mixture of chloride and iodide.
9. Determination of sulpha drugs by potentiometry using $NaNO_2$ and iodometric assay of penicillin.
10. Polarographic determination of copper and zinc in brass.
11. Determination of iron in mustard seeds and phosphorus in peas by spectrophotometry.
12. Determination of manganese in steel by extraction-free spectrophotometry and molybdenum in steel by extractive spectrophotometry

Part IV

1. Analysis of waste waters for DO and COD by titrimetry.
2. Analysis of a ground water sample for sulphate by titrimetry (EDTA) and turbidimetry.
3. Determination of chromium(III) and iron(III) in a mixture by kinetic masking methods.
4. Photometric and potentiometric titration of copper with EDTA.
5. Analysis of brackish water for chloride content by a**)** spectrophotometry (mercuric thiocyanate method), b) conductometry (silver nitrate) and c**)** potentiometry (silver nitrate).
6. Conductometric titration of sodium acetate with HCl and NH_4Cl with NaOH.
6. Spectrophotometric determination of iron in natural waters using thiocyanate and 1,10-phenanthroline as reagents.
7. Determination of fluoride in drinking water/ground water by spectrophotometry (alizarin red lake method).
8. Analysis of waste water for a) phosphate by molybdenum blue method b) ammoni-nitrogen by Nessler's method c) nitrite-nitrogen by NEDA method
9. Analysis of a soil sample for a) Calcium carbonate and organic carbon by titrimetry b) calcium and magnesium by EDTA titration
10. Analysis of a soil sample for a) Available phosphorus by spectrophotometry b) Nitrate-nitrogen/nitrite nitrogen/ammonia nitrogen by spectrophotometry c) sodium and potassium by flame photometry.
11. Analysis of urine for a) urea and uric acid by titrimetry snd spectrophotometry b) Sulphate by precipitation titration after ion-exchange separation c) sugar by Benedict' s reagent

100% practical

References

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern Analytical chemistry	David Harvey	Mcgraw Hill	1st	2000
4	Analytical Chemistry 2.0	David Harvey	McGraw-Hill Companies, Depauw	1st	2018
5	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6th	2008
6	Vogel's Textbook of Quantitative Chemical Analysis	J. Mendham, R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd., New Delhi	6th	2003.
7	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	malaya publishing House,	5th	2002.
8	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
9	Instant Notes of Analytical Chemistry by	Kealey and Haines,	va Books Pvt. Ltd.,		2002.
10	Principles and Practice of Analytical Chemistry	by F. W. Fifield and Kealey,	Blackwell Sci., Ltd. Malden, USA.	5th	2000
11	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
12	Biochemical analysis	Irwin H Segel	Wiley	2nd	2004
13	Spectroscopy	H. Kaur`	Pragathi prakashan	11th	2016
14	Instrumental Methods of Analysis	Supriya S. Mahajan	Popular Prakashan Private Ltd	1st	2010
15	Analytical Chemistry	S. M. Khopkar	New age international publishers	1st	2013
16	Elements of Spectroscopy	Guptha, Kumar, Sharma	Pragati Edition	28th	2016
17	Calculations in Chemistry	John olusina obimakinde, Samuel Oluwaseun	IK international publishing house	1st	2014

		obimakinde			

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THIRD SEMESTER
PRACTICALS

Course	INORGANIC CHEMISTRY PRACTICALS							
Course Type	Soft Core-Practicals		Total Hours	96	Hours/Week	04	Credits	02
Course Code	DP320	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	To introduce multistep inorganic compound synthesis and its spectroscopic analysis, separation and estimation of organic compounds.
CO-2	Compostion of metal and ligand by spectrophotometric method.
CO-3	To acquire knowledge of laboratory techniques , inorganic synthesis and characterization.
CO-4	To impart the knowledge about redox, complexometric and precipitation reactions.
CO-5	Determination of stability constants of inorganic compounds.

Course Learning Outcome:

CLO No.	Course learning outcome	PSOs Addressed	Cognitive Level
CLO-1	To acquire knowledge of laboratory techniques , inorganic synthesis and characterization.	PSO-3	Analyze,Appl Evaluate
CLO-2	Estimate the ions present in the sample by different techniques.	PSO-3	Analyze,Appl Evaluate
CLO-3	Flame photometric determination.	PSO-3	Analyze,Appl Evaluate
CLO-4	Determination of stability constants of inorganic compounds.	PSO-3	Analyze,Appl Evaluate
CLO-5	To impart the knowledge about redox, complexometric and precipitation reactions.	PSO-3	Analyze,Appl Evaluate

PART – III

1. Determination of bismuth, cadmium and lead in a mixture: Analysis of a low melting alloy (Wood's alloy).
2. Simultaneous spectrophotometric determination of chromium and manganese in a steel solution.
3. Quantitative analysis of copper(II) and iron(II) in a mixture:
   i. Copper gravimetrically as CuSCN and
   ii. Iron volumetrically using cerium(IV) solution
4. Determination of chromium(III) and iron(III) in a mixture: Kinetic masking method.
5. Electrogravimetric determination of:
   a) Copper in copper sulphate
   b) Nickel in nickel suphate
   c) Copper and nickel in alloy solution
   d) Lead in lead nitrate.
6. Flame photometric determination of the following metal ions from different samples:
   a) sodium b) potassium c) calcium d) lithium and d) sodium and potassium in a mixture.
7. Polarographic estimation of cadmium and zinc.
8. Determination of iron as the 8-hydroxyquinolate by solvent extraction method.
9. Quantitative determination of nickel using dithizone and 1,10-phenanthroline by synergistic extraction.
10. Spectrophotometric determination of the *p*Ka value of methyl red.

PART – IV

1. Preparation and characterization of:
   a) Chloropentammine cobalt(III) chloride
   b) Estimation of chloride in a complex by potentiometric or ion-exchange method
   c) Record the electronic absorption spectrum of a complex and verify Tanabe Sugano diagram
2. Preparation of *cis*- and *trans*- dichlorobis(ethylenediammine) cobalt(III)chloride. Record the UV-Vis spectra and compare it with *cis*-form. Measure the molar conductance.
3. Preparation of hexamine cobalt(III) chloride and estimate cobalt ion.
4. Determination of magnetic susceptibility of any two compounds/complexes by Gouy method.
5. Determination of the composition of iron-phenanthroline complex by:
   (a) Job's method
   (b) mole-ratio method and
   (c) slope-ratio method.
6. Determine the stability constant of iron-tiron/iron-phenanthroline by Turner-Anderson method.
7. Preparation of tris(oxalate)ferrate(III) and estimate the metal ion.
8. Using chloropentamine cobalt(III) chloride, prepare nitro and nitritopentamine cobalt(III) chloride. Record the IR spectra of the isomers and interpret.
9. Estimate the chloride ion in a given complex by silver nitrate titration after ion-exchangeseparation.
10. Demonstration Experiments:
    (a) Recording and interpretation of IR and NMR spectra of complexes.
    (b) Spectrochemical series - Evaluation of Dq value.
    (c) DNA interaction with metal complexes by UV-visible absorption and viscosity methods.

References

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6th	2008
2	Vogel's Textbook of Quantitative Chemical Analysis	J. Mendham, R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd., New Delhi	6th	2003.
3	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
4	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
5	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
6	Quantitative Chemical Analysis	Daniel C. Harris,	-	7th	2006
7	PracticalInorganic chemistry	Gurdeep Raj	GOEL Publishing house	26th	2015

Course Title	**ORGANIC CHEMISTRY PRACTICALS**								
Course Type	**Soft Core-Practicals**		**Total Hours**	**96**	**Hours/Week**		**04**	**Credits**	**02**
Course Code	DP330	Evaluation	Internal	C1+C2 = 15+15			30 Marks		100
			External	Duration	03Hr	C3	70 Marks		

CO No.	Course Objectives
CO-1	To understand the concept of extraction, isolation and Estimation techniques
CO-2	To understand th e concepts of reaction mechanism and its applications
CO-3	To Identify suitable reagent for various chemical conversion
CO-4	Develop basic skills for the multi-step synthesis of organic compounds
CO-5	To Practice various Chromatographic Techniques

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level

CLO-1	The ability to apply reagent knowledge Practically	**PSO-3**	Understand
CLO-2	Understanding the solution of problems related isolation and Purification	**PSO-3**	Apply and Analyse
CLO-3	Practical skills in handling Chromatographic Techniques	**PSO-3**	Apply
CLO-4	Evaluate the risks associated with an experiment	**PSO-3**	Analyse and

COURSE CONTENTS

PART – III		**48hrs**
1. 1	Oxidation of cyclohexanol to adipic acid via cyclohexanone	
1.2	Esterification: Preparation of benzoccaine from *p*-nitrotoluene	
1.3	Diazotization (Sandmeyer's reaction): Preparation of *p*-chlorobenzoic acid from *p*toluidine	
1.4	Molecular rearrangement: i. Preparation of o-chlorobenzoic acid from phthalic anhydride ii. Preparation benzilic acid from benzaldehyde iii. Preparation of *o*-hydroxy benzophenone from phenyl benzoate via Fries rearrangement iv. Prepartion of benzanilide from benzophenone oxime via Beckmann rearrangement.	
1.5	Grignard reaction: Prepearation of triphenyl carbinol	
1.6	Preparation of luminol from phthalic anhydride	
1.7	Synthesis of isoxazolines and pyrazolines via 1,3-dipolar cycloaddition.	
1.8	Solvothermal synthesis	
1.9	Synthesis of tetralones starting from aryl aldehydes.	
1.20	Synthesis of m-chloroiodobenzene from m-dinitrobenzene ***Key words- Multistep Synthesis and Analysis***	

60% Problems (in the form of Predicting products) and 40% Theory

REFERENCES

Sl. No	**Title of the book**	**Name of the author**	**Name of the publisher**	**Edition**	**Year of publication**
1.	Comprehensive practical organic chemistry	V. K. Ahluwalia and Renu Agarwal	Universities Press	-	2001
2.	An Introduction to Practical Organic Chemistry	Caroline Frances Cornwallis	BiblioBazaar	-	2009
3.	Advanced Practical Organic Chemistry	By John Leonard, Barry Lygo, Garry Procter	Stanley Thrones Publishing	2nd	1994
4.	Manual of Organic Chemistry	Dey and Seetharaman	Allied Publishers,	-	1957
5.	Laboratory Manual of Organic Chemistry	R.K. Bansal	New Age International (P) Ltd. London	3rd	1996.

6.	A Text Book of Practical Organic Chemistry by Vol.III	A.I. Vogel,	Longman Scientific & Technical	-	1989
RECOMMENDED BOOKS					
1.	Practical Organic Chemistry	Mann & Saunders	Orient Longman	4th	2004
2.	Semi micro Qualitative Organic Analysis	Cheronis, Entrikin and Hodnet.	R.E. Krieger Publishing Company,	3rd	1983

Part IV	**Isolation of natural products**	**48 hrs**
2.1	Fractional crystallization: separation of mixture of naphthalene and biphenyl	
2.2	Fractional distillation: Separation of mixture of hexane and toluene.	
2.3	Thin layer chromatography: Separation of plant pigments	
2.4	Column chromatography: Separation of mixture of *o* and *p*-nitro anilines	
2.5	Isolation of piperine from pepper	
2.6	Isolation of caffeine from tea	
2.7	Isolation of azeleic acid from castor oil	
2.8	Estimation of sugars by Fehlings method	
2.9	Determination of enol content by Meyer's method	
2.10	Estimation of ketones by haloform reaction	
2.11	Estimation of sugars by Bertrand's method	
2.12	Estimation of nitro groups ***Key words- Chromatographic Separation, Analysis and Estimation***	

PRACTICALS

Course Title	**PHYSICAL CHEMISTRY PRACTICALS**							
Course Type	**Hard Core-Theory**		**Total Hours**	**48**	**Hours/Week**	**04**	**Credits**	**02**
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To demonstrate an ability to conduct experiments in the above sub-disciplines wi mastery of appropriate techniques and proficiency using core chemical instrumentation and modeling methods
CO-2	To develop skills in quantitative modeling of static and dynamic chemical system
CO-3	To develop laboratory competence in relating chemical structure to spectroscopic phenomena

CO-4	To demonstrate the ability to perform accurate quantitative measurements with an understanding of the theory and use of contemporary chemical instrumentation, interpret experimental results, perform calculations on these results and draw reasonable, accurate conclusions
CO-5	To demonstrate broad knowledge of descriptive Chemistry

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Think critically and analyze chemical problems.	PSO-3	Understand, Apply and Analyse
CLO-2	Present scientific and technical information resulting from laboratory experimentation in both written and oral formats	PSO-3	Apply and understand
CLO-3	Work effectively and safely in a laboratory environment.	PSO-3	Apply, Analyse and Evaluate
CLO-4	Use technologies/instrumentation to gather and analyze data.	PSO-3	Understand, Apply and evaluate
CLO-5	Work in teams as well as independently	PSO-3	Understand and Analyse
CLO-6	Apply modern methods of analysis to chemical systems in a laboratory setting	PSO-3	Analyse and understand

COURSE CONTENTS

1. Kinetics of reaction between sodium formate and iodine, determination of energy of activation.
2. To study the kinetics of saponification of ethyl acetate by conductivity method, determination of the energy of activation.
3. To study the kinetics of reaction between acetone and iodine - determination of order of reaction with respect to iodine and acetone.
4. Determination of mean ionic activity coefficient of a weak electrolyte (acetic acid) by conduct metric measurements.
5. To study the acid catalyzed kinetics of oxidation of glycine by chloramine-T (CAT) - determination of order of reaction with respect to [CAT] and [glycine].
6. To determine the eutectic point of a two component system (Naphthalene-*m*-dinitrobenzene)
7. Conduct metric method of determination of solubility of sparingly soluble salt.
8. Potentiometric titration of mixture of KCl + KBr + KI against$AgNO_3$.
9. Study of phase diagram of a three component system (acetic acid-chloroform and water)

1. Determination of energy of activation and thermodynamic parameters (ΔH, ΔS, ΔG) for reaction between sodium format and iodine.
2. To study the kinetics of reaction between acetone and iodine-determination of order of reaction with respect to iodine and H_2SO_4.
3. Determination of mean ionic activity coefficient of a weak electrolyte (formic acid) by conduct metric measurements.
4. To study the acid catalyzed kinetics of oxidation of glycine by chloramine-T (CAT)-determination of order of reaction with respect to [CAT] and $[H^+]$.
5. Kinetics of decomposition of benzene diazonium chloride, determination of energy of activation and thermodynamic parameters.
6. To determine the eutectic point of a two component system (Naphthalene-biphenyl).
7. Conduct metric method of determination of solubility of sparingly soluble salt.
8. Study of phase diagram of a three component system (benzene-alcohol-water).
9. Kinetics of saponification of ethyl acetate by conductivity method and study the effect of dielectric constant of the medium (using CH_3OH).
10. Determination of pKa value of an indicator (methyl orange).

60% Theory and 40% Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Practical Physical Chemistry	-	-	-	-
2	Experimental Physical Chemistry	-	-	-	-
3	Experiments in Physical Chemistry	Yadav	Geol Publishing House	-	-
RECOMMENDED BOOKS					
1	Experiments in Chemistry	D.V. Jahagirdar	Himalaya Publishing House	-	-
2	Experimental Physical Chemistry	R.C. Das and B. Behera	Tata Mc Graw Hill	-	-

THIRD SEMESTER

OPEN ELECTIVE

Course Title	**SEPARATION AND PURIFICATIONS TECHNIQUES**							
Course Type	**Soft Core- Theory**		Total Hours	32	Hours/Week	02	Credits	02
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

CO No.	Course Objectives
CO-1	Understand the theory behind the chromatographic separations
CO-2	Separation based on the nature of the stationary and mobile phase
CO-3	Electrophoretic methods as tool is separation of charged substances
CO-4	Principles and objectives of analytical sedimentation and membrane separation

MAPPING CLO'S WITH PSO's AND CD's:

CLO No.	Course Outcomes	PSOs Addressed	Cognitive Level
CLO-1	Chromatographic separation- mathematical equations	PSO-5	Remember, Understand, Apply , Analyse and evaluate
CLO-2	Separation of synthetic and natural samples by chromatographic methods	PSO-5	Apply, Analyse and Evaluate
CLO-3	Separation of samples-based on charge, molecular mass, lock-key mechanism	PSO-5	Apply, Analyse and Evaluate
CLO-4	Centrifugation, analytical sedimentation, Membrane separation	PSO-5	Remember, Understand, Apply and Analyse
CLO-5	Electrophoretic methods-separation based on electrophoresis and electro-osmotic flow	PSO-5	Understand, Apply, Analyse and evaluate
CLO-6	Problems related to the chromatographic separation	PSO-5	Evaluate

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1	**Principles of chromatography:** Distribution coefficients, modes of chromatography, Selection of stationary and mobile phases, analyte development and elution. Chromatographic parameters - Retention time and volume, capacity factor, plate	**2 hrs**
1.2	**Gas-liquid chromatography:** Principle. Apparatus-columns, sample application, mobile phase, stationary phases, detectors. Applications.	**2 hrs**
1.3	**HPLC:** Principles,Instrumentation - columns, stationary phase and matrices, sample application, mobile phases, pumps, detectors. Fast protein liquid chromatography [FPLC].	**2 hrs**
1.4	**Ion-exchange chromatography:** Principle, materials and applications.	**1 hr**
1.5	**Molecular exclusion chromatography:** Principle, materials. Applications - purification, relative molecular mass determination, solution concentration, desalting.	**2 hrs**
1.6	**Affinity chromatography:** Principle, materials-matrix, liquid, practical procedure, applications, Lectin affinity chromatography, metal-chelate chromatography, covalent chromatography.	**3 hrs**
1.7	**Thin layer chromatography:** Principle, apparatus - preparation of plates, sample application, plate development, detection of analytes. Applications. Selection of chromatographic systems.	**4 hrs**

Module-2 **16 hrs**

2.1 **Electrophoretic techniques:** Electrophoresis & Capillary Electrophoresis**:** Theory - **e**lectrophoretic mobility, electroosmotic mobility, electroosmotic flow velocity, total mobility, migration time, efficiency, selectivity and resolution. Instrumentation - capillary tubes, hydrodynamic and electrokinetic methods of sample injection, applying electric field and detectors. Capillary electrophoresis methods - capillary zone electrophoresis, micellar electrokinetic capillary chromatography, capillary gel electrophoresis and capillary electrochromatography. Concepts of distillation, crystallization, evaporation, dialysis, electrodialysis, osmosis and reverse osmosis. **11hrs**

2.2 **Centrifugation:** Centrifugal force, centrifugal sedimentation, centrifugal decantation, Centrifuges, selection of centrifuge tubes. Preparative, density gradient and isopycnic centrifugation. Applications. **2hrs**

2.3 **Analytical sedimentation:** Sedimentation coefficient, sedimentation velocity, applications in biological separations. **2hrs**

2.4 **Membrane separation:** Principles and applications. **1hr**

Keywords: Electrophoretic methods, centrifugaton, sedimentation, membrane separation

60 % Theory and 40% problems

References:

Sl.No	Title of the book	Name of the author	Name of publisher the	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Modern chemistry Analytical	David Harvey	Mcgraw Hill	1st	2000
4	Analytical 2.0 Chemistry	David Harvey	McGraw-Hill Companies, Depauw	1st	2018
6	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
7	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th edition,	2005

8	Instant Notes of Analytical Chemistry	Kealey and Haines,	Viva Books Pvt. Ltd.,		2002.
9	Principles and Practice of Analytical Chemistry	by F. W. Fifield and Kealey,	Blackwell Sci., Ltd. Malden, USA,	5th edition,	2000
10	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th edition	2007

OPEN ELECTIVE

Course Title	**BIOLOGICAL INORGANIC CHEMISTRY**								
Course Type	**Soft Core- Theory**		**Total Hours**	**32**	**Hours/Week**		**02**	**Credits**	**02**
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks		100
			External	Duration	03Hrs	C3	70 Marks		

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	Inorganic composition of the living cell
CO-2	Knowledge about the oxygen carriers, ion transportation across the membrane, iron sulfur proteins
CO-3	Discuss how different elements are taken up selectively by different cells and intracellular compartments and the various ways they are exploited.
CO-4	Discuss the structure and functions of complexes and materials that are formed in the biological environment.
CO-5	Knowledge about the essential, trace and toxic elements known to us and role played by them and elaborates about the toxicity and deficiency of different elements known to us

MAPPING CLO'S WITH PSO's AND CD's:

CLO No.	Course Outcomes	PSOs Addressed	Cognitive Level
CLO-1	Learn about the various diseases due to deficiency and excess of different elements in our body	**PSO-1**	Understand, remember
CLO-2	Know the mechanism of the toxicity of various metals in our body	**PSO-1**	Understand, remember
CLO-3	Know about the various chelating agents and their action over the metal poisoning	**PSO-1**	Understand, remember
CLO-4	Know the mechanism of cisplatin in the fight against cancer	**PSO-1**	Understand, remember
CLO-5	Importance of biological oxygen carriers in living organisms	**PSO-1**	Understand, remember
CLO-6	The use of metal complexes as drugs and	**PSO-1**	Understand,

	chemotherapeutic agents		remember

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **An overview of metals in Biology**: Introduction, the element content of living systems, biological chemistry of hydrogen, the economical use of resources-abundance and availability. Biological need for and the behaviors of inorganic elements. **4 hrs**

1.2 **Basic coordination chemistry for biologists:** Introduction, ionic bonding, covalent bonding, Hard and soft ligands, the chelate effect, coordination geometry, crystal field and ligand field theory. **5 hrs**

1.3 **Metal assimilation pathways:** Introduction, metal assimilation in bacteria, plants, fungi and in mammals (iron, copper and zinc). **3 hrs**

1.4 **Storage and homeostasis of metal ions:** Introduction, metal storage and homeostasis in bacteria, plants, fungi and in mammals (iron, copper and zinc). **4 hrs**

Key words: Metals in biology, co-ordination chemistry, assimilation process

MODULE-2 **16 hrs**

2.1 **Sodium and potassium-channels and pumps:** Introduction, transport across membranes, potassium channels, sodium channels, the sodium-potassium ATPase, active transport driven by Na+ gradients, sodium/proton exchangers, other roles of intracellular K+**.** **3 hrs**

2.2 **The biological chemistry of magnesium and calcium:** Introduction, magnesium chemistry, magnesium-dependent enzymes, phosphoryl group transfer - kinases and phosphatases. Comparison of calcium and magnesium. Protein ligands for calcium.Calcium triggering calmodulins. **4 hrs**

2.3 **Iron - Essential for almost all life:** Introduction, iron and oxygen, the biological importance of iron. Biological functions of iron-containing proteins. Haemoproteins, iron-sulfur proteins, iron regulation - relationship to genes and evolution. **3 hrs**

2.4 **Metals in brain and their role in various neuro-degenerative diseases:** **4 hrs**
roduction, metals in brain, calcium, zinc, copper. Disorders of copper metabolism-Wilson's and Menkes diseases. Iron, redox metal ions, oxidative stress and neuro-degenerative diseases (PD and AD).

2.5 **Metals in medicine:** Introduction, *cis*-platin, contrast agents for MRI, radioactive pharmaceuticals, lithium compounds in therapy. **2 hrs**

Keyword: Sodium-potassium channels, Biology of Mg and Ca, Iron in human life, metal-degenrative diseases, Metals in medicine

100 % Theory

References:

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
01	Biological Chemistry of Elements. The Inorganic Chemistry of Life	J.J.R. Frausto da silva and R.J.P.	Oxford University Press,	1st	2001

		Williams,	New York		
02	Bioinorganic Chemistry: A Survey	EiichiroOchiai,	Elsevier, USA	2nd	2008
RECOMMANDED BOOKS					
02	Biological Inorganic Chemistry: An Introduction	Robert R. Crichton,	Elsevier, USA	1st	2008

OPEN ELECTIVE

Course Title	**APPLICATIONS OF SYNTHETIC PRODUCTS**							
Course Type	Soft Core- Theory		Total Hours	64	Hours/Week	04	Credits	04
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To impart knowledge in advanced concepts and applications in various fields of Chemistry
CO-2	To provide wide choice of elective subjects with updated and new areas in various branches of Chemistry to meet the needs of all students
CO-3	To convey the concepts of dyes and drugs
CO-4	Understand the concepts and principals of polymers, soaps, cosmetics and insecticides.

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	The adoption of the multistep synthetic approaches in the manufacture of dyes, drugs and polymers	PSO-1	Apply Analyse And Evaluate
CLO-2	Describe methods of drug development including design and discovery.	PSO-1	Apply and Analyse
CLO-3	Explain the relationship between drug's chemical structure and its therapeutic Properties.	PSO-1	Understand, Apply and Analyse
CLO-4	Predict a drugs properties based on its structure	PSO-1	Understand and Apply

COURSE CONTENTS

MODULE 1		**32 hrs**
1.1	**Dyes**: Color and constitution, classification, dyeing method and their industrial importance	**4 hrs**

1.2	**Drugs**: Basic concepts, classification, sources, the requirement of an ideal drug.	**2 hrs**
1.3	**Synthetic drugs**: Structure and medicinal properties: Sulphanilamide – an example of sulpha drug-paracetamol, aspirin, oil of wintergreen; Mephensin – a muscle relaxant; Ibuprofen – an anti-inflammatory drug; L-dopa – cures Parkinson's disease; Chloroquine – an antimalarial drug; Chlorpromazine – an antipsychotic agent; Phenobarbital – a barbiturate; Omeprazole – an drug; Ciprofloxacin – an antibacterial drug; Formulation of drugs – introduction and classification.	**16 hrs**
1.4	**Polymers**: Introduction, biodegradable and non-biodegradable polymers and their industrial importance, plastics (uses and effects on enviornment), natural and synthetic rubbers, polyamides and poly esters like nylon, decron, terelyne. Thermoplastics - poly carbonates, poly acrylates in lens applications, polyurethanes and conducting polymers.	**10 hrs**
MODULE 2		**32 hrs**
2.1	**Soaps and detergents**: Production and their cleansing action.	**4hrs**
2.2	**Liquid crystals** and their applications	**2 hr**
2.3	**Surfactants**	**2hr**
2.4	**Cosmetics**: Detailed study of formulations and manufacturing of cream and lotions, lipstick and nail polish, shampoos, hair dyes and tooth pastes.	**4hrs**
2.5	**Flavours:** Natural flavouring materials and classification	**4hrs**
2.6	**Sweeteners**: Natural and synthetic sweeteners.	**4hrs**
2.7	**Insecticides**: Introduction, classification, applications and their effect on environment.	**4hrs**
2.8	**Pheromones**: Introduction, Sources, biological importance.	**4hrs**
2.9	**Explosives**: Introduction, RDX, Gun powder.	**4hrs**

100% Theory

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
13.	Organic Chemistry ELBS, Vol. I & II,.	I.L. Finar	Longmann	-	1984
14.	Principles of Biochemistry	K. Albert, L. Lehninger, D. L. Nelson, M. M. Cox	CBZ publishers	1st	1993
15.	Harper's Biochemistry	R. Harper	Prentice Hall Press, New York,	22nd	1990.
RECOMMENDED BOOKS					
1	Encyclopedia of Chemical technology	Kirck-Othmer series	-	-	-
2	Harper's Review of Biochemistry	P.W. Martin, P.A. Mayer & V.W. Rodfwell	Maurzen Asian Edition, California	15th	1981

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OPEN ELECTIVE

Course Title	GENERAL ASPECTS OF PHYSICAL CHEMISTRY							
Course Type	Soft Core- Theory		Total Hours	64	Hours/Week	04	Credits	04
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To know the concepts of enzyme kinetics, pharmacokinetics, polymers.
CO-2	To know the basic concepts of corrosion, electrode process and photodegradation
CO-3	To deliver the concepts of Polymer Science
CO-4	Explain the theories of Nanomaterials

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Synthesize of nanoparticles and its applications.	PSO-5	Apply and Analyse
CLO-2	Execute enzyme assisted chemical reactions	PSO-5	Apply and Analyse
CLO-3	The learner will be able to apply theories in electrochemistry to analyze electrode kinetic	PSO-5	Understand, Apply and Analyse
CLO-4	Apply basic chemical concepts to analyze chemical processes involved in different environmental problems	PSO-5	Understand and Apply

COURSE CONTENTS

MODULE 1		32hrs
1.1	**Enzyme kinetics:** Effect of substrate concentration (Michaelis Menton equation), Effect of *p*H, effect of catalysts and inhibitors (substrate, zeolite, Cr^{3+}, Fe^{2+} - ZnO, U.V light), effect of temperature.	8hrs
1.2	**Pharmacokinetics**: Plasma level-time curve, significance of measuring plasma drug concentrations. Drug Bioavailability: Factors affecting the drug bioavailability.	4hrs
1.3	**Polymers:** Molecular weight and size. Polydispersion. Average molecular weight concepts -number weight and viscosity average molecular weight. Principles of determination of molecular weights - End group analysis, viscosity, osmometry, cryoscopy and ebulliometry method.	8hrs
1.4	**Chemistry of Nanomaterials**: Nanoparticles. Synthesis - Laser abalation, chemical vapour transportor method (CVT) and sol-gel method. Carbon nanotube, Carbon nanowires and its composites. Synthesis of metal oxides	12hrs

	and its composite nanoparticles by solvothermal and hydrothermal method. Inorganic and organic nano porous aerogels.	
MODULE 2		**32 hrs**
2.1	**Corrosion:** Manifestations of corrosion, types of corrosion, basis of electrochemical corrosion, theories of corrosion. Local cell theory (Wagner and Traud theory) Current – potential relations (Evan diagram) in corrosion cells. Effect of *p*H, nature of metal and dissolved oxygen (principle of differential aeration) on corrosion. Corrosion inhibition and prevention by painting, phosphating and by using biomaterials, corrosion protection by anodic (passivation) and cathodic protection.	**18 hrs**
2.2	**Irrversible electrode process:** Introduction, reversible and irreversible electrodes reversible and irreversible cells. Polarization, overvoltage - ohmic overvoltage, concentration overvoltage activation overvoltage, experimental determination of overvoltage. Polarography and its application.	**8 hrs**
2.3	**Photodegradation:** Photocatalyst – ZnO, TiO_2, principle, application of ZnO/TiO_2 in the photo degradation of dyes (IC), pesticides (DDT) and in industrial effluents. Effect of photo degradation on COD value.	**6 hrs**

80% Theory 20 % Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Essentials of Physical Chemistry and Pharmacy	H.J. Arnikar, S.S. Kadam, K.N. Gujan	Orient Longman, Bombay		1992
2.	Applied Biopharmacokinetics and Pharmacokinetics	Leon Shargel, Andrew Yu	Prentice Hall International, Inc.	4th	
3.	Introduction to Statistical Thermodynamics	M. Dole	Prantice Hall		1962
4.	Nanometerials	A.K. Bandopadhyay	New age International	2nd	-
5.	Hand Book of Nanotechnology	Bharat Bhushan	Springer	-	-
RECOMMENDED BOOKS					
1	Nanotechnolog y - Importance and Applications Nanotechnology	Richard Booker and Earl Boysen	Wiley	-	-
2	Theoretical Chemistry	S. Glasstone	East West Press	-	1973

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FOURTH SEMESTER

Course Title	BIOINORGANIC CHEMISTRY							
Course Type	Hard Core- Theory		Total Hours	48	Hours/Week	03	Credits	03
Course Code	D0210	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	Inorganic composition of the living cell
CO-2	Knowledge about the oxygen carriers, ion transportation across the membrane, iron sulfur proteins
CO-3	Discuss how different elements are taken up selectively by different cells and intracellular compartments and the various ways they are exploited.
CO-4	Discuss the structure and functions of complexes and materials that are formed in the biological environment.
CO-5	Knowledge about the essential, trace and toxic elements known to us and role played by them and elaborates about the toxicity and deficiency of different elements known to us

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Learn about the various diseases due to deficiency and excess of different elements in our body	PSO-1	Understand, remember
CLO-2	Know the mechanism of the toxicity of various metals in our body	PSO-1	Understand, remember
CLO-3	Know about the various chelating agents and their action over the metal poisoning.	PSO-1	Understand, remember
CLO-4	Know the mechanism of cisplatin in the fight against cancer	PSO-1	Understand, remember
CLO-5	Importance of biological oxygen carriers in living organisms	PSO-1	Understand, remember
CL O-6	The use of metal complexes as drugs and chemotherapeutic agents.	PSO-1	Understand, remember

MODULE – 1		16 hrs
1.1	**Structural and molecular biology:** Introduction, Essential and trace metal ions in biological process, The structural building blocks of proteins, the structural building block of nucleic acids. Metal ion interactions with nucleosides and nucleotides. General features of DNA - metal complex interaction.	3 hrs
1.2	**Ion transport across cell membrane:** Structure and function of biological membranes, concentration of metal ions outside and inside cells, mechanism of ion transport across cell membrane; ionophores-channel forming and carrier ionophores; active and passive transport,Na^+/K^+ pump- importance and mechanism of action.	3 hrs

1.3	**Biochemistry of calcium:** Introduction - comparison of Ca^{2+} and Mg^{2+}. Biological roles of calcium, binding sites of calcium and proteins, storage of calcium, calcium in muscle contraction, calcium in blood clotting process.	**3 hrs**
1.4	**Metal ion transport and storage:** Iron storage and transport: Transferrin, ferratin, phosvitin and gastroferrin. Iron transport in microbes: siderophores, *in vivo* microbial transport of iron	**3 hrs**
1.5	**Bioenergetics:** Introduction, Redox reactions in metabolism, the central role of ATP in metabolism. Kinetic stability of ATP, Mitochondrial flow of electrons from NADH to O_2. Oxidative phosphorylation and respiratory chain.	**2 hrs**
1.6	**Vitamin B_{12} and Coenzymes:** Structural feature, names of different forms, chemistry of cobalamin, biochemical functions of cobalamins, model compounds. Special characteristics of B12 co-enzyme. ***Keywords: Ion transport, Bioenergetics, Vitamin B_{12}***	**2 hrs**
MODULE – 2		**16 hrs**
2.1	**Oxygen transport and oxygen uptake proteins:** Properties of dioxygen (O_2): Thermodynamic and kinetic aspects of dioxygen as an oxidant, activation of dioxygen through complexation with metal ions. Haemoglobin (Hb) and Myoglobin (Mb) in oxygen transport mechanism: Introduction to porphyrin system, substituent effects on porphyrin rings, functions of Hb and Mb. Characteristics of O_2- binding interaction with Hb and Mb. Model compounds for oxygen carriers (Vaska's complex and cobalt(III) – Schiff base complexes). non-porphyrin systems- Hemerythrin and hemocyanin.	**6 hrs**
2.2	**Electron transport proteins and redox enzymes:** Iron – sulfur proteins (rubredoxins and ferredoxins) and cytochromes.	**3 hrs**
2.3	Zinc enzymes- nature's choice of Zn(II) for non-redox enzymes, structure of active site and function of Carboxypeptidase-A, Carbonic anhydrase, alcohol dehydrogenase, Copper enzyme- Superoxide dismutase: Structure and reactivity. Molybdenum containing enzymes: Aspects of molybdenum chemistry, Xanthine oxidase, aldehyde oxidase, sulfite oxidase, nitrogenase and nitrite reductase. Iron enzyme-catalase, peroxidase and cytochrome p-450. ***Keywords:*** *Oxygen transport proteins, Zinc containing enzymes.*	**7 hrs**
MODULE - 3		**16hrs**
3.1	**Therapeutic uses of Metals - Metals in medicine:** Introduction, metals and human biochemistry, general requirements.	**3 hrs**
3.2	**Disease due to metal deficiency and treatment:** Iron, zinc, copper, sodium, potassium, magnesium, calcium and selenium.	**3 hrs**
3.3	**Metal complexes as drugs and therapeutic agents:** Introduction, antibacterial agents, antiviral agents, metal complexes in cancer therapy, metal complexes for the treatment of rheumatoid arthritis, vanadium diabetes, metal complexes as radio diagnostic agents.	**5 hrs**
3.4	**Treatment of toxicity due to inorganics:** General aspects of mechanism of metal ion toxicity, (i) Mechanism of antidote complex with poison, rendering it inert: arsenic, lead, mercury, iron, copper (ii) Antidote accelerated metabolic conversion of poison to non-toxic product: cyanide and carbon monoxide. ***Keywords: Therapatic agents, metal ion toxicity***	**5 hrs**

References:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Bioinorganic Chemistry	A.K. Das	Books and Allied (P) Ltd,		2007
2	Bioinorganic Chemistry	K. Hussain Reddy,	New Age International Ltd.,		2003
3	Bioinorganic Chemistry	R.M. Roat-Malone	Wileyinter science,	2nd	2007
4	Transition Metal Complexes as Drugs and chemotherapeutic Agents	N. Farrell, Kluwer	Academic publisher		1989
5	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
6	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
7	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004

RECOMMANDED BOOKS

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publicatio
1	Inorganic Chemistry	James E. Huheey, Ellen A. Keiter,	Pearson Education	4th	2013
2	Inorganic chemistry	Mark Weller, Tina Overton	Oxford Uiversity press	6th	2010
3	Bioinorganic Chemistry	A.K. Das	Books and Allied (P) Ltd,		2007
4	Bioinorganic Chemistry	K. Hussain Reddy,	New Age International		2003
5	Inorganic Chemistry	C.E. Housecroft and A.G. Sharpe	Pearson Education Ltd	2nd	2005
6	Inorganic Chemistry	Gary L. Miessler, Donald A. Tarr	Pearson Education	3rd	2004

Course Title	**MOLECULAR SYMMETRY AND GROUP THEORY; STATISTICAL MECHANICS, PHASE RULE STUDIES; RADIATION CHEMISTRY, POLYMERS AND ELECTROPHORESIS (HC)**							
Course Type	Hard Core- Theory		Total Hours	48	Hours/Week	03	Credits	03
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	arks	

COURSE OBJECTIVES:

CO No.	Course Objectives

CO-1	Symmetry elements and symmetry operations,
CO-2	Matrix representation of symmetry operations and applications
CO-3	Microwave rotational and vibrational (IR and Raman) spectroscopy
CO-4	Electronic transitions-UV Vis spectroscopy and application in structure elucidation
CO-5	Basics of ESR and its application to free radical system

MAPPING CLO'S WITH PSO'S AND CD'S:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Molecular symmetry and group theory	PSO-1	Understand, Apply
CLO-2	Application of group theory to vibrational spectroscopy	PSO-1	Apply, Analyse and Evaluate
CLO-3	Introduction to microwave spectroscopy	PSO-1	Understand
CLO-4	Vibrational spectroscopy towards structural elucidation	PSO-1	Understand, Apply and Analyse
CLO-5	UV-Vis spectroscopy introduction and contribution for structure elucidation	PSO-1	Understand, Apply and Analyse
CLO-6	ESR: Basics and application	PSO-1	Understand

COURSE CONTENTS:

MODULE -1 **16 hrs**

1.1 Molecular symmetry and group theory: Symmetry elements and symmetry operations. Concept of a group, definition of a point group. Classification of molecules into point groups. Subgroups, Multiplication tables (C_n , C_{2V} and C_{3V}). Matrix notation for the symmetry elements. Classess and similarity transformation. Representation of groups: The Great Orthogonality theorem and its consequences. Character tables (C s , C i , C 2 , C 2v , C 2h and C 3v). **10 hrs**

1.2 Applications of group theory: Group theory and hybrid orbital. Group theory to Crystal field theory and Molecular orbital theory (octahedral and tetrahedral complexes). Determining the symmetry groups of normal modes (both linear and non-linear molecules).theoretical selection rules: Electronic transitions (Carbonyl chromophore by taking formaldehyde as example), IR spectra, Raman spectra. **06 hrs**

Keywords: Group theory, point groups, Great orthogonality theorem, IR and Raman activity

MODULE -2 **16 hrs**

2.1 Statistical mechanics: Introduction, thermodynamic probability relation between entropy and thermodynamic probability. Partition function - translational, rotational, vibrational and excitation partition functions. 04 hrs

2.2 Phase rule studies: Application of phase rule to the two component systems – compound formation with congruent melting point and incongruent melting points, Application of phase rule to three component systems. Systems of three liquids 07hrs

2.3 Energetic of cell reactions: Effect of temperature, pressure and concentration on energetic of cell reactions (calculation of ΔG, ΔH and ΔS). Electrochemical energy sources – batteries, classification, characteristics, and primary, secondary and lithium batteries 05 hrs

Keyword: Probability, partition functions, cell reactions

MODULE -3 **16 hrs**

3.1 **Radiation chemistry**: Introduction, MODULEs, interaction of electromagnetic radiation with matter, -value, LET of radiation, Fricke dosimeter. Radiolysis - cysteine and biphenyl. Radioisotopes as tracers, use of isotopic tracers in the elucidation of reaction mechanism, structure determination and solubility of sparingly soluble substances. ^{14}C dating, medical applications of isotopic tracers. Hazards in radiochemical work and radiation protection. **04 hrs**

3.2 **Polymers**: Fundamentals of polymers - monomers, repeat MODULEs, degree of polymerization. Linear, branched and network polymers. Classification of polymers. Polymerization - Condensations, addition, free radical, ionic, co-ordination polymerization and ring opening polymerization. Molecular weight and size. Polydispersion. Average molecular weight concepts – number, weight and viscosity average molecular weight. Determination of molecular weights - viscosity method, osmotic pressure method, sedimentation and light **05 hrs**

3.3 **Electrophoresis** - principles of free electrophoresis, zone electrophoresis, gel electrophoresis and its applications in qualitative and quantitative study of proteins. Determination of isoelectric point of a protein. Electro osmosis and streaming potential and its biological significance. Biological significance of Donnan membrane phenomenon. Micelles and its involvement during digestion and absorption of dietary lipids. Diffusion of solutes across bio membranes and its application in the mechanism of respiratory exchange. "Salting In" and "Salting out" of proteins. Osmotic behaviour of cells and osmo-regulation and its application in the evolution of excretory systems of *organisms.* **07hrs**

Keyword: isotopic tracers, Osmosis, electrophoresis.

60% theory and 40 % application

REFERENCES:

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1.	Group theory in Chemistry	MS Gopinathan, V. Ramakrishnan	Vishal publishing Co.,	2nd	2016
2.	Organic Spectroscopy	W. Kemp	Pagrave Publishers, New York	3rd	1991
	Biophysical Chemistry, Principle and Technique	A. Upadhyay, K. Upadhyay and N.	Himalaya Publishing House,	-	1998.
3.	Elements of Physical Chemistry	Lewis and Glass tone	--	--	--

FOURTH SEMESTER

Course Title	KINETICS, AUTOMATED AND RADIOMETRIC METHODS							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	2	Credits	02
Course Code	D0310	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

CO No.	Course Objectives
CO-1	Kinetic methods of analysis for enzymes
CO-2	Automated methods of analysis by flow injection analysis
CO-3	Radio analysis methods
CO-4	Radio chemical analysis
CO-5	Fluorescence immune assay methods

MAPPING CLO'S WITH PSO's AND CD's

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Enzymatic methods of analysis	PSO-1	Understand, Apply and Analyse
CLO-2	Automated and flow injection methods of analysis	PSO-5	Apply, and Analyse
CLO-3	Radiochemical analytical methods	PSO-5	Apply, Analyse
CLO-4	Radio immunoassay	PSO-5	Understand, Apply , Analyse
CLO-5	Isotopic dilution methods	PSO-1	Understand, Apply

COURSE CONTENTS:

MODULE-1 **16 hrs**

1.1 **Kinetic methods of analysis:** Introduction, basis of kinetic methods, rate law expressions. Classifying chemical kinetic methods – direct-computation integral methods, direct-computation rate methods, curve-fitting methods. Instrumentation. Quantitative applications – enzyme catalyzed reactions, non-enzyme catalyzed reactions, non-catalytic reactions. Determining V_{max}, Km for enzyme catalyzed reactions. Elucidating mechanism for the inhibition of enzyme catalysis. Determination of enzymes, LDH, GOT and GPT. Determination of substrates – urea, uric acid, blood glucose and blood alcohol. Analysis of closely related compounds - neglect of reaction of slow reacting component method and logarithmic extrapolation method. **8 hrs**

1.2 **Automated methods of analysis:** An overview. Principles of automation. Automated instruments: process control. Continuous analyzers. Discrete autoanalyzers. Instruments used in automated process control. Automatic instruments - discrete and continuous flow sampling instruments. Flow injection analysis – principles - dispersion co-efficient. Factors affecting peak height, sample volume, channel length and flow rate, and channel geometry. Applications
-limited dispersion applications, medium dispersion applications, stopped flow methods and flow injection titrations. Discrete automatic systems - centrifugal fast scan analyzer, automatic organic elemental analyzers. Analysis based on multilayer films-general principles, film structures, instrumentation, performance and applications – blood urea nitrogen, blood glucose and potassium. **8 hrs**

Key word: Enzymatic methods of analysis, Flow injection systems

MODULE - 2 **16hrs**

2.1 **Radiometric methods:** Radioactive isotopes. Nuclear emissions - a andb-particles, neutrons, gamma rays and miscellaneous nuclear particles. Nuclear reactions, radiochemical decay and activity. Instrumentation and measurement of radioactivity. Radiation detectors - gas ionization, scintillation and semiconductor detectors. Pulse height analysis. Autoradiography. Statistics of **6 hrs**

radioactive measurements

2.2 **Radiochemical analysis**: **Neutron activation methods** - neutrons and their sources. Interaction of neutrons with matter. Theory, experimental considerations and applications. **3 hrs**

2.3 **Isotope dilution methods -** direct isotope dilution and inverse isotope dilution methods and their applications. Radiometric titrations. Radiorelease methods. Radioactive tracers. **3 hr**

2.4 **Radio immunoassay:** Principles of immunoassay. Specificity of immuno assays. Preparation of the antibody, incubation period for the assay, separation of the bound and free antigen. Fluorescence immunoassay. Enzyme immunoassay. **4 hrs**

Key word: Radiometric analytical methods

100% theory

References:

Sl.No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Analytical chemistry	G. D. Christian	Wiley publication	6th	2013
2	Fundamentals of Analytical Chemistry	Skoog, West, Holler, Crouch	Saunders College Publishing, New York,	8th	2004
3	Quantitative Analysis	R.A. Day and A.L. Underwood,	Prentice Hall, Inc. New Delhi,	6th	2008
4	Vogel's Textbook of Quantitative Chemical Analysis	J. Mendham, R.C. Denney, J.D. Barnes and M.J.K. Thomas	Pearson Education Pvt. Ltd., New Delhi	6th	2003.
5	Instrumental methods of chemical analysis	G. R. Chatwal, S. K. Anand	Himalaya publishing House,	5th	2002.
6	Fundamental of Analytical Chemistry	D.A. Skoog, D.M. West, Holler and Crouch	Saunders College Publishing, New York,	8th	2005
7	Principles of instrumental analysis	Holler, crouch and Skoog	Cengage Learning	6th	2007
8	Clinical chemistry and molecular diagnostics	Carl A Burtis and David E Bruns	Elsevier Inc	7th	2015

FOURTH SEMESTER

Course Title	**STATISTICAL MECHANICS AND PHASE RULE STUDIES; SOLID STATE CHEMISTRY (SC)**							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code		Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	02Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	To discuss the fundamental topics of solid state chemistry and electrokinetics
CO-2	To understand various concepts of conductors and its applications
CO-3	Exploration of the fundamental relationship between electronic structure, chemical bonding, and atomic order
CO-4	Utilize band theory to describe the operation of modern semiconductor devices.
CO-5	Specify atomic planes, directions, and families of planes and directions within a given crystal structure using Miller indices

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Will learn the unique properties organic and inorganic solids have with respect to compounds in solution or in the gas phase	PSO-5	Understand, Apply and Analyse
CLO-2	Will learn the methods used to prepare, purify, and crystallize that are specific for organic and inorganic solids.	PSO-5	Apply and understand
CLO-3	Use of spectroscopic, diffraction, microscopic, thermal, and magnetic methods to characterize organic and inorganic solids	PSO-5	Apply, Analyse and Evaluate
CLO-4	Describe solid state phase relations, their chemical synthesis, and thermodynamical and kinetic parameters reaction kinetics as well as characterization methods	PSO-5	Understand, Apply and evaluate
CLO-5	Design and development of materials with pre-required properties based on the structure of solids.	PSO-5	Understand and Analyse
CLO-6	To identify the compounds being oxidized and reduced.	PSO-5	Analyse and understand

COURSE CONTENTS

MODULE-1 **16 hrs**

1.1 **Statistical mechanics**: Introduction, thermodynamic probability relation between entropy and thermodynamic probability. Partition function - translational, rotational and vibrational partition functions **4 hrs**

1.2 **Phase rule studies:** Application of phase rule to the two component systems – compound formation with congruent melting point and incongruent melting points, Roozeboom's classification. Application of phase rule to three component systems. Systems of three liquids **6 hrs**

1.3 **Energetics of cell reactions:** Effect of temperature, pressure and concentration on energetics of cell reactions (calculation of ΔG, ΔH and ΔS). Electrochemical energy sources – batteries, classification, characteristics, primary, secondary and lithium batteries **6 hr**

Key words : partition functions, component, phase rule

MODULE-2 **16 hrs**

2.1	Fundamentals of X-ray crystallography, law of interfacial angles, laws of symmetry, Miller indices, Bragg equation (no derivation), Experimental methods – powder and rotating crystal methods, indexing of powder and rotating crystal photographs. Atomic scattering factor, structure factor, Fourier synthesis and electron density diagrams. Electron diffraction of gases, experimental technique, Scattering-Intensity curves, Wierl equation (no derivation), Radial distribution method determination of bond lengths and bond angles.	**16 hrs**

Key words: Miller indices, Electron diffraction, crystallography

60% Theory and 40% Problems

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Fundamentals of Crystallography	D.D. Sood, A.V.R. Reddy and N. Ramamoorthy	----	--	-
2	Chemical Kinetics	K.J. Laidler	Pearson	3rd	2003
3	Physical Chemistry	G.M. Barrow	Macmillan Publishers	--	1998
RECOMMENDED BOOK					
1	Essentials of Physical Chemistry and Pharmacy	H.J. Arnikar, S.S. Kadam, K.N. Gujan	Orient Longman, Bombay	---	1992

FOURTH SEMESTER

Course Title	**STRUCTURAL METHODS IN INORGANIC CHEMISTRY**							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code	DP320	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

Course Objectives:

CO No.	Course Objectives
CO-1	To introduce multistep inorganic compound synthesis and its spectroscopic analysis, separation and estimation of organic compounds.
CO-2	Composition of metal and ligand by spectrophotometric method.
CO-3	To acquire knowledge of laboratory techniques, inorganic synthesis and characterization.
CO-4	To impart the knowledge about redox, complexometric and precipitation reactions.
CO-5	Determination of stability constants of inorganic compounds.

Course Learning Outcome:

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	To acquire knowledge of laboratory techniques , inorganic synthesis and characterization.	PSO-3	Analyze, Apply Evaluate
CLO-2	Estimate the ions present in the sample by different techniques.	PSO-3	Analyze, Apply Evaluate
CLO-3	Flame photometric determination.	PSO-3	Analyze, Apply Evaluate
CLO-4	Determination of stability constants of inorganic compounds.	PSO-3	Analyze, Apply Evaluate
CLO-5	To impart the knowledge about redox, complexometric and precipitation reactions.	PSO-3	Analyze, Apply Evaluate

MODULE -1		**16 hrs**
1.1	**NMR spectroscopy:** Basic principles, chemical shift and factors affecting it, coupling constants. 19F, 31P - NMR and NMR of paramagnetic complexes. Double resonance technique, The Nuclear Overhauser Effect, Magnetic susceptibility measurements b y Evan's method. NMR to solids.	**5 hrs**
1.2	**ESR spectroscopy:** Theory, presentation of the spectrum, hyperfine coupling, the g value and factors affecting the magnitude of the g value. Zero-field splitting and Kramers' degeneracy. Application to simple inorganic and organic free radicals and to metal complexes.	**5 hrs**
1.3	**NQR spectroscopy:** Theory, energies of the quadrapole transitions, instrumentation, effect of magnetic field on the spectra, relationship between electric field gradient and molecular structures. Applications - interpretation of e2Qq data, structural information from NQR data	**6 hrs**
MODULE -2		**16 hrs**
2.1	**Vibrational spectroscopy:** Introduction, theory of infrared absorption, theoretical group frequencies, correlation chart. Applications to coordination compounds - aquo, amine, urea, DMSO, *cis* and *trans* metal complexes. Change in spectra accompanying change in symmetry upon coordination (nitrite, sulphate, nitrate,	**6 hrs**
2.2	**Mossbauer spectroscopy:** Theoretical basis, interpretation of Mossbauer spectra - isomer shift, quadrupole splitting and magnetic hyperfine structures. Application: $I2Br_2Cl_4$, $Fe_3(CO)_{12}$, Prussian blue, nitroprusside, hexacyanoferrate.	**5 hrs**

2.3	**Photoelectron spectroscopy:** Introduction, principles, chemical shifts, photoelectron spectra of simple molecules, X-ray photoelectron and Auger electron spectroscopy. Applications.	
2.4	**Mass spectrometry:** Theory, experimental techniques, molecular ions, fragmentation and ion reaction, Applications to coordination compounds.	**5 hrs**

References:

1	Physical Methods in Inorganic Chemistry	R.S. Drago, Saunders Publishers		1966
2	. Structural Methods in Inorganic Chemistry	E.A.V. Ebsworth, D.W.H. Ranklin and Cradock	Blackwell Scientific Publications	1988
3	Electronic Absorption Spectroscopy and Related Techniques	D.N. Sathyanarayana	Universities Press	2001

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FOURTH SEMESTER

Course Title	**PHOTOCHEMISTRY, PERICYCLIC REACTIONS AND ORGANOMETALLIC CHEMISTRY**							
Course Type	Soft Core- Theory		Total Hours	32	Hours/Week	02	Credits	02
Course Code	D0340	Evaluation	Internal	C1+C2 = 15+15			30 Marks	100
			External	Duration	03Hrs	C3	70 Marks	

COURSE OBJECTIVES:

CO No.	Course Objectives
CO-1	The concepts involved in understanding organic photochemical reactions, their mechanisms and applications in organic synthesis will be presented.
CO-2	Synthetic applications and mechanisms of various pericyclic reactions will be discussed.
CO-3	Different methods of analysis of pericyclic reactions to arrive at the Woodward-Hoffmann rules will be presented
CO-4	Develop basic skills for the multi-step synthesis of organic compounds
CO-5	Understanding the solution of problems related to the synthesis of organic target molecules

MAPPING CLO'S WITH PSO'S AND CD'S

CLO No.	Course Learning Outcomes On Completion of the Course the Students will Learn to	PSOs Addressed	Cognitive Level
CLO-1	Able to predict the stereochemistry & products of the Pericyclic reactions	PSO-1	Apply and Analyse
CLO-2	Envisage an organic photochemical reaction and identify the product with the type of functional group present on the molecule	PSO-5	Understand and Apply
CLO-3	Predict the course of Apply photochemistry concepts, plan and program molecules for photochemical application of specific interest	PSO-5	Understand, Apply and Analyse

CLO-4	Rationalisation of the outcome of organic reactions using mechanistic reasoning	PSO-5	Understand and Apply
CLO-5	Acquire Knowledge about various new modern organic synthetic reactions and their mechanisms involving in the formation of C-C, C-X, C=C bonds.	PSO-5	Understand and Apply

COURSE CONTENTS

MODULE- 1 **16 hrs**

1.1 Photochemistry and concerted reactions: Introduction, light absorption and electronic transitions, Jablonski diagram, intersystem crossing, energy transfer, sensitizers, quenchers. Photochemistry of olefins, conjugated dienes, aromatic compounds, ketones, enones, photooxidations, photoreductions, Norrish type I and II reactions, Paterno-Buchi reaction, Barton reaction, Di-pi-rearrangements. **6 hrs**

1.2 Electrocyclic reactions: Stereochemistry, symmetry and Woodward-Hofmann rules for electrocyclic reactions, FMO theory of electrocyclic reactions, correlation diagram for cyclobutadiene and cyclohexadiene systems. **3 hrs**

1.3 Cycloaddition reactions: Classification, analysis by FMO and correlation diagram method. 1,3-dipolar cycloadditions: involving nitrile oxide, nitrile imine, nitrile ylide cycloaddition. Intra and intermolecular 3+2 cycloaddition and their application in organic synthesis. **3 hrs**

1.4 Cycloaddition reactions: Deils-Alder reacion, hetero Diels-Alder reaction and their applications. Sigmatropic reactions: Classification, stereochemistry and mechanisms. suprafacial and antarafacial shifts of H, sigmatropic shifts involving carbon moieties. [3,3] and [5,5]- sigmatropic rearrangement, Claisen, Cope and aza-Cope rearrangement. **4 hrs**

Keywords:Organic photochemistry, Concerted Reactions

MODULE- 2 **16 hrs**

2.1 Chemistry of organometallic compounds: Synthesis and reactions of organolithium (n-BuLi, PhLi), organomagnesium (Grignard reagent). **2 hrs**

2.2 Organoaluminium reagents: Preparation, site selective and stereoselctive additions of nucleophiles mediated by organoaluminum reagents, reaction with acid chlorides, allyl vinyl ethers, 1,2-addition to imines and application in the synthesis of natural products. **3 hrs**

2.3 Organopalladium compounds: Suzuki coupling, Heck reaction. **1 hr**

2.4 Organotin reagents: Barton decarboxylation reaction, Barton deoxygenation reaction, Stille coupling, Stille-Kelley coupling reactions, Barton McCombie reaction, Keck stereoselective allylation and other applications. **3 hrs**

2.5 Organoboron compounds: Introduction and preparations. Hydroboration and its applications. Reactions of organoboranes: isomerization reactions, oxidation, protonolysis, carbonylation, cyanidation. Reaction of nonallylic boron stabilized carbanions: alkylation reactions, acylation reaction, Reactions with aldehydes or ketones (E and Z-alkenes). **4 hrs**

2.6 Organosulphur compounds: Introduction. Preparations, reactions, mechanism and synthetic applications of important sulphur containing reagents like dithiane, sulphur ylides etc. Organosilicon compounds: Introduction, preparations and reactions, Peterson reaction. **3 hrs**

Keywords: Named Reactions, Catalytic Cycle

REFERENCES

Sl. No	Title of the book	Name of the author	Name of the publisher	Edition	Year of publication
1	Pericyclic Reactions – A textbook.	S. Sankararaman,	Wiley-VCH	-	2005
2	Advanced Organic Chemistry	J. March	Wiley	7th	2013
3	Advanced Organic Chemistry	F. A. Carey and R. J. Sundberg,	Plenum	3rd	1990
4	Principles of Organic Synthesis,	ROC Norman and J. M. Coxon	CRC Press	6th	1996
5	Photochemistry and Pericyclic Reactions	Jagdamba Singh, Jaya Singh	New Academic Science	-	2009
RECOMMENDED BOOKS					
1	Comprehensive Organic Synthesis	B.M. Trost and I. Fleming	Pergamon Press	-	1991
2	Advanced General Organic Chemistry	S.K. Ghosh	Book and Allied	6th	1998